Maulana Iskak
17/413560 /TK /46000
PSI TF-B

1) Tujuan ekonomis FCC

Operasi optimal pada suatu pabrik ditunjukkan oleh persamaan:

J = penggunaan bahan baku + penggunaan energi − produksi wet gas
− (penggunaan bahan baku − produksi wet gas) + penggunaan energi

Pada proses di FCC, energi diperlukan untuk memanaskan feed pada furnace dan untuk menggerakkan kompresor lift air, combustion air, wet gas. Sumber bahan bakar yang digunakan untuk pemanasan feed di furnace, pembangkitan listrik untuk motor combustion air dan wet gas, serta produksi steam untuk turbin lift air berasal dari hasil produksi hidrokarbon FCC (tidak membeli bahan bakar), sehingga profit dari efisiensi energi tidak signifikan secara ekonomi. Oleh karena itu dipilih cara kedua yaitu memaksimalkan produksi pada FCC agar profit maksimal.

Tujuan utama dari strategi kontrol untuk memaksimalkan produksi adalah dengan cara memaksimalkan total feed flow rate FCC dengan tetap mempertahankan suhu riser dan peralatan lainnya berada pada batasan kondisi operasinya. Suhu perengkahan pada riser biasanya berada pada rentang 925-1050°F. Akan tetapi karena ada batasan suhu metalurgi riser, yaitu 995°F, suhu riser dijaga pada rentang 925-995°F. Karena reaksi bersifat endotermik, untuk mendapatkan rentang suhu riser tersebut suhu feed diatur pada 400-700°F (dengan mengatur pembakaran pada furnace) dan suhu katalis diatur pada 1250-1350°F (dengan mengatur pembakaran pada regenerator).
Ketika feed flow rate dinaikkan, tekanan reaktor akan meningkat karena produksi gas hasil perengkahan. Kenaikan tekanan reaktor dapat mengurangi perengkahan produk beroktan tinggi sehingga perlu dijaga tekanannya. Reaktor mempunyai batas tekanan maksimal 49,2 psig.
Kemudian ketika tekanan reaktor naik, tekanan regenerator juga perlu dinaikkan agar beda tekanan reaktor − regenerator berada pada rentang (-5) - 2 psi. Beda tekanan ini mempengaruhi sirkulasi katalis yang akan mempengaruhi proses perengkahan. Regenerator mempunyai tekanan maksimum 39,7 psig.
Selain itu, dalam memaksimalkan feed flow rate juga terdapat batasan maksimum untuk wash oil, diesel oil, gas oil, slurry oil masing-masing sebesar 17 lb/s, 16 lb/s, 144 lb/s, 10 lb/s

Secara singkat, tujuan strategi kontrol FCC adalah :
- Memaksimalkan feed flow rate dengan batasan maksimum wash oil, diesel oil, gas oil dan slurry oil berturut-turut sebesar 17 lb/s, 16 lb/s, 144 lb/s, dan 10 lb/s. Dalam memaksimalkan feed flow rate perlu dipertimbangkan suhu riser dan beda tekanan reaktor-regenerator yang masing-masing berada pada rentang 925-995 °F dan (-5)-2 psi agar proses perengkahan tetap berlangsung optimal meskipun feed flow rate bertambah sehingga didapatkan produksi wet gas yang maksimal.
- Selain batasan-batasan di atas, terdapat pula batasan lainnya :
  * Batasan keselamatan equipment
    - Tekanan maksimum reaktor, $P_{4,max} = 49,2$ psig
    - Tekanan menara fraksinasi maksimum, $P_{5\,max} = 39,7$ psig
    - Tekanan maksimum regenerator, $P_{6\,max} = 39,7$ psig
    - Suhu cyclone maksimum, $T_{cyc\,max} = 1310$ °F
    - Suhu regenerator-bed : $1265 \leq T_{reg} \leq 1310$ °F
    - Suhu furnace maksimum, $T_{3\,max} = 1700$ °F
    - Ketinggian katalis : $0 \leq L_{sp} \leq 20$ ft
    - Beda suhu cyclone-regenerator : $0 \leq \Delta T \leq 20$ °F
    - Flow rate fuel gas maksimum : $F_{5\,max} = 40$ scf/s
  * Batasan keselamatan lingkungan
    - Konsentrasi O_2 minimum, $C_{O_2,min} = 1,5$ mol %
    - Konsentrasi CO maksimum, $C_{CO,max} = 350$ ppm
- Dari tujuan strategi kontrol didapatkan batasan untuk kondisi ekonomis
  - Flow rate wash oil maksimum, $F_{1\,max} = 17$ lb/s
  - Flow rate diesel oil maksimum, $F_{2\,max} = 16$ lb/s
  - Flow rate gas oil maksimum, $F_{3\,max} = 144$ lb/s
  - Flow rate slurry oil maksimum, $F_{4\,max} = 10$ lb/s
  - Suhu riser : $925 \leq T_r \leq 995$ °F
  - Beda tekanan reaktor-regenerator : $(-5) \leq \Delta P_{RR} \leq 2$ psi
  - Suhu feed preheater : $400 \leq T_2 \leq 700$ °F
  - Suhu regenerated katalis : $1250 \leq T_{regc.} \leq 1350$ °F

<<requirement>>
Fluid Catalytic Cracking
id = "1"
Text = "Kebutuhan FCC"

<<requirement>>
Ekonomis
Id = "1.1"
Text = "maksimalkan feed flow rate"

<<derive req>>

<<requirement>>
Batasan ekonomis
Id = "1.1.1"
Text = " $F_1 \leq 17$ lb/s
$F_2 \leq 16$ lb/s
$F_3 \leq 144$ lb/s
$F_4 \leq 10$ lb/s
$925 \leq T_r \leq 995\,°F$
$(-5) \leq \Delta P_{RR} \leq 2$ psi
$400 \leq T_2 \leq 700\,°F$
$1250 \leq T_{reg,c} \leq 1350\,°F$ "

<<requirement>>
keselamatan peralatan
Id = "1.2.1"
Text = " $F_5 \leq 40$ scf/s
$P_4 \leq 49,2$ psig
$P_5 \leq 39,7$ psig
$P_6 \leq 39,7$ psig
$T_{cyc} \leq 1310\,°F$
$1265 \leq T_{reg} \leq 1310\,°F$
$T_3 \leq 1700\,°F$
$0 \leq L_{sp} \leq 20$ ft
$0 \leq \Delta T \leq 20\,°F$ "

<<requirement>>
Keselamatan
Id = "1.2"
Text = " Batasan keselamatan"

<<requirement>>
Keselamatan lingkungan
Id = "1.2.2"
Text = " $C_{O_2} \geq 1,5$ mol%
$C_{CO} \leq 350$ ppm"

2) Analisis dan perhitungan letak katup kontrol → faktor friksi diabaikan

a) Wash oil flow (V_1)

- control valve dipasang pada F_1 untuk memanipulasi laju aliran wash oil pada FCC.
- Kondisi operasi :

$P_2 = 260$ psig → emerson source book : refinery.

$P_1 = 270$ psig → asumsi $\Delta P = 10$ psig untuk kontrol yang baik

$\Delta P = 145$ psig

$\dot{m}_m = 17$ lb/s

$\dot{m}_n = 13,8$ lb/s

$SG = 0,916$ → dari API gravity $= 23$ → $SG = \frac{141,5}{131,5 + 23} = 0,916$

$\rho = 62,4 \times SG = 62,4 \times 0,916 = 57,16$ lb/ft³

- Untuk discharge pompa, kecepatan fluida dapat didekati dengan:

$u = (5 + D/3)$ ft/s

$q = A \cdot u$

$q = \frac{\pi}{4} D^2 \left(5 + \frac{D}{3}\right)$

$\frac{4q}{\pi} = 5D^2 + \frac{D^3}{3}$

untuk q_m, maka :

- $q_m = \frac{\dot{m}_m}{\rho} = \frac{17}{57,16} = 0,297$ ft³/s $= 2,22$ gal/s
- $\frac{4 \times 0,297}{\pi} = 5D^2 + \frac{D^3}{3}$

$D = 0,27$ ft

$= 3,24'' \approx 4''$ pipa → NPS 4 valve

- Tidak mengkonsumsi tekanan yang besar → dipilih linear control valve
- Ketika terjadi kegagalan diharapkan feed wash oil berhenti → dipilih air to open control valve
- control valve untuk mengatur laju aliran (throttling), tight shut off, → dipilih single port, sliding stem globe valve, class IV shutoff
- $C_v = \frac{q_m}{f(e)} \sqrt{\frac{SG}{\Delta P}} = \frac{133,2}{1} \sqrt{\frac{0,916}{10}} = 40,31$

- $q_m = 2,22$ gal/s $= 133,2$ gal/m
- faktor friksi diabaikan

$l_n = 81,18\%$ $\approx 80\%$

- Digunakan aktuator dengan diafragma karena sinyal pneumatic 3-15 psi
- Dipilih : Emerson easy-e EAD & 667 actuator

CV → 100% = 41, 80% = 35,7, 10% = 2,95, NPS 2"

b) Diesel oil flow valve (V_2)

- Digunakan untuk memanipulasi laju aliran diesel oil pada feed FCC
- Tidak mengonsumsi tekanan yang besar → linear control valve
- Ketika terjadi kegagalan, feed diesel oil berhenti → air to open control valve
- Digunakan untuk mengatur laju alir (throttling), tight shut off → single port, sliding stem globe valve dengan Class IV shut off
- Kondisi operasi :

$P_o = 260$ psig — $\dot{m}_m = 16$ lb/s — $\rho = 62,4 \times 0,916 = 57,16$ lb/ft³

$P_i = 270$ psig (asumsi u/ kontrol yg baik) — $\dot{m}_n = 0$ lb/s

$\Delta P = 10$ psig — SG = 0,916 (API : 23)

- Kecepatan fluida discharge pump :

$u = (5 + D/3)$ ft/s → $\dfrac{4 q_m}{\pi} = 5D^2 + \dfrac{D^3}{3}$

$\dfrac{4 \times 0,28}{\pi} = 5D^2 + \dfrac{D^3}{3}$

$D = 0,26$ ft
$= 3,12$ "
≈ 4 " → NPS 4 -pipa

$q_m = \dfrac{16}{57,16} = 0,28$ ft³/s
$= 2,09$ gal/s
$= 125,4$ gpm

- $C_V = \dfrac{125,4}{1}\sqrt{\dfrac{0,916}{10}} = 37,95$ | $l_n = 0$ %

- Digunakan aktuator tipe diafragma karena sinyal pneumatic 3-15 psi
- Dipilih : Emerson easy-e ETAD & 667 actuator → CV : 100 % : 41, 10 % = 2,95 | NPS = 2"

c) Total fresh feed flow valve (V_3)

- Digunakan untuk memanipulasi total laju aliran fresh feed di FCC
- Tidak mengonsumsi banyak tekanan → linear control valve
- Ketika fail, fresh feed pada FCC harus di stop agar produksi berhenti → air to open control valve
- Untuk mengatur laju alir (throttling) dengan tight shut off → single port, sliding stem globe valve dengan Class IV shut off
- Kondisi operasi :

emerson source book refining.

$P_o = 100$ psig — $\Delta P = 160$ psig — $\dot{m}_n = 126$ lb/s — $\rho = 57,16$ lb/ft³

$P_i = 260$ psig — $\dot{m}_m = 177$ lb/s — SG = 0,916 (API : 23)

- Kecepatan fluida discharge pump

$u = (5 + D/3)$ ft/s

$\dfrac{4 q_m}{\pi} = 5D^2 + \dfrac{D^3}{3}$

$\dfrac{4 \times 3,096}{\pi} = 5D^2 + \dfrac{D^3}{3}$

$D = 0,86$ ft
$= 10,32$ "
≈ 11 " → NPS 11 -pipa

$q_m = \dfrac{177}{57,16} = 3,096$ ft³/s $= 23,16$ gal/s
$= 1389,6$ gpm

• $Cv = \frac{1389,6}{1} \sqrt{\frac{0,916}{160}} = 105,14$ | $ln = 71,19\% \approx 70\%$

• Digunakan aktuator tipe diafragma karena sinyal pneumatic 3-15 psi
• Dipilih Emerson easy e ET & 667 actuator → CV : 100% = 108,70% = 72,9
10% = 7,02, NPS = 4"

d) Slurry oil flow valve (V_4)
- Untuk memanipulasi laju aliran slurry recycle oil feed FCC
- Tidak mengonsumsi banyak tekanan → linear control valve
- Ketika fail, slurry oil feed flow harus dihentikan agar produksi berhenti → air to open control valve
- Untuk mengatur laju aliran (throttling) dengan tight shut off → single port, sliding stem globe valve dengan class IV shut off
- Kondisi operasi : → Terdapat pompa untuk memompa slurry

$P_o = 110$ psig (asumsi) | $\Delta P = 10$ psig | $\dot{m}_m = 10$ lb/s
$P_i = 100$ psig (sama dengan fresh feed) | $SG = 0,916$ (API : 23) | $\dot{m}_n : 5,25$ lb/s
$\rho = 57,16$ lb/ft³

- Kecepatan discharge slurry dapat didekati dengan : $u = (5 + D/3)$ ft/s :

$$\frac{4 q_m}{\pi} = 5D^2 + \frac{D^3}{3}$$

$$\frac{4 \times 0,175}{\pi} = 5D^2 + \frac{D^3}{3}$$

$D = 0,21$ ft
$= 2,52''$ → NPS 3 → pipa.

$q_m = \frac{10}{57,16} = 0,175$ ft³/s $= 1,31$ gal/s $= 78,6$ gpm

- $Cv = \frac{78,6}{1} \sqrt{\frac{0,916}{10}} = 23,79$ | $ln = 52,5\% \approx 50\%$

- Digunakan aktuator tipe diafragma karena sinyal pneumatic 3-15 psi
- Dipilih Emerson easy e ET & 667 actuator → CV : 24,2 (100%), 3,97 (10%)
12,9 (50%), NPS : 1"

e) Fuel gas flow valve (V_5) → fuel gas : natural gas
- untuk memanipulasi temperatur furnace pre heater dengan memanipulasi fuel flow rate
- Furnace mengonsumsi banyak tekanan → equal precentage control valve
- Ketika fail, fuel gas harus dihentikan agar pembakaran pada furnace berhenti → air to open control valve
- Untuk mengatur laju aliran (throttling) dengan tight shut off → single port slide stem globe valve dengan class IV/V shut off
- Kondisi operasi :

emerson sourcebook refining { $P_i = 190$ psig $= 204,7$ psia ; $P_o = 100$ psig $= 114,7$ psia }
$\Delta P = 90$ psia
$x = \Delta P / P_i = 0,47$

$q_m = 40$ scf/s
$q_n = 34$ scf/s
$SG = 0,7$

$T = 105°F$
$M = 17,79$ lb/lbmol
$X_T = 0,72$ (to close)
$F_\gamma = 0,907$
$Z = 1$ (gas ideal)

$$Y = 1 - \frac{x}{3 F_\gamma X_T} = 1 - \frac{0,47}{3 \cdot 0,907 \cdot 0,72} = 0,76$$

- Fuel gas dikompresi, kecepatan discharge kompresor :

$u = 20D$ ft/s

$q = A \cdot u$

$q = \frac{\pi}{4} D^2 \cdot 20D$

$\frac{q}{\pi} = D^3$

$D = \sqrt[3]{\frac{q}{5\pi}}$

untuk q_m :

$D = \sqrt[3]{\frac{q_m}{5\pi}}$

$= \sqrt[3]{\frac{3,12}{5\pi}}$

$= 0,58$ ft

$= 6,96$ "

≈ 7 " → NPS 7 → pipa

$q_m = 40$ scf/s

$q_m (cf/s) = \frac{14,696}{204,7} \times \frac{(105+460)}{519,7} \times 40$

$q_m = 3,12$ cf/s

- $C_V = \frac{Q}{N P_1 Y f(\ell)} \sqrt{\frac{MTZ}{x}} = \frac{149.000}{7320 \times 204,7 \times 0,76 \times 10} \sqrt{\frac{17,79 \times 565 \times 1}{0,47}}$

$= 18,47$

$q_m = 40$ scf/s $= 149.000$ scf/h

→ saat q_n → $f(\ell) = 0,85$ → $\ell = 95\% \approx 100\%$

- Digunakan actuator tipe diafragma karena sinyal pneumatic 3-15 psi
- Dipilih Emerson easy-e ETAD & 677 actuator → $C_V = 100\% = 18,5$

$10\% = 1,02$; NPS = 1"

f) wet gas valve (V_{11}) → propana

- Untuk memanipulasi laju aliran wet gas produk yang dapat mempengaruhi tekanan menara fraksinasi
- Menara fraksinasi mengonsumsi banyak tekanan → equal precentage
- Ketika fail, valve dibuka penuh agar tidak terjadi over pressure pada menara fraksinasi → air to close control valve
- untuk mengatur laju aliran (throttling) dengan tight shut off → single port, sliding stem globe valve dengan class IV shut off.
- Kondisi operasi :

$P_o = P_7 = 22,68$ psia

$P_i = P_5 = 23,52$ psia

$\Delta P = 0,84$ psig

$R = 20$ (asumsi)

$P_{VRU} = 101$ psia

$V_{11} = 0,95$

$K_{11} = 1,5$ mol/s $\sqrt{psia}$

$x = 0,84 / 23,52 = 0,036$

$M = 44,097$ lb/lbmol (propana)

$SG = 1,5$

$T = 100,4$ °F

- $F_{V11} = K_{11} \cdot f_{pp}(V_{11}) \sqrt{P_5 - P_7}$

$= 1,5 \cdot 0,83 \sqrt{0,84}$

$= 1,14$ mol/s $= 2,51 \times 10^{-3}$ lbmol/s

$f_{pp}(0,95) = \exp(2 \ln(0,15)(1-0,95))$

$= 0,83$

- $q = \frac{F_{V11} \cdot R \cdot T}{144 \cdot P} = \frac{2,51 \cdot 10^{-3} \times 1545 \times (100,4+460)}{144 \cdot 22,68} = 0,665$ ft³/s

$q (scf/s) = \frac{P}{14,696} \times \frac{519,7}{T} \times q = \frac{22,68}{14,696} \times \frac{519,7}{560,4} \times 0,665 = 0,95$ scf/s

$= 3420$ scf/h

- Kecepatan suction kompresor → $U = 20D$ ft/s

$q = Au$

$D = \sqrt[3]{\frac{q}{5\pi}} = \sqrt[3]{\frac{0,665}{5\pi}} = 0,35$ ft $= 4,2$ " ≈ 5 " → NPS 5 → pipa

- $X_T = 1$
- $F_\gamma = 0,821$

$Y = 1 - \frac{0,036}{3 \times 0,821 \times 0,72} = 0,98$

VISION

- $Cv = \dfrac{3420}{20^{-0,05} \times 23,52 \times 7320 \times 0,98} \sqrt{\dfrac{44,097 \times 560,9}{0,036}} = 19,51$ | $ln = 95\% \approx 100\%$
- Digunakan aktuator tipe diafragma → sinyal pneumatic 3-15 psi
- Dipilih : Emerson easy-e EAD & 667 actuator → CV : 100% = 18,5, 10% = 1,02

g) Flue gas valve (V_{19}) | NPS = 1"

- Untuk memanipulasi laju flue gas yang dapat mempengaruhi tekanan regenerator
- Regenerator mengonsumsi banyak tekanan → equal precentage control valve
- Ketika fail, valve dibuka penuh agar tidak terjadi over pressure pada regenerator → air to close control valve
- Untuk mengatur tekanan regenerator, secara akurat dengan mengatur laju alir → single port, sliding stem globe valve → Emerson easy-e ET & 677 actuator
- Kondisi operasi :

$P_i = P_6 = 29,64$ psia | $R = 20$ (asumsi) | $X = 14,94/29,64 = 0,5$
$P_o = 14,7$ psia | $T = T_{cyc} = 1283,6\ °F$ | $Mr = 28,97$ lb/lbmol (udara)
$\Delta P = 14,94$ psia | $V_{19} = 0,61$ | $SG = 1$
$F_{sg} = 2,60$ mol/s = 0,0057 lbmol/s

* asumsi : $U = 20$ ft/s

$D = \sqrt[3]{\dfrac{q}{5\pi}} = \sqrt[3]{\dfrac{3,598}{5\pi}} = 0,61$ ft $= 7,32"$ → NPS 8

$q_{sg} = \dfrac{0,0057 \times 1545 \times (1743,6)}{144 \times 29,64} = 3,598$ cf/s

$X_T = 0,72$, $F\gamma = 1$ → $Y = 1 - \dfrac{0,5}{3 \times 1 \times 0,72} = 0,768$

- q_{sg} (scf/s) $= \dfrac{29,64}{14,696} \times \dfrac{519,7}{1743,6} \times 3,598 = 2,163$ scf/s $= 7786,8$ scf/h

- $Cv = \dfrac{7786,8}{20^{-0,39} \times 7320 \times 29,64 \times 0,72} \sqrt{\dfrac{28,97 \times 1743,6}{0,5}} = 50,96$ | $ln = 61\% \approx 60\%$

- Digunakan aktuator tipe diafragma (pneumatic 3-15 psi)

* Dipilih easy e ES valve, 667 actuator → 100% = 56,2
60% : 16,3, 10% : 1,74 "

h) Lift air valve (V_{lift})

- untuk memanipulasi laju udara pada regenerator agar terjadi pembakaran sempurna
- Regenerator mengonsumsi banyak tekanan → equal precentage
- Ketika fail, valve ditutup penuh agar katalis berhenti diregenerasi dengan menghentikan lift air → air to open control valve
- Untuk mengatur kecepatan regenerasi (mendorong katalis) dengan mengatur laju alir, tight shut off, beda tekanan rendah → butterfly valve, class IV shut off
- Kondisi operasi

$P_i = P_3 = 40,5$ psia | $T = 270\ °F$ | $SG = 1$
$P_o = P_6 = 29,64$ psia | $V_{lift} = 0,424$ | $X = 10,86/40,5 = 0,27$
$\Delta P = 10,86$ psia | $M = 28,97$ lb/lbmol | $X_T = 0,55$ (to open)
$Fg = 14,51$ lb/s | $F\gamma = 1$ | $Y = 1 - \dfrac{0,27}{3 \times 1 \times 0,55} = 0,836$

$$q_g = \frac{Fg \cdot RT}{144 \cdot P \cdot M} = \frac{14{,}51 \times 1545 \times 730}{144 \times 40{,}5 \times 28{,}97} = 98{,}86\ ft^3/s$$

$$q_g\,(scf/s) = \frac{40{,}5}{14{,}696} \times \frac{519{,}7}{730} \times 98{,}86 = 193{,}96\ scf/s = 698.256\ scf/h$$

$$Cv = \frac{698.256}{20^{0{,}576} \times 7320 \times 40{,}8 \times 0{,}836} \sqrt{\frac{28{,}97 \times 730}{0{,}27}} = 4395{,}26$$ $ln = 42{,}9\% \approx 40\%$

- Kecepatan discharge kompresor : $u = 200\ ft/s$

$$D = \sqrt[3]{\frac{q}{5\pi}} = \sqrt[3]{\frac{98{,}86}{5\pi}} = 1{,}85\ ft = 22{,}2'' \approx 24'' \rightarrow \text{NPS } 24 \rightarrow \text{pipa}$$

- Digunakan aktuator tipe diafragma → sinyal pneumatic 3-15 psi
- Dipilih Fisher 8590 & 667 actuator (Emerson) → CV : 100% : 13565, 40% : 3998, 10% : 823, NPS : 24"

7) Spill air valve (V_9)
   - untuk manipulasi laju alir spill air
   - regenerator konsumsi banyak tekanan → equal precentage
   - Hanya digunakan ketika pembersihan → air to open
   - karena digunakan ketika pembersihan, maka control valve selalu ditutup 100% → tidak bisa dimampulasi (tidak masuk degree of freedom) sehingga spill air valve tidak dianalisis & tidak dibuat spesifikasinya. Akan tetapi tetap digambar di P&ID

J) Combustion air valve (V_6)
   - untuk mampulasi combustion air
   - combustion air valve dibuka 100% untuk menjaga agar terjadi pembakaran sempurna di regenerator. Oleh karena itu combustion air valve tidak dapat dimanipulasi (tidak masuk degree of freedom) sehingga tidak dianalisis & tidak dibuat spesifikasinya, akan tetapi tetap digambar di P&ID

* Total derajat kebebasan = 8

ISA S20.50, Rev. 1

CONTROL VALVE DATA SHEET

Second Printing

PROJECT 2020/FCC/02	DATA SHEET 01 of 29
UNIT #1	SPEC PSI-2020-01
P.O. PO. 12345	TAG FV-01
ITEM 1	DWG PSI-FCC-2020
CONTRACT 2020-psi-Maulana	SERVICE R-01 feed
*MFR. SERIAL D100 390 X 012	

			Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
1		Fluid				Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	lb/s	17	13,8	0	–	
3		Inlet Pressure	psig	270	270	270	-	
4		Outlet Pressure	psig	260	260	260		
5		Inlet Temperature	°F	460,9	460,9	460,9		
6		Spec Wt/Spec Grav/Mol Wt		0,916	0,916	0,916	–	
7		Viscosity/Spec Heats Ratio		-	-	-	–	
8		Vapor Pressure P_V		-	-	-	–	
9		*Required C_V		41	35,7	2,95	–	
10		*Travel	%	100	80	10	0	
11		Allowable/*Predicted SPL	dBA	/	/	/	–	
12								

13	LINE	Pipe Line Size & Schedule In 4"	53	ACTUATOR	*Type Diaphragm
14		Out 4"	54		*Mfr & Model Emerson / Fisher 667
15		Pipe Line Insulation rockwool	55		*Size 30 Eff Area 96 inch²
16	VALVE BODY/BONNET	*Type Globe	56		On/Off – Modulating modulating
17		*Size 2" ANSI Class IV	57		Spring Action Open/Close open
18		Max Press/Temp 300 psig / 500 °F.	58		*Max Allowable Pressure 15 psig
19		*Mfr & Model Emerson / Fisher EAD	59		*Min Required Pressure 3 psig
20		*Body/Bonnet Matl WCC steel	60		Available Air Supply Pressure:
21		*Liner Material/ID S31600	61		Max 20 psig Min 0 psig
22		End Connection In raised face flanged	62		*Bench Range 3 / 15 psig
23		Out raised face flanged	63		Actuator Orientation vertical up
24		Flg Face Finish CL600	64		Handwheel Type side-mounted
25		End Ext/Matl Graphite	65		Air Failure Valve YES Set at 20 psig
26		*Flow Direction flow to open	66		
27		*Type of Bonnet plain bonnet	67		Input Signal
28		Lub & Iso Valve Yes Lube silicone.	68	POSITIONER	*Type
29		*Packing Material Double PTFE	69		*Mfr & Model
30		*Packing Type Enviro-seal PTFE	70		*On Incr Signal Output Incr/Decr
31			71		Gauges By-pass
32	TRIM	*Type single seat cage guided	72		*Cam Characteristic
33		*Size 3/8" Rated Travel 0,75"	73		
34		*Characteristic linear	74	SWITCHES	Type Quantity
35		*Balanced/Unbalanced balanced	75		*Mfr & Model
36		*Rated C_V 35,7 F_L 0,66 x_T 0,72	76		Contacts/Rating
37		*Plug/Ball/Disk Material S31600	77		Actuation Points
38		*Seat Material S31600	78		
39		*Cage/Guide Material 316 SST	79	AIR SET	*Mfr & Model
40		*Stem Material S20910	80		*Set Pressure
41			81		Filter Gauge
42			82		
43	SPECIALS/ACCESSORIES	NEC Class Group Div	83	TESTS	*Hydro Pressure
44			84		ANSI/FCI Leakage Class
45			85		
46			86		

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CONTROL VALVE DATA SHEET

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PROJECT 2020 / FCC / 02	DATA SHEET 02 of 24
UNIT #1	SPEC PSI - 2020 - 02
P.O. PO - 12345	TAG FV - 02
ITEM 1	DWG PSI - FCC - 2020
CONTRACT 2020 - PSI - Maulana	SERVICE R-01 feed
*MFR. SERIAL D100390X012	

No.	Section	Item	Units	Max Flow	Norm Flow	Min Flow	Shut-Off
1		Fluid			Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	lb/s	16	0	0	
3		Inlet Pressure	psig	270	270	270	—
4		Outlet Pressure	psig	270	270	270	
5		Inlet Temperature	°F	460,9	460,9	460,9	
6		Spec Wt/Spec Grav/Mol Wt		0,916	0,916	0,916	
7		Viscosity/Spec Heats Ratio		—	—	—	—
8		Vapor Pressure P_V		—	—	—	—
9		*Required C_V		41	2,95	295	—
10		*Travel	%	100	10	10	0
11		Allowable/*Predicted SPL	dBA	/	/	/	—
12							

No.	Section	Item	No.	Section	Item
13	LINE	Pipe Line Size In 4"	53	ACTUATOR	*Type diaphragm
14		& Schedule Out 4"	54		*Mfr & Model Emerson / Fisher 667
15		Pipe Line Insulation rockwool	55		*Size 30 Eff Area 46 inch²
16	VALVE BODY/BONNET	*Type Globe	56		On/Off — Modulating modulating
17		*Size 2" ANSI Class IV	57		Spring Action Open/Close open
18		Max Press/Temp 300 psig / 00 °F	58		*Max Allowable Pressure 15 psig
19		*Mfr & Model Emerson / Fisher EAD	59		*Min Required Pressure 0 psig
20		*Body/Bonnet Matl WCC steel	60		Available Air Supply Pressure:
21		*Liner Material/ID S31600	61		Max 20 psig Min 0 psig
22		End Connection In raised face flanged	62		*Bench Range 3 / 15 psig
23		End Connection Out raised face flanged	63		Actuator Orientation vertical up
24		Flg Face Finish CL 600	64		Handwheel Type side mounted
25		End Ext/Matl Graphite	65		Air Failure Valve yes Set at 20 psig
26		*Flow Direction flow to open	66		
27		*Type of Bonnet plain bonnet	67	POSITIONER	Input Signal
28		Lub & Iso Valve yes Lube silicone	68		*Type
29		*Packing Material double PTFE	69		*Mfr & Model
30		*Packing Type Enviro - seal PTFE	70		*On Incr Signal Output Incr/Decr
31			71		Gauges By-pass
32	TRIM	*Type single seat cage guided	72		*Cam Characteristic
33		*Size 3/8" Rated Travel 0,75"	73		
34		*Characteristic linear	74	SWITCHES	Type Quantity
35		*Balanced/Unbalanced balanced	75		*Mfr & Model
36		*Rated C_V 2,95 F_L 0,66 X_T 0,72	76		Contacts/Rating
37		*Plug/Ball/Disk Material S31600	77		Actuation Points
38		*Seat Material S31600	78		
39		*Cage/Guide Material 316 SST	79	AIR SET	*Mfr & Model
40		*Stem Material S20910	80		*Set Pressure
41			81		Filter Gauge
42			82		
43	SPECIALS/ACCESSORIES	NEC Class Group Div	83	TESTS	*Hydro Pressure
44			84		ANSI/FCI Leakage Class
45			85		
46			86		

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CONTROL VALVE DATA SHEET

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PROJECT 2020/FCC/02	DATA SHEET 03 of 29
UNIT #1	SPEC PSI-2020-03
P.O. PO.12345	TAG FV-03
ITEM 1	DWG PSI-FCC-2020
CONTRACT 2020-PSI-Maulana	SERVICE R-01 feed
*MFR. SERIAL D100390X012	

1 Fluid

Crit Press PC

#	SERVICE CONDITIONS	Units	Max Flow	Norm Flow	Min Flow	Shut-Off
2	Flow Rate	lb/s	177	126	0	–
3	Inlet Pressure	psig	260	260	260	
4	Outlet Pressure	psig	100	100	100	
5	Inlet Temperature	°F	460,9	460,9	460,9	
6	Spec Wt/Spec Grav/Mol Wt		0,916	0,916	0,916	–
7	Viscosity/Spec Heats Ratio		–	–	–	–
8	Vapor Pressure P_V		–	–	–	–
9	*Required C_V		108	84,6	7,02	–
10	*Travel	%	100	80	10	0
11	Allowable/*Predicted SPL	dBA	/	/	/	–
12						

LINE

13 Pipe Line Size In 11"
14 & Schedule Out 11"
15 Pipe Line Insulation rockwool

VALVE BODY/BONNET

16 *Type Globe
17 *Size 4" ANSI Class IV
18 Max Press/Temp 300 psig / 500 °F
19 *Mfr & Model Emerson / Fisher EAD
20 *Body/Bonnet Matl WCC steel
21 *Liner Material/ID S31600
22 End Connection In raised-face flanged
23 Out raised-face flanged
24 Flg Face Finish CL 600
25 End Ext/Matl Graphite
26 *Flow Direction flow to open
27 *Type of Bonnet plain bonnet
28 Lub & Iso Valve yes Lube silicone
29 *Packing Material double PTFE
30 *Packing Type Enviro-seal PTFE
31

TRIM

32 *Type Single seat cageguided
33 *Size 0,5" Rated Travel 1,125"
34 *Characteristic linear
35 *Balanced/Unbalanced balanced
36 *Rated C_V 84,6 F_L 0,76 X_T 0,72
37 *Plug/Ball/Disk Material S31600
38 *Seat Material S31600
39 *Cage/Guide Material 316 SST
40 *Stem Material S20910
41
42

SPECIALS/ACCESSORIES

43 NEC Class ____ Group ____ Div ____
44
45
46
47
48
49
50
51
52

ACTUATOR

53 *Type diaphragm
54 *Mfr & Model Emerson / Fisher 667
55 *Size 45 Eff Area 105 inch²
56 On/Off – Modulating modulating
57 Spring Action Open/Close open
58 *Max Allowable Pressure 15 psig
59 *Min Required Pressure 3 psig
60 Available Air Supply Pressure:
61 Max 20 psig Min 0 psig
62 *Bench Range 3 / 15 psig
63 Actuator Orientation vertical up
64 Handwheel Type side mounted
65 Air Failure Valve yes Set at 20 psig
66

POSITIONER

67 Input Signal
68 *Type
69 *Mfr & Model
70 *On Incr Signal Output Incr/Decr
71 Gauges ____ By-pass ____
72 *Cam Characteristic
73

SWITCHES

74 Type ____ Quantity ____
75 *Mfr & Model
76 Contacts/Rating
77 Actuation Points
78

AIR SET

79 *Mfr & Model
80 *Set Pressure
81 Filter ____ Gauge ____
82

TESTS

83 *Hydro Pressure
84 ANSI/FCI Leakage Class
85
86

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CONTROL VALVE DATA SHEET

Second Printing

PROJECT 2020 / FCC Y, 02	DATA SHEET 04 of 24
UNIT # 1	SPEC PSI - 2020 - 04
P.O. PO. 12345	TAG FV - 04
ITEM 1	DWG PSI - FCC - 2020
CONTRACT 2020 - PSI - Maulana	SERVICE R-01 feed
•MFR. SERIAL D100 340X012	

No.	Section	Item	Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
1		Fluid				Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	lb/s	10	5,25	0	–	
3		Inlet Pressure	psig	110	110	110		
4		Outlet Pressure	psig	100	100	100		
5		Inlet Temperature	°F	460,9	460,9	460,9		
6		Spec Wt/Spec Grav/Mol Wt		0,916	0,916	0,916	–	
7		Viscosity/Spec Heats Ratio		–	–	–	–	
8		Vapor Pressure P_V		–	–	–	–	
9		*Required C_V		24,2	12,9	3,97	–	
10		*Travel	%	100	50	10	0	
11		Allowable/*Predicted SPL	dBA	/	/	/	–	
12								

No.	Section	Item	No.	Section	Item
13	LINE	Pipe Line Size & Schedule In 3"	53	ACTUATOR	*Type diaph
14		Out 3"	54		*Mfr & Model Emerson / Fisher 657
15		Pipe Line Insulation rockwool	55		*Size 30 Eff Area 46 inch²
16	VALVE BODY/BONNET	*Type Globe	56		On/Off – Modulating modulating
17		*Size 1½ ANSI Class IV	57		Spring Action Open/Close open
18		Max Press/Temp	58		*Max Allowable Pressure 15 psig
19		*Mfr & Model Emerson / Fisher EAS	59		*Min Required Pressure 3 psig
20		*Body/Bonnet Matl WCC steel	60		Available Air Supply Pressure:
21		*Liner Material/ID S31600	61		Max 20 psig Min 0 psig
22		End Connection In raised-face flanged	62		*Bench Range 3/15 psig
23		Out raised-face flanged	63		Actuator Orientation vertical up
24		Flg Face Finish CL 600	64		Handwheel Type side mounted
25		End Ext/Matl graphite	65		Air Failure Valve yes Set at 20 psig
26		*Flow Direction flow to open	66		
27		*Type of Bonnet plain bonnet	67	POSITIONER	Input Signal
28		Lub & Iso Valve yes Lube silicone	68		*Type
29		*Packing Material double PTFE	69		*Mfr & Model
30		*Packing Type Enviro Seal PTFE	70		*On Incr Signal Output Incr/Decr
31			71		Gauges By-pass
32	TRIM	*Type Single seat cage guided	72		*Cam Characteristic
33		*Size 3/8" Rated Travel 0,75"	73		
34		*Characteristic linear	74	SWITCHES	Type Quantity
35		*Balanced/Unbalanced balanced	75		*Mfr & Model
36		*Rated C_V 12,9 F_L 0,92 X_T 0,72	76		Contacts/Rating
37		*Plug/Ball/Disk Material S31600	77		Actuation Points
38		*Seat Material S31600	78		
39		*Cage/Guide Material 316SST	79	AIR SET	*Mfr & Model
40		*Stem Material S20910	80		*Set Pressure
41			81		Filter Gauge
42			82		
43	SPECIALS/ACCESSORIES	NEC Class Group Div	83	TESTS	*Hydro Pressure
44			84		ANSI/FCI Leakage Class
45			85		
46			86		
47					
48					
49					
50					
51					
52					

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CONTROL VALVE DATA SHEET

Second Printing

PROJECT 2020 / FCC / 02	DATA SHEET 05 of 29
UNIT # 1	SPEC PSI - 2020 - 05
P.O. PO. 12345	TAG FV - 05
ITEM 1	DWG PSI - FCC - 2020
CONTRACT 2020 - PSI - Maulana	SERVICE ~~21~~-01 fuel gas
•MFR. SERIAL D100390X012	

			Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
1		Fluid				Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	scf/s	40	34	0	–	
3		Inlet Pressure	psig	190	190	190		
4		Outlet Pressure	psig	100	100	100		
5		Inlet Temperature	°F	105	105	105		
6		Spec Wt/Spec Grav/Mol Wt		0,7	0,7	0,7	–	
7		Viscosity/Spec Heats Ratio		–	–	–	–	
8		Vapor Pressure P_V		–	–	–	–	
9		•Required C_V					–	
10		•Travel	%	100	95	10	0	
11		Allowable/•Predicted SPL	dBA	/	/	/	–	
12								

13	LINE	Pipe Line Size & Schedule In	7"
14		Out	7"
15		Pipe Line Insulation	rockwool
16	VALVE BODY/BONNET	•Type	Globe
17		•Size 1" ANSI Class	IV
18		Max Press/Temp	250 psig / 200 °F
19		•Mfr & Model	Emerson / Fisher EAD
20		•Body/Bonnet Matl	WCC steel
21		•Liner Material/ID	S31600
22		End Connection In	raised face flanged
23		Out	raised face flanged
24		Flg Face Finish	CL600
25		End Ext/Matl	Graphite
26		•Flow Direction	flow to open
27		•Type of Bonnet	plain bonnet
28		Lub & Iso Valve	Yes Lube silicone
29		•Packing Material	double PTFE
30		•Packing Type	Enviro - Seal PTFE
31			
32	TRIM	•Type	Single seat cage guided
33		•Size	3/8" Rated Travel 0,75"
34		•Characteristic	equal precentage
35		•Balanced/Unbalanced	balanced
36		•Rated C_V	18,5 ? F_L 0,93 X_T 0,72
37		•Plug/Ball/Disk Material	S31600
38		•Seat Material	S31600
39		•Cage/Guide Material	316 SST
40		•Stem Material	S20910
41			
42			
43	SPECIALS/ACCESSORIES	NEC Class ___ Group ___ Div ___	
44			
45			
46			
47			
48			
49			
50			
51			
52			

53	ACTUATOR	•Type	diaphragm
54		•Mfr & Model	Emerson / Fisher 667
55		•Size	30 Eff Area 46 inch²
56		On/Off — Modulating	modulating
57		Spring Action Open/Close	Open
58		•Max Allowable Pressure	15 psig
59		•Min Required Pressure	0 psig
60		Available Air Supply Pressure:	
61		Max 20 psig Min	0 psig
62		•Bench Range	3,15 psig
63		Actuator Orientation	vertical up
64		Handwheel Type	side mounted
65		Air Failure Valve	Yes Set at 20 psig
66			
67	POSITIONER	Input Signal	
68		•Type	
69		•Mfr & Model	
70		•On Incr Signal Output Incr/Decr	
71		Gauges ___ By-pass	
72		•Cam Characteristic	
73			
74	SWITCHES	Type ___ Quantity	
75		•Mfr & Model	
76		Contacts/Rating	
77		Actuation Points	
78			
79	AIR SET	•Mfr & Model	
80		•Set Pressure	
81		Filter ___ Gauge	
82			
83	TESTS	•Hydro Pressure	
84		ANSI/FCI Leakage Class	
85			
86			

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PROJECT 2020 /FCC /02	DATA SHEET 06 of 24
UNIT #1	SPEC PSI - 2020 - 06
P.O. PO 12345	TAG PV - 04
ITEM 1	DWG PSI - FCC . 2020
CONTRACT 2020 - PSI - Maulana	SERVICE F - 01 fractionation
•MFR. SERIAL D1003 90X012	

1 Fluid — Crit Press PC

	SERVICE CONDITIONS	Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
2	Flow Rate	scf/s	0,95	0,95	0,41	–	
3	Inlet Pressure	psia	23,52	23,52	23,52		
4	Outlet Pressure	psia	22,68	22,68	22,68		
5	Inlet Temperature	°F	100,9	100,9	100,9		
6	Spec Wt/Spec Grav/Mol Wt		1,5	1,5	1,5	–	
7	Viscosity/Spec Heats Ratio					–	
8	Vapor Pressure P_V					–	
9	•Required C_V		18,5	18,5	0,783	–	
10	•Travel	%	100%	100%	100%	0	
11	Allowable/•Predicted SPL	dBA	/	/	/	–	
12							

LINE

13 Pipe Line Size In 5"
14 & Schedule Out 5"
15 Pipe Line Insulation rockwool

VALVE BODY/BONNET

16 •Type Globe
17 •Size 1" ANSI Class IV
18 Max Press/Temp 39,2 psi / 200 °F
19 •Mfr & Model Emerson / Fisher ED
20 •Body/Bonnet Matl WCC steel
21 •Liner Material/ID S31600
22 End Connection In raised flanged face
23 Out raised face flanged
24 Flg Face Finish CL600
25 End Ext/Matl graphite
26 •Flow Direction flow to close
27 •Type of Bonnet plain bonnet
28 Lub & Iso Valve yes Lube silicone
29 •Packing Material double PTFE
30 •Packing Type Enviro-seal PTFE
31

TRIM

32 •Type single seat cage guided
33 •Size 3/8" Rated Travel 0,75"
34 •Characteristic equal precentage
35 •Balanced/Unbalanced balanced
36 •Rated C_V 18,5 F_L 0,93 X_T 0,72
37 •Plug/Ball/Disk Material S31600
38 •Seat Material S31600
39 •Cage/Guide Material 316 SST
40 •Stem Material S 20910
41
42

SPECIALS/ACCESSORIES

43 NEC Class ______ Group ______ Div ______
44
45
46
47
48
49
50
51
52

ACTUATOR

53 •Type diapraghm
54 •Mfr & Model Emerson / Fisher 667
55 •Size 30 Eff Area 46 inch²
56 On/Off – Modulating modulating
57 Spring Action Open/Close close
58 •Max Allowable Pressure 15 psig
59 •Min Required Pressure 3 psig
60 Available Air Supply Pressure:
61 Max 20 psig Min 0 psig
62 •Bench Range 3 / 15 psig
63 Actuator Orientation vertical up
64 Handwheel Type side mounted
65 Air Failure Valve yes Set at 20 psig
66

POSITIONER

67 Input Signal
68 •Type
69 •Mfr & Model
70 •On Incr Signal Output Incr/Decr
71 Gauges ______ By-pass ______
72 •Cam Characteristic
73

SWITCHES

74 Type ______ Quantity ______
75 •Mfr & Model
76 Contacts/Rating
77 Actuation Points
78

AIR SET

79 •Mfr & Model
80 •Set Pressure
81 Filter ______ Gauge ______
82

TESTS

83 •Hydro Pressure
84 ANSI/FCI Leakage Class
85
86

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CONTROL VALVE DATA SHEET

Second Printing

PROJECT 2020/ECC/02
UNIT #1
P.O. PO 12345
ITEM 1
CONTRACT 2020 - PSI - Maulana
•MFR. SERIAL D100 390 X012

DATA SHEET 07 of 29
SPEC PSI - 2020 - 07
TAG PDV - 06
DWG PSI - ECC - 2020
SERVICE E-01 regenerator

1 Fluid — Crit Press PC

SERVICE CONDITIONS

		Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
2	Flow Rate	SCFS	2,22	2,163	0,14	–	
3	Inlet Pressure	psia	29,64	29,64	29,64		
4	Outlet Pressure	psia	14,7	19,7	14,7		
5	Inlet Temperature	°F	1283,6	1283,6	1283,6		
6	Spec Wt/Spec Grav/Mol Wt		1	1	1	–	
7	Viscosity/Spec Heats Ratio					–	
8	Vapor Pressure P_V					–	
9	•Required C_V		56,2	16,3	1,74	–	
10	•Travel	%	100	60	10	0	
11	Allowable/•Predicted SPL	dBA	/	/	/	–	
12							

LINE

13 Pipe Line Size & Schedule — In 8"
14 Out 8"
15 Pipe Line Insulation rockwool

VALVE BODY/BONNET

16 •Type Globe
17 •Size 2" ANSI Class IV
18 Max Press/Temp 39,2psia / 1310 °F
19 •Mfr & Model Emerson / Fisher ES
20 •Body/Bonnet Matl WCC Steel
21 •Liner Material/ID S31600
22 End Connection — In raised face flanged
23 Out raised face flanged
24 Flg Face Finish CL 600
25 End Ext/Matl graphite
26 •Flow Direction flow to close
27 •Type of Bonnet plain bonnet
28 Lub & Iso Valve yes Lube silicone
29 •Packing Material double PTFE
30 •Packing Type enviro seal PTFE
31

TRIM

32 •Type single seat cage guided
33 •Size 3/8" Rated Travel 0,75"
34 •Characteristic equal precentage
35 •Balanced/Unbalanced balanced
36 •Rated C_V 16,3 F_L 0,92 x_T 0,72
37 •Plug/Ball/Disk Material S31600
38 •Seat Material S31600
39 •Cage/Guide Material 316SST
40 •Stem Material S20910
41
42

SPECIALS/ACCESSORIES

43 NEC Class ____ Group ____ Div ____
44
45
46
47
48
49
50
51
52

ACTUATOR

53 •Type diaphragm
54 •Mfr & Model emerson / fisher 67
55 •Size 30 Eff Area 46 inch²
56 On/Off ____ Modulating ____
57 Spring Action Open/Close open
58 •Max Allowable Pressure 15 psig
59 •Min Required Pressure 3 psig
60 Available Air Supply Pressure:
61 Max 20 psig Min 0 psig
62 •Bench Range 3 / 15 psig
63 Actuator Orientation vertical up
64 Handwheel Type side mounted
65 Air Failure Valve yes Set at 20 psig
66

POSITIONER

67 Input Signal
68 •Type
69 •Mfr & Model
70 •On Incr Signal Output Incr/Decr
71 Gauges ____ By-pass ____
72 •Cam Characteristic
73

SWITCHES

74 Type ____ Quantity ____
75 •Mfr & Model
76 Contacts/Rating
77 Actuation Points
78

AIR SET

79 •Mfr & Model
80 •Set Pressure
81 Filter ____ Gauge ____
82

TESTS

83 •Hydro Pressure
84 ANSI/FCI Leakage Class
85
86

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PROJECT 2020/FCC/02
UNIT IZ I
P.O. PO 12345
ITEM 1
CONTRACT 2020 - PSI - Maulana
•MFR. SERIAL D1003960X012

DATA SHEET 08 of 24
SPEC PSI-2020-08
TAG FV-09
DWG PSI - FCC-2020
SERVICE E-01 regenerator

			Units	Max Flow	Norm Flow	Min Flow	Shut-Off	
1		Fluid				Crit Press PC		
2	SERVICE CONDITIONS	Flow Rate	SCFS	1089.19	193.36	54.5	–	
3		Inlet Pressure	psia	40.5	40.5	40.5		
4		Outlet Pressure	psig	29.64	29.64	29.64		
5		Inlet Temperature	°F	270 °F	270	270		
6		Spec Wt/Spec Grav/Mol Wt		1	1	1	–	
7		Viscosity/Spec Heats Ratio					–	
8		Vapor Pressure P_V					–	
9		*Required C_V		13565	3998	823	–	
10		*Travel	%	100%	40%	10%	0	
11		Allowable/*Predicted SPL	dBA	/	/	/	–	
12								

LINE

13 Pipe Line Size In 24"
14 & Schedule Out 24"
15 Pipe Line Insulation rockwool

VALVE BODY/BONNET

16 *Type butterfly
17 *Size 24" ANSI Class IV
18 Max Press/Temp 100 psia / 300 °F
19 *Mfr & Model Emerson / Fisher 8590
20 *Body/Bonnet Matl WCC steel
21 *Liner Material/ID
22 End Connection In raised face flanged
23 Out raised face flanged
24 Flg Face Finish CL 600
25 End Ext/Matl graphite
26 *Flow Direction flow to open
27 *Type of Bonnet plain bonnet
28 Lub & Iso Valve yes Lube silicone
29 *Packing Material double PTFE
30 *Packing Type Enviro-Seal PTFE
31

TRIM

32 *Type single seat cage guided
33 *Size 4" Rated Travel 8"
34 *Characteristic linear
35 *Balanced/Unbalanced balanced
36 *Rated C_V 3998 F_L 0.9 x_T 0.72
37 *Plug/Ball/Disk Material S41600
38 *Seat Material S41600
39 *Cage/Guide Material WCC
40 *Stem Material S31600
41
42

SPECIALS/ACCESSORIES

43 NEC Class ______ Group ______ Div ______
44
45
46
47
48
49
50
51
52

ACTUATOR

53 *Type Diaphragm
54 *Mfr & Model Emerson / Fisher 667
55 *Size 100 Eff Area 450 inch²
56 On/Off — Modulating modulating
57 Spring Action Open/Close close
58 *Max Allowable Pressure 15 psig
59 *Min Required Pressure 3 psig
60 Available Air Supply Pressure:
61 Max 20 psig Min 0 psig
62 *Bench Range 3 / 15 psig
63 Actuator Orientation vertical up
64 Handwheel Type side mounted
65 Air Failure Valve yes Set at 20 psig
66

POSITIONER

67 Input Signal
68 *Type
69 *Mfr & Model
70 *On Incr Signal Output Incr/Decr
71 Gauges ______ By-pass ______
72 *Cam Characteristic

SWITCHES

73
74 Type ______ Quantity ______
75 *Mfr & Model
76 Contacts/Rating
77 Actuation Points
78

AIR SET

79 *Mfr & Model
80 *Set Pressure
81 Filter ______ Gauge ______
82

TESTS

83 *Hydro Pressure
84 ANSI/FCI Leakage Class
85
86

Rev	Date	Revision	Orig	App

*Information supplied by manufacturer unless already specified

ISA FORM S20.50, Rev.1

3) Analisis controlled variable

a) Keterkaitan dengan tujuan proses

Tujuan : memaksimalkan laju produksi dengan cara memaksimalkan laju feed sesuai dengan yang telah ditentukan

b) Keselamatan operasional

o> Tekanan reaktor (P_4)

- Tekanan reaktor mempunyai nilai maksimum sehingga harus dikendalikan agar tidak terjadi ledakan. Selain itu tekanan reaktor dapat mempengaruhi proses perengkahan
- range : 0-49,2 psig → pressure gauge dengan diafragma
- span : 49,2 psig
- akurasi: ±0,01-0,2% span
- rangeability : 150-200 : 1
- karakteristik linier

- output : 4-20 mA
- suhu : 925-995 °F → material stainless steel
- Voltage input : 24 VDC
- Proses di reactor flammable → explosion proof

nya.

* Dipilih pressure gauge transmitter diafragma → Emerson Rosemount 2088G

o> Suhu riser-reaktor (T_r)

- Suhu riser mempunyai nilai maksimum sehingga harus dikendalikan agar riser tidak meleleh. Selain itu, proses perengkahan akan optimal pada suhu tertentu sehingga perlu dikendalikan.
- range : 925-995 °F
- span : 70 °F
- akurasi ±0,01-0,2% span
- karakteristik linier

→ RTD dengan material platinum (linier untuk T < 1112 °F)

- tekanan : 0-49,2 psig.
- voltage input : 24 VDC
- proses di reactor flammable → explosion proof
- output : 4-20 mA

* Dipilih RTD dengan material platinum → Emerson Rosemount 214C-RH

=> Tekanan regenerator (P_6)

- Tekanan regenerator mempunyai nilai maksimal sehingga harus dikendalikan. Selain itu, tekanan regenerator dapat mempengaruhi laju sirkulasi katalis

- range : 0-39,7 psig → pressure gauge dengan diafragma	- suhu : 1265 - 1310 °F → material stainless steel
- span : 39,7 psig	- output : 4-20 mA
- akurasi : ±0,01-0,2 % span	- Voltage input : 24 VDC
- rangeability : 150-200 : 1	- proses pembakaran di regenerator
- karakteristik linier	→ explosion proof

* Dipilih pressure gauge transmitter diafragma → Emerson Rosemount 2088 G

=> Suhu regenerator (Treg)

Suhu regenerator mempunyai nilai maksimal sehingga harus dikendalikan agar regenerator tidak meleleh. Selain itu suhu regenerator dapat mempengaruhi suhu katalis yang akhirnya akan mempengaruhi proses perengkahan.

- range = 1265-1310 °F → Thermocouple tipe K, (Ni-Cr/Ni-Al)	- Tekanan : 0-39,7 psig
- span = 45 °F	- voltage input : 12 24 VDC
- akurasi ±0,01-0,2% span	- proses pembakaran terjadi di regenerator → explosion proof
- karakteristik linier	- output : 4-20 mA

* Dipilih thermocouple tipe K → Emerson Rosemount 214C-TK

=> Tekanan menara fraksinasi (P_5)

- Menara fraksinasi mempunyai tekanan maksimum sehingga harus dikendalikan agar tidak meledak. Selain itu, tekanan menara fraksinasi dapat mempengaruhi tekanan reaktor sehingga dapat mempengaruhi proses perengkahan

- range : 0-39,7 psig → pressure gauge dengan diafragma	- output : 4-20 mA
- span = 39,7 psig	- suhu bagian atas : ±100 °F → material stainless steel
- akurasi = ±0,01-0,2% span	- voltage input : 24 VDC
- rangeability : 150-200 : 1	- fluida flammable → explosion proof
- karakteristik linier	

* Dipilih presture gauge transmitter diafragma → Emerson Rosemount 2088 G

c) Laju produksi

=> Wet gas flow rate (F_{v11})

- wet gas flow rate merupakan laju produksi dari FCC. Dengan mengubah laju produksi dapat mengubah tekanan reaktor yang mempengaruhi laju produksi sehingga perlu dikendalikan.

- range : 0 - 0,95 scf/s	- suhu : 460,9 °F → stainless steel
- span : 0,95 scf/s	- output : 4-20 mA
- Tekanan : 0 - 39,7 psig	- voltage input : 24 VDC
- akurasi : ±1-2 % span	- wet gas flammable → explosion proof
- rangeability : 4-8 : 1	* Dipilih orifice flow meter
- karakteristik linier	→ Emerson Rosemount 3051 CFC

∘> Beda tekanan reaktor - regenerator (ΔP_{RR})

- Beda tekanan reaktor-regenerator harus dijaga agar sirkulasi katalis tetap lancar sehingga produksi tidak terganggu

- range : (-5) - 2 psig → pressure gauge	dengan diafragma	- suhu : 925 - 1310 °F →
- span : 7 psig		- voltage input = 24 VDC
- akurasi : ±0,01 - 0,2 % span		- output : 4-20 mA
- rangeability : 150-200 : 1		- proses flammable → explosion proof
- karakteristik linier		

* Dipilih differential pressure transmitter diafragma → Emerson Rosemount 3051 S

∘> Suhu fresh feed setelah preheater (T_2)

- Suhu fresh feed akan mempengaruhi produksi karena proses reaksi endotermik sehingga diperlukan fresh feed dengan suhu yang cukup

- range : 400-700°F → RTD	dengan material platinum	voltage input : 24 VDC
- span : 300 °F		output : 4 - 20 mA
- akurasi : ±0,01 - 0,2 % span		fluida flammable → explosion proof
- tekanan : 80-120 psig		
- karakteristik linier		

* Dipilih RTD dengan material platinum → Emerson Rosemount 214C-RT

∘> Lift air flow rate (F_9)

- Lift air digunakan untuk mendorong katalis di regenerator sehingga perlu dikontrol karena akan mempengaruhi pembakaran coke di regenerator yang kemudian akan mempengaruhi suhu katalis. Suhu katalis mempengaruhi proses perengkahan.

- range : 14,51 lb/s → 0 - 30 lb/s	- rangeability : 4-8 : 1
- span : 30 lb/s → saat membuka 0,424	- karakteristik linier
- tekanan : 40,5 psia (0,424) → 0 - 100 psig	- Voltage input : 24 VDC
- suhu : 270° F → stainless steel	- output : 4-20 mA
- akurasi : ±1-2 % span	- non flammable → general purpose

* Dipilih orifice flow meter → Emerson rosemount 3051 CFC

d) Kesetimbangan komponen

∘> Suhu flue gas (T_{sg})

Suhu flue gas dapat digunakan untuk mengukur kadar O_2 pada flue gas. Kadar O_2 (excess oxygen) perlu dikendalikan agar terjadi pembakaran sempurna di regenerator. Selain itu dengan mengatur kadar O_2, kadar CO juga akan mengikuti kadar O_2 sehingga flue gas tidak merusak lingkungan. Jika kadar O_2 diatur sesuai nilai optimalnya, suhu regenerator akan berada pada suhu optimal sehingga suhu katalis akan optimal pula dan suhu perengkahan pada riser akan berada pada suhu optimal → produksi maksimal

3) Analisis controlled variable

a) Keterkaitan dengan tujuan proses

Tujuan : memaksimalkan laju produksi dengan cara memaksimalkan laju feed sesuai dengan yang telah ditentukan

o> Wash oil flow rate (F_1)

- wash oil flow rate diatur pada flow maksimalnya agar produksi maksimal akan tetapi dengan tetap memperhatikan variabel proses lainnya.
- range : 0-17 lb/s → span = 17 lb/s
- Temperatur : 460,9 °F
- Tekanan : 260 psig
- diameter pipa : NPS 4
- akurasi : ±1-2 % dari flow rate span
- karakteristik linier
- repeatability turn down : 4-8 : 1
- NPS : 4"
- output : 4-20 mA
- Voltage : 24 VDC
- wash oil flamable → explosion proof
- Dari temperatur kerja → material stainless steel

* Dipilih orifice flow meter → Emerson Rosemount 3051 CFC

o> Diesel oil flow rate (F_2)

- diesel oil diatur pada flow maksimal agar produksi maksimal dengan tetap memperhatikan variabel proses lain
- range = 0-16 lb/s → span : 16 lb/s
- akurasi : ±1-2 % span
- rangeability turndown : 4-8 : 1
- karakteristik linier
- Suhu : 460,9 °F → material stainless steel
- tekanan : 260 psig
- voltage input : 12-32 VDC
- diesel oil flamable → explosion proof
- output : 4-20 mA
- NPS : 4"

* Dipilih orifice flow meter → Emerson rosemount 3051 CFC

o> Total fresh feed flow rate (F_3)

- total fresh feed diatur pada flow maksimal agar produksi maksimal dengan tetap memperhatikan variabel proses lain
- range : 0-149 lb/s
- span : 149 lb/s
- akurasi : ±1-2 % span
- rangeability turn down : 4-8 : 1
- karakteristik linier
- output : 4-20 mA - NPS : 11"
- suhu : 460,9 °F → material stainless steel
- tekanan : 260 psig
- voltage input : 24 VDC
- fresh feed flamable → explosion proof

* Dipilih orifice flow meter → Emerson Rosemount 3051 CFC

o> slurry recycle oil flow rate (F_4)
- slurry recycle oil diatur pada flow maksimal agar produksi maksimal dengan tetap memperhatikan variabel proses lain

- range : 0-10 lb/s	- suhu : 460,9 °F.
- span : 10 lb/s	→ material stainless steel
- akurasi : ± 1-2 %	- tekanan : 260 psig → 150-350 psig
- rangeability turndown : 4-8 : 1.	- voltage input : 24 VDC
- karakteristik linier	- slurry recycle flammable → explosion proof
- output : 4-20 mA	- NPS : 3"
* Dipilih orifice flow meter	- dapat terjadi erosi akibat katalis
→ Emerson Rosemount 3051CFC	→ harus sering dipantau

o> Fuel gas flow rate (F_5)
Fuel gas flow rate diatur untuk menghasilkan pembakaran pre heater sesuai dengan yang diinginkan sehingga didapatkan suhu feed yang optimal

- range : 0-40 scf/s	suhu : 405°F → 0-200°F → stainless steel
- span : 40 scf/s	tekanan : 0-190 psig
- akurasi : ±1-2 % span	voltage input : 24 VDC
- rangeability : 4-8 : 1	output : 4-20 mA
- karakteristik linier	flammable fluid → explosion proof
	* orifice flow meter → Emerson Rosemount 3051CFC

d) Kesetimbangan komponen
o> Suhu flue gas (T_{sg})
Suhu flue gas dapat digunakan untuk mengukur kadar O_2 pada flue gas. Kadar O_2 (excess oxygen) perlu dikendalikan agar terjadi pembakaran sempurna di regenerator. Selain itu dengan mengatur kadar O_2, kadar CO juga akan mengikuti kadar O_2 sehingga flue gas tidak merusak lingkungan. Jika kadar O_2 diatur sesuai nilai optimalnya, suhu regenerator akan berada pada suhu optimal sehingga suhu katalis akan optimal pula dan suhu perengkahan pada riser akan berada pada suhu optimal → produksi maksimal

DIFFERENTIAL PRESSURE INSTRUMENTS				SHEET 19 OF 24	
NO	BY	DATE	REVISION	SPEC. NO. PSI-2020-99	REV.
				CONTRACT 2020-PSI-maulana	DATE 22/10/20
				REQ.	P.O. 12345
				BY	CHK'D / APPR.

Section	No.	Item	Specification
	1	Tag No.	-FT-01 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If ☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ Is ☐ Ds ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 psi Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At:
	22		Elevation Suppression
	23	Process Data	Fluid wash oil Max Temp. 500°F Max. Press. 300 psig
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material
			Temp. Element ☐ Range Type
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts
			Valve Manifold
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator
			Other
	28	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes:

ISA Form S20.20a

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 10 OF 26	
		SPEC. NO. B1-2020-10	REV.
	NO | BY | DATE | REVISION	CONTRACT 2020-B1-Moulding	DATE 22/10/20
		REQ. P.O. 12345	
		BY | CHK'D	APPR.

	1	Tag No.	FT-02 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size ______ Color: MFR STD ☐ Other ______
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ______
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other ______
	6	Power Supply	117V 60 Hz ☐ Other ac ______ dc ☑ 24 Volts ______
	7	Chart	12 in. Circ. ☐ Other ______ Range ______ No. ______
	8	Chart Drive	24 hr Other ______ Elec. ☐ Spring ☐ Other ______
	9	Scale	Type ______ Range: 1 ______ 2 ______ 3 ______
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______
			For Receiver, See Spec Sheet ______
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If ☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ I_s ☐ D_s ☐
			Other ______
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ______
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other ______
	15	Manual Reg.	None ☐ MFR STD ☑ Other ______
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other ______
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other ______
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 psi Body Rating ______ psig
	21	Diff. Range	Fixed ☑ Adj. Range ______ Set At ______
	22		Elevation ______ Suppression ______
	23	Process Data	Fluid diesel oil Max Temp. 500 F Max. Press. 300 psig
	24	Process Conn.	½ in. NPT ☐ Other ______
	25	Alarm Switches	Quantity ______ Form ______ Rating ______
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ______ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ______ Material ______
			Temp. Element ☐ Range ______
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ______ Charts
			Valve Manifold ______
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator ______
			Other ______
	29	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes.

ISA Form S20.20a

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 11 OF 26	
		SPEC. NO. B1-2020-10	REV.
	NO | BY | DATE | REVISION	CONTRACT 2020-B1-manual	DATE 22/10/20
		REQ. P.O. 12345	
		BY | CHK'D	APPR.

	No.	Item	Specification
	1	Tag No.	FT-03 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other ____
	6	Power Supply	117V 60 Hz ☐ Other ac ____ dc ☑ 24 Volts ____
	7	Chart	12 in. Circ. ☐ Other ____ Range ____ No. ____
	8	Chart Drive	24 hr Other ____ Elec. ☐ Spring ☐ Other ____
	9	Scale	Type ____ Range: 1 ____ 2 ____ 3 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
			For Receiver, See Spec Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ I_s ☐ D_s ☐
			Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☑ Other ____
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other ____
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other ____
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 psi Body Rating ____ psig
	21	Diff. Range	Fixed ☑ Adj. Range ____ Set At ____
	22		Elevation ____ Suppression ____
	23	Process Data	Fluid 99S oil Max Temp. 500 F Max. Press. 300 psig
	24	Process Conn.	½ in. NPT ☐ Other ____
	25	Alarm Switches	Quantity ____ Form ____ Rating ____
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ____ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ____ Material ____
			Temp. Element ☐ Range ____ Type ____
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ____ Charts
			Valve Manifold ____
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator ____
			Other ____
	28	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes:

ISA Form S20.20a

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 12 OF 26
		SPEC. NO. PSI-2020-12 / REV.
	NO / BY / DATE / REVISION	CONTRACT 2020-PI-mau / DATE 22/10/20
		REQ. / P.O.
		BY / CHK'D / APPR.

	1	Tag No.	FT-04 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other ______
	3	Case	MFR STD ☐ Nom Size ______ Color: MFR STD ☐ Other ______
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ______
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other ______
	6	Power Supply	117V 60 Hz ☐ Other ac ______ dc ☑ 24 Volts ______
	7	Chart	12 in. Circ. ☐ Other ______ Range ______ No. ______
	8	Chart Drive	24 hr Other ______ Elec. ☐ Spring ☐ Other ______
	9	Scale	Type ______ Range: 1 ______ 2 ______ 3 ______
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______
			For Receiver, See Spec Sheet ______
CONTROLLER	11	Control Modes	P-Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s-Slow, f=Fast If☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ I_s ☐ D_s ☐
			Other ______
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ______
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other ______
	15	Manual Reg.	None ☐ MFR STD ☑ Other ______
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other ______
	18	Element Type	Diaphragm ☐ Bellows ☐ Mercury ☐ Other ______
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 psi Body Rating ______ psig
	21	Diff. Range	Fixed ☑ Adj. Range ______ Set At ______
	22		Elevation ______ Suppression ______
	23	Process Data	Fluid recycle oil Max Temp. 500 F Max. Press. 300 psig
	24	Process Conn.	½ in. NPT ☐ Other ______
	25	Alarm Switches	Quantity ______ Form ______ Rating ______
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ______ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ______ Material ______
			Temp. Element ☐ Range ______ Type ______
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ______ Charts
			Valve Manifold ______
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator ______
			Other ______
	28	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes:

ISA Form S20.20a

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 13 OF 26
		SPEC. NO. PSI-2020-13 REV
	NO / BY / DATE / REVISION	CONTRACT 2020-PSI-maa1 DATE 27/10/20
		REQ. P.O. 12345
		BY / CHK'D / APPR.

	1	Tag No.	FT-05 Service PF-01 fuel gas
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☑ Weather proof ☐ Explosion proof ☐ Class
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ I_s ☐ D_s ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 psi Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At
	22		Elevation Suppression
	23	Process Data	Fluid natural gas Max Temp. 200 F Max. Press. 200 psig
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material
			Temp. Element ☐ Range Type
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts
			Valve Manifold
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator
			Other
	28	MFR & Model No.	Emerson / Rosemount 3051 SFC

Notes:

ISA Form S20.20a

TEMPERATURE INSTRUMENTS (FILLED SYSTEM)				SHEET 14 OF 26	
NO	BY	DATE	REVISION	SPEC. NO. PSI-2020-14	REV.
				CONTRACT 2020-PSI-maul	DATE 22/10/20
				REQ. - P.O. 12345	
				BY / CHK'D	APPR.

	1	Tag No.	TT-01 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☐ Class For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac ____ dc ☑ 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks
	8	Chart Drive	Speed Power
	9	Scales	Type Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ I_s ☐ D_s ☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal 460.9 °F Max 900 F Max. Press. 300 PSI
	19	Range	Fixed ☑ Adj. Range Set At Overrange Protection to
	20	Bulb	Type Rigid Mtl. 321 SST. Extension: Length Type Size: Diameter 10mm Length 50 mm Insertion 30mm Conn: Location Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☐ Length 50 mm Mtl. Pt-100 Armor 316SST
	22	Well	Mtl. 321 SST. Insertion 80 mm Lag Ext. Conn. Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To ____ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Other
	26	Mfr. & Model No.	Emerson / Rosemount 214C-RH

Notes:

ISA FORM S20.11a

		TEMPERATURE INSTRUMENTS (FILLED SYSTE	SHEET 16 OF 26
			SPEC. NO. PI-2020-15 REV.
		NO / BY / DATE / REVISION	CONTRACT 2020-PSI-mau DATE 22/10/20
			REQ. P.O. 12345
			BY / CHK'D / APPR.

	1	Tag No.	TT-02 Service R-01 feed
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ___ Color: MFR STD ☐ Other ___
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ___
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I For Use in Intrinsically Safe System ☐ Other ___
	6	Power Supply	117 V 60Hz ☐ Other ac ___ dc ☑ 24 Volts
	7	Chart	Strip ☐ ___ Roll ☐ ___ Fold ☐ ___ Circular ___ Time Marks ___
	8	Chart Drive	Speed ___ Power ___
	9	Scales	Type ___ Range 1 ___ 2 ___ 3 ___ 4 ___
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ___ For Receiver See Spec. Sheet ___
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☑ If ☐ Df ☐ Is ☐ Ds ☐ Other ___
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ___
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ___
	15	Manual Reg.	None ☐ MFR STD ☐ Other ___
	16	Output	4 20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ___
ELEMENT	17	Fill	SAMA Class III A Compensation ___
	18	Process Data	Temp: Normal 225°C Max 500 F Max. Press. 300 psig
	19	Range	Fixed ☑ Adj. Range ___ Set At ___ Overrange Protection to ___
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length ___ Type ___ Size: Diameter 10 mm Length 5 mm Insertion 30 mm Conn: ___ Location ___ Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. Pt-100 Armor 316 SST
	22	Well	Mtl. 321 SST Insertion 80 mm Lag Ext. ___ Conn. ___ Const: Drilled ☑ Built Up ☐ Other ___
	23	Alarm Switches	Quantity ___ Form ___ Rating ___
	24	Function	Temp ☐ Deviation ☐ Contacts To ___ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ ___ Charts Other
	26	Mfr. & Model No.	Emerson / Rosemount 214C RTD

Notes:

ISA FORM S20.11a

TEMPERATURE INSTRUMENTS (FILLED SYSTEM)				SHEET 16 OF 26	
NO	BY	DATE	REVISION	SPEC. NO. PSI-2020-16	REV.
				CONTRACT 2020-PSI-MCW	DATE 22/10/20
				REQ.	P.O. 12345
				BY / CHK'D	APPR.

Section	No.	Item	Specification
	1	Tag No.	TT-03 Service H-01 furnace
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ___ Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I; For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac ___ dc ☑ 24 Volts
	7	Chart	Strip ☐ ___ Roll ☐ ___ Fold ☐ ___ Circular ___ Time Marks ___
	8	Chart Drive	Speed ___ Power ___
	9	Scales	Type ___; Range 1 ___ 2 ___ 3 ___ 4 ___
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ___; For Receiver See Spec. Sheet ___
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast; P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐; Other ___
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ___
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ___
	15	Manual Reg.	None ☐ MFR STD ☐ Other ___
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ___
ELEMENT	17	Fill	SAMA Class IIIA Compensation ___
	18	Process Data	Temp: Normal ___ Max 1900 °F Max. Press. ___
	19	Range	Fixed ☑ Adj. Range ___ Set At ___; Overrange Protection to ___
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length ___ Type ___; Size: Diameter 15 mm Length 50 mm Insertion 30 mm; Conn: ___ Location ___ Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. K type Armor 316 SST
	22	Well	Mtl. 321 SST Insertion 80 mm Lag Ext. ___ Conn. ___; Const: Drilled ☑ Built-Up ☐ Other ___
	23	Alarm Switches	Quantity ___ Form ___ Rating ___
	24	Function	Temp ☐ Deviation ☐ Contacts To ___ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts; Other ___
	26	Mfr. & Model No.	Emerson / Rosemount 214CFK

Notes.

ISA FORM S20.11a

	TEMPERATURE INSTRUMENTS (FILLED SYSTEM)	SHEET 17 OF 24	
		SPEC. NO PSI-2020-17	REV.
	NO | BY | DATE | REVISION	CONTRACT PSI20-PSI-mail	DATE 22/10/20
		REQ. - P.O. 12345	
		BY | CHK'D	APPR.

	No.	Item	Specification
	1	Tag No.	TT-04 Service R-01 reactor
GENERAL	2	Function	Record ☐ Indicate ☑ Control _ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I For Use in Intrinsically Safe System ☐ Other ____
	6	Power Supply	117 V 60Hz ☐ Other ac ____ dc ☑ 24 Volts
	7	Chart	Strip ☐ ____ Roll ☐ ____ Fold ☐ ____ Circular ____ Time Marks ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____ Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____ For Receiver See Spec. Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate), Sub: s = Slow f = Fast P ☐ PI ☐ PD ☐ PID ☐ I_f ☐ D_f ☐ I_s ☐ D_s ☐ Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Fill	SAMA Class III A Compensation ____
	18	Process Data	Temp: Normal 925 F Max 995 °F Max. Press. 49,2 psig
	19	Range	Fixed ☑ Adj. Range ____ Set At ____ Overrange Protection to ____
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length ____ Type ____ Size: Diameter 10 mm Length 50 mm Insertion 30 mm Conn: ____ Location ____ Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☐ Length 50 mm Mtl. Pt-100 Armor 316 SST
	22	Well	Mtl. 321 SST Insertion 80 mm Lag Ext. ____ Conn. ____ Const: Drilled ☑ Built-Up ☐ Other ____
	23	Alarm Switches	Quantity ____ Form ____ Rating ____
	24	Function	Temp ☐ Deviation ☐ Contacts To ____ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ ____ Charts Other ____
	26	Mfr. & Model No.	Emerson / Rosemount 214C RTD

Notes.

ISA FORM S20 11a

	PRESSURE INSTRUMENTS				SHEET 18 OF 26	
					SPEC. NO. 951-2020-18	REV.
	NO	BY	DATE	REVISION	CONTRACT 2020-951-Maul	DATE 22/10/20
					REQ. P.O. 12345	
					BY CHK'D	APPR.

	No.	Item	
	1	Tag No.	- PT - 04 Service R 01 reactor
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For Use In Intrin. Safe System ☐ Other
	6	Power Supply	117V 60Hz ☐ Other ac dc 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks Range Number
	8	Chart Drive	Speed Power
	9	Scales	Type Range 1 2 3 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain) I=Integral (Auto-Reset) D=Derivative (Rate) Sub: s=Slow f=Fast P ☐ PI ☑ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐ Other
	12	Action	On Meas. Increase Output: Increases ☑ Decreases
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20mA ☑ 10-50mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Service	Gage Press. ☑ Vacuum ☐ Absolute ☐ Compound ☐
	18	Element Type	Diaphragm ☑ Helix ☐ Bourdon ☐ Bellows ☐ Other
	19	Material	316 SS ☐ Ber. Copper ☐ Other 316 SST
	20	Range	Fixed ☑ Adj. Range Set at Overrange protection to
	21	Process Data	Press: Normal Max 49.2 psig Element Range
	22	Process Conn.	¼ in. NPT ☐ ½ in. NPT ☑ Other Location: Bottom ☐ Back ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Press ☐ Deviation ☐ Contacts To on Inc Press.
OPTIONS	25	Options	Filt-Reg. ☐ Sup Gage ☐ Output Gage ☐ Charts Diaph Seal ☐ Type Diaph Bot Bowl Conn Capillary: Length Mtl. Other
	26	MFR & Model No.	Emerson / Rosemount 2088G

Notes:

ISA Form S20.40a

		PRESSURE INSTRUMENTS	SHEET 19 OF 36
			SPEC. NO. PS1-2020-19 — REV.
	NO / BY / DATE / REVISION		CONTRACT 2020-PS1-mau) — DATE 22/10/20
			REQ. — P.O. 12345
			BY — CHK'D — APPR.

	No.	Item	
	1	Tag No. PT-05	Service E-01 fractination
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I For Use In Intrin. Safe System ☐ Other ____
	6	Power Supply	117V 60Hz ☐ Other ac ____ dc 24 Volts
	7	Chart	____ Strip ☐ ____ Roll ☐ ____ Fold ☐ Circular ____ Time Marks ____ Range ____ Number ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____ Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____ For Receiver See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain) I=Integral (Auto-Reset) D=Derivative (Rate) Sub: s=Slow f=Fast P ☐ PI ☐ PD ☐ PID ☐ If ☐ Df ☐ I_s ☐ D_s ☐ Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20mA ☐ 10-50mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Service	Gage Press. ☑ Vacuum ☐ Absolute ☐ Compound ☐
	18	Element Type	Diaphragm ☑ Helix ☐ Bourdon ☐ Bellows ☐ Other ____
	19	Material	316 SS ☐ Ber. Copper ☐ Other 316 SST
	20	Range	Fixed ☑ Adj. Range ____ Set at ____ Overrange protection to ____
	21	Process Data	Press: Normal ____ Max 39,7 psig Element Range ____
	22	Process Conn.	¼ in. NPT ☐ ½ in. NPT ☑ Other ____ Location: Bottom ☐ Back ☐ Other ____
	23	Alarm Switches	Quantity ____ Form ____ Rating ____
	24	Function	Press ☐ Deviation ☐ Contacts To ____ on Inc Press.
OPTIONS	25	Options	Filt-Reg. ☐ Sup Gage ☐ Output Gage ☐ ____ Charts Diaph Seal ☐ Type ____ Diaph ____ Bot Bowl ____ Conn ____ Capillary: Length ____ Mtl. ____ Other ____
	26	MFR & Model No.	Emerson / Rosemount 2088 G

Notes:

ISA Form S20.40a

	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 20 OF 26
	NO / BY / DATE / REVISION	SPEC. NO. ps1-2020-20 REV.
		CONTRACT 2020-PR-mau DATE 27/10/20
		REQ. P.O.
		BY CHK'D APPR.

	1	Tag No.	FT-11 Service F-01 gas outlet
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☐ PD ☐ PID ☐ Is ☐ Ds ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At
	22		Elevation Suppression
	23	Process Data	Fluid Propane Max Temp. Max. Press. 30 psia
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material
			Temp. Element ☐ Range Type
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts
			Valve Manifold
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator
			Other
	28	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes:

ISA Form S20.20a

		DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 27 OF 26
			SPEC. NO. PSI-2020-21 REV.
		NO / BY / DATE / REVISION	CONTRACT 2020-PSI-mau DATE 22/10/20
			REQ. P.O. 12345
			BY / CHK'D / APPR.

	1	Tag No.	PDT-06 Service R-01 – E-01
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P-Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate) Sub: s-Slow, f-Fast If☐ Df☐ P☐ PI☑ PD☐ PID☐ Is☐ Ds☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☐ Level ☐ Diff. Pressure ☑ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At
	22		Elevation Suppression
	23	Process Data	Fluid hydrocarbon Max Temp. 1310 F Max. Press. 2 psi
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material Temp. Element ☐ Range Type Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Valve Manifold Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐ Integrator Other
	28	MFR & Model No.	Emerson / Rosemount 3051S

Notes:

ISA Form S20.20a

DIFFERENTIAL PRESSURE INSTRUMENTS				SHEET 22 OF 26	
NO	BY	DATE	REVISION	SPEC. NO. PSI-2020-22	REV.
				CONTRACT 2020-PSI-mail	DATE 22/10/20
				REQ.	P.O. 12345
				BY	CHK'D / APPR.

	No.	Item	Specification
	1	Tag No.	LT - 0 Service E-01 regenerator
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size ______ Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ______
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I For use in Intrinsically Safe System ☐ Other ______
	6	Power Supply	117V 60 Hz ☐ Other ac ______ dc ☑ 24 Volts ______
	7	Chart	12 in. Circ. ☐ Other ______ Range ______ No. ______
	8	Chart Drive	24 hr Other ______ Elec. ☐ Spring ☐ Other ______
	9	Scale	Type ______ Range: 1 ______ 2 ______ 3 ______
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______ For Receiver, See Spec Sheet ______
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate) Sub: s=Slow, f=Fast If☐ Df ☐ P ☐ PI ☐ PD ☐ PID ☐ I_s ☐ D_s ☐ Other ______
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ______
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ______
	15	Manual Reg.	None ☐ MFR STD ☐ Other ______
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ______
UNIT	17	Service	Flow ☐ Level ☑ Diff. Pressure ☐ Other ______
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other ______
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange ______ Body Rating ______ psig
	21	Diff. Range	Fixed ☑ Adj. Range ______ Set At ______
	22		Elevation ______ Suppression ______
	23	Process Data	Fluid catalyst Max Temp. 1310 F Max. Press. 39.7 psig
	24	Process Conn.	½ in. NPT ☐ Other ______
	25	Alarm Switches	Quantity ______ Form ______ Rating ______
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To ______ on Inc. Meas.
	27	Options	Pressure Element ☐ Range ______ Material ______ Temp. Element ☐ Range ______ Type ______ Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ ______ Charts Valve Manifold ______ Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐ Integrator ______ Other ______
	28	MFR & Model No.	Emerson / Rosemount 3051L

Notes:

ISA Form S20.20a

TEMPERATURE INSTRUMENTS (FILLED SYSTEM)		SHEET 23 OF 26	
NO	BY	DATE	REVISION
SPEC. NO. PSI-2020-23	REV.		
CONTRACT PD20-PSI-maul	DATE 22/10/20		
REQ.	P.O. 12395		
BY	CHK'D	APPR.	

	1	Tag No.	TT-06 Service E-01 regenerator
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size ____ Color: MFR STD ☐ Other ____
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other ____
	5	Enclosure Class	General Purpose ☐ Weather proof ☐ Explosion proof ☑ Class I
			For Use in Intrinsically Safe System ☐ Other ____
	6	Power Supply	117 V 60Hz ☐ Other ac ____ dc ☑ 24 Volts
	7	Chart	Strip ☐ ____ Roll ☐ ____ Fold ☐ ____ Circular ____ Time Marks ____
	8	Chart Drive	Speed ____ Power ____
	9	Scales	Type ____
			Range 1 ____ 2 ____ 3 ____ 4 ____
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
			For Receiver See Spec. Sheet ____
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate),
			Sub: s = Slow f = Fast
			P ☐ PI ☐ PD ☐ PID ☐ I_f ☐ D_f ☐ I_s ☐ D_s ☐
			Other ____
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other ____
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other ____
	15	Manual Reg.	None ☐ MFR STD ☐ Other ____
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other ____
ELEMENT	17	Fill	SAMA Class III A Compensation ____
	18	Process Data	Temp: Normal ____ Max 1310 °F Max. Press. 39,2 PSIG
	19	Range	Fixed ☑ Adj. Range ____ Set At ____
			Overrange Protection to ____
	20	Bulb	Type rigid Mtl. 321SST Extension: Length ____ Type ____
			Size: Diameter 15mm Length 50 mm Insertion 30 mm
			Conn: ____ Location 1' Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. type K Armor 316 SST
	22	Well	Mtl. 321 SST Insertion 80 mm Lag Ext. ____ Conn. ____
			Const: Drilled ☑ Built-Up ☐ Other ____
	23	Alarm Switches	Quantity ____ Form ____ Rating ____
	24	Function	Temp ☐ Deviation ☐ Contacts To ____ On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ ____ Charts
			Other ____
	26	Mfr. & Model No.	Emerson / Rosemount 214C . TK

Notes:

ISA FORM S20.11a

TEMPERATURE INSTRUMENTS (FILLED SYSTEM)	SHEET 24 OF 26	
	SPEC. NO. PSI-2000-24	REV.
NO / BY / DATE / REVISION	CONTRACT 2010-PSI-maul	DATE 22/10/10
	REQ. P.O 12345	
	BY / CHK'D	APPR.

	1	Tag No.	TT-14 Service E-01 regenerator flue gas
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose Weather proof ☐ Explosion proof ☑ Class I
			For Use in Intrinsically Safe System ☐ Other
	6	Power Supply	117 V 60Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	Strip ☐ Roll ☐ Fold ☐ Circular Time Marks
	8	Chart Drive	Speed Power
	9	Scales	Type
			Range 1 2 3 4
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver See Spec. Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate),
			Sub: s = Slow f = Fast
			P ☐ PI ☑ PD ☐ PID ☐ If ☐ Df ☐ Is ☐ Ds ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☑ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
ELEMENT	17	Fill	SAMA Class III A Compensation
	18	Process Data	Temp: Normal Max 1310 °F Max. Press. 39,2 psig
	19	Range	Fixed ☑ Adj. Range Set At
			Overrange Protection to
	20	Bulb	Type rigid Mtl. 321 SST Extension: Length Type
			Size: Diameter 15 mm Length 50 mm Insertion 30 mm
			Conn: Location Ft. Above ☐ Below ☐ Instr.
	21	Capillary	MFR STD ☑ Length 50 mm Mtl. type K Armor 316 SST
	22	Well	Mtl. 321 SST Insertion Lag Ext. Conn.
			Const: Drilled ☑ Built-Up ☐ Other
	23	Alarm Switches	Quantity Form Rating
	24	Function	Temp ☐ Deviation ☐ Contacts To On Temp. Increase
	25	Options	Filt-Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts
			Other
	26	Mfr. & Model No.	Emerson / Rosemount 244C, TK

Notes:

ISA FORM S20.11a

	DIFFERENTIAL PRESSURE INSTRUMENTS				SHEET 25 OF 26
	NO	BY	DATE	REVISION	SPEC. NO. psi-2020-26 REV.
					CONTRACT 2020-psi-maul DATE 22/10/20
					REQ. P.O.
					BY CHK'D APPR.

	No.	Item	Specification
	1	Tag No.	-FT-09 Service E-01 lift air
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☑ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ✓ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☑ Weather proof ☐ Explosion proof ☐ Class
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
			For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P=Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate)
			Sub: s=Slow, f=Fast If ☐ Df ☐ P ☐ PI ☑ PD ☐ PID ☐ Is ☐ Ds ☐
			Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☑
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other cascade
	15	Manual Reg.	None ☐ MFR STD ☑ Other
	16	Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 PSI Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At
	22		Elevation Suppression
	23	Process Data	Fluid air Max Temp. 300 F Max. Press. 30 psia.
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material
			Temp. Element ☐ Range Type
			Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts
			Valve Manifold
			Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐
			Integrator
			Other
	28	MFR & Model No.	Emerson / Rosemount 3051 CFC

Notes:

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	DIFFERENTIAL PRESSURE INSTRUMENTS	SHEET 26 OF 36
		SPEC. NO. PSI-2020-26 REV.
	NO / BY / DATE / REVISION	CONTRACT 2020-PSI-mcau DATE 22/10/20
		REQ. P.O. 12345
		BY / CHK'D / APPR.

	1	Tag No.	FT-13 Service E-0 combustion air
GENERAL	2	Function	Record ☐ Indicate ☑ Control ☐ Blind ☐ Trans ☐ Integ ☐ Other
	3	Case	MFR STD ☑ Nom Size Color: MFR STD ☐ Other
	4	Mounting	Flush ☑ Surface ☐ Yoke ☐ Other
	5	Enclosure Class	General Purpose ☑ Weather proof ☐ Explosion proof ☐ Class
			For use in Intrinsically Safe System ☐ Other
	6	Power Supply	117V 60 Hz ☐ Other ac dc ☑ 24 Volts
	7	Chart	12 in. Circ. ☐ Other Range No.
	8	Chart Drive	24 hr Other Elec. ☐ Spring ☐ Other
	9	Scale	Type Range: 1 2 3
XMTR	10	Transmitter Output	4-20 mA ☑ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other For Receiver, See Spec Sheet
CONTROLLER	11	Control Modes	P-Prop (Gain), I=Integral (Auto Reset), D=Derivative (Rate) Sub: s-Slow, f-Fast If☐ Df ☐ P ☐ PI ☐ PD ☐ PID ☐ I_s ☐ D_s ☐ Other
	12	Action	On Meas. Increase Output: Increases ☐ Decreases ☐
	13	Auto-Man Switch	None ☐ MFR STD ☐ Other
	14	Set Point Adj.	Manual ☐ External ☐ Remote ☐ Other
	15	Manual Reg.	None ☐ MFR STD ☐ Other
	16	Output	4-20 mA ☐ 10-50 mA ☐ 21-103 kPa (3-15 psig) ☐ Other
UNIT	17	Service	Flow ☑ Level ☐ Diff. Pressure ☐ Other
	18	Element Type	Diaphragm ☑ Bellows ☐ Mercury ☐ Other
	19	Material	Body 316 SST Element 316 SST
	20	Rating	Overrange 36 PSI Body Rating psig
	21	Diff. Range	Fixed ☑ Adj. Range Set At
	22		Elevation Suppression
	23	Process Data	Fluid air Max Temp. 300 F Max. Press. 30 psia.
	24	Process Conn.	½ in. NPT ☐ Other
	25	Alarm Switches	Quantity Form Rating
	26	Function	Meas. Var. ☐ Deviation ☐ Contacts To on Inc. Meas.
	27	Options	Pressure Element ☐ Range Material Temp. Element ☐ Range Type Filt Reg. ☐ Sup. Gage ☐ Output Gage ☐ Charts Valve Manifold Cond. Pots ☐ Adj. Damp ☐ Integral Sq. Rt. Ext. ☐ Integrator Other
	28	MFR & Model No.	Emerson / Rosemount CFC

Notes:

ISA Form S20.20a

4) Pasangan CV-MV

* orifice flow meter → Emerson Rosemount 3051CFC

a) Tujuan proses : memaksimalkan produk dengan memaksimalkan feed dengan tetap memperhatikan variabel proses lain → tidak bisa sebanyak-banyak

b) ◦ Manipulated variable

- wash oil valve (V_1)
- diesel oil valve (V_2)
- total feed valve (V_3)
- slurry oil valve (V_4)
- Fuel gas valve (V_5)
- wet gas valve (V_{11})
- Flue gas valve (V_{14})
- Lift air valve (V_{lift})
- Spill air valve (V_9)
- Combustion air valve (V_6)

DOF = 10

* Spill air hanya dipakai ketika pembersihan → ditutup, bukaan 0%
* Combustion air dibuka 100% agar menjamin pembakaran sempurna di regenerator.

⇒ spill air & combustion air tak bisa dimanipulasi → DOF = 8

◦ Controlled variable

- wash oil flow rate (F_1)
- diesel oil flow rate (F_2)
- Total fresh feed flow rate (F_3)
- slurry oil flow rate (F_4)
- Fuel gas flow rate (F_5)
- wet gas flow rate (F_{V11})
- Flue gas flow rate (F_{sg})
- Lift air flow rate (F_9)
- Tekanan reaktor (P_4)
- Suhu reaktor (T_r)
- Tekanan regenerator (P_6)
- Suhu regenerator (T_{reg})
- Tekanan menara fraksinasi (P_5)
- Suhu feed outlet preheater (T_2)
- Suhu flue gas (T_{sg})

c) Keselamatan operasional

o) CV: tekanan reaktor (P_4) – MV: wet gas flow valve (V_{11})

- Tekanan reaktor dipengaruhi oleh tekanan menara fraksinasi dan feed flow rate. Akan tetapi tekanan menara fraksinasi yang lebih mempengaruhi karena beda tekanan reaktor dan menara fraksinasi harus selalu konstan agar produk perengkahan dapat mengalir secara spontan ke menara fraksinasi.
- Tekanan menara fraksinasi dipengaruhi oleh laju aliran wet gas. Laju wet gas merupakan produk FCC. Oleh karena itu dengan mengatur tekanan reaktor sekaligus akan membuat produksi maksimal
- Pasangan : tekanan reaktor (P_4) – Wet gas flow valve (V_{11}), DOF=7
- Wet gas flow rate → measured variable untuk memantau laju produk
- Tekanan menara fraksinasi → measured variable untuk memantau tekanan

d) Laju produksi

o) CV: beda tekanan reaktor - regenerator (ΔP_{RR}) – MV: Flue gas valve (V_{14})

Perubahan tekanan pada reaktor akan mempengaruhi tekanan regenerator sehingga beda tekanan reaktor-regenerator juga akan berubah. Hal ini akan menyebabkan laju sirkulasi katalis juga berubah. Beda tekanan dipilih sebagai controlled variable karena dapat digunakan untuk mengatur tekanan regenerator sekaligus untuk memastikan beda tekanan reaktor-regenerator pada rentang yang optimal agar produksi maksimal.

Beda tekanan reaktor-regenerator dipengaruhi oleh :
- Tekanan reaktor → sudah dikontrol
- Tekanan regenerator : - Flue gas
  - Lift air
  - Combustion air → tidak dapat dikontrol

Flue gas lebih mempengaruhi tekanan regenerator daripada lift air karena komponen terbesar yang menekan regenerator

- Pasangan : Beda tekanan reaktor -regenerator – Flue gas valve (V_{14})
- Tekanan regenerator tidak menjadi measured variable karena dapat dikalkulasi dari tekanan reaktor (P_4) dan beda tekanan reaktor-regenerator (ΔP_{RR})
- DOF = 7-1 = 6

o) CV: Suhu feed outlet preheater (T_2) – MV: fuel gas valve (V_5)

suhu feed akan mempengaruhi suhu reaktor. Selain itu, karena reaksi endotermik, suhu feed perlu diatur agar produksi maksimal. Suhu feed dipengaruhi oleh suhu furnace preheater yang dipengaruhi oleh laju fuel gas untuk pembakaran. oleh karena itu suhu feed diatur dengan mengatur

laju fuel gas pembakaran. Selain itu, karena kontrol suhu mempunyai respon yang lama, maka loop kontrol suhu di-cascade dengan loop kontrol fuel gas flow dimana kontrol suhu sebagai master yang memberi set point kepada kontrol flow (slave) sesuai dengan yang dibutuhkan. *
- Pasangan : suhu feed (T_2) - fuel gas valve (V_5)
- DOF = 6-1 = 5

o> CV = wash oil flow rate (F_1) - MV = wash oil valve (V_1)
  - wash oil flow rate perlu dikontrol dan diatur pada flow optimalnya agar produksi maksimal
  - Wash oil flow rate diatur menggunakan wash oil valve
  - Pasangan : wash oil flow rate (F_1) - wash oil valve (V_1)
  - DOF = 5-1 = 4

o> CV = diesel oil flow rate (F_2) - diesel oil valve (V_2)
  - diesel oil flow rate perlu dikontrol dan diatur pada flow optimalnya agar produksi maksimal
  - diesel oil flow rate diatur dengan diesel oil valve
  - pasangan : diesel oil flow rate (F_2) - diesel oil valve (V_2)
  - DOF = 4-1 = 3

o> CV : total fresh feed flow rate (F_3) - total fresh feed valve (V_3)
  - total fresh feed flow rate harus dikontrol dan diatur pada flow optimalnya agar produksi maksimal dengan mengatur total fresh feed valve (V_3)
  - pasangan : total fresh feed flow rate (F_3) - total fresh feed valve (V_3)
  - DOF = 3-1 = 2

o> CV : slurry recycle oil flow rate (F_4) - slurry recycle oil valve (V_4)
  - slurry recycle oil flow rate harus dikontrol dan diatur pada flow optimalnya agar produksi maksimum dengan mengatur slurry recycle oil valve
  - pasangan : slurry recycle oil flow rate (F_4) - slurry recycle oil valve (V_4)
  - DOF = 2-1 = 1

e) Kesetimbangan komponen

o> CV: suhu flue gas (T_{sg}) - MV: lift air valve (V_{lift})
  - Suhu flue gas digunakan untuk mengukur kadar O_2 pada flue gas. Suhu flue gas dipengaruhi oleh lift air dan combustion air flow. Akan tetapi karena combustion air flow valve selalu terbuka 100% maka lift air dipilih sebagai manipulated variable.
  - Pasangan : suhu flue gas (T_{sg}) - lift air valve (V_{lift})
  - kontrol suhu mempunyai respon yang lama sehingga di cascade dengan kontrol flow lift air
  - DOF = 1-1 = 0

* Measured variable

- Tekanan menara fraksinasi (P_5)
  Digunakan untuk memantau tekanan menara fraksinasi
  - Analisis dan perhitungan dari no. 3
- Wet gas flow rate (V_u)
  Untuk memantau laju produksi dari FCC
  - Analisis dan perhitungan dari no 3
- Suhu reaktor
  Untuk memantau suhu riser-reaktor karena suhunya tidak diatur secara langsung melainkan tergantung pada suhu feed (T_2) dan suhu katalis (suhu regenerator)
  - Analisis dan perhitungan dari no. 3
- Suhu regenerator
  Untuk memantau suhu regenerator karena suhunya tidak diatur secara langsung melainkan tergantung pembakaran di regenerator sempurna / tidak sempurna.
  - Analisis dan perhitungan dari no. 3
- Suhu feed sebelum preheater → disturbance (T_1)
  - Untuk memantau suhu feed sebelum preheater karena merupakan disturbance
  - range : 460,9 °F → 300 - 600 °F → RTD dengan material platinum
  - span : 300 °F
  - akurasi : ± 0,01 - 0,2 % span
  - karakteristik linier
  - Tekanan : 0 - 200 psig
  - voltage input : 24 VDC
  - output : 4 - 20 mA
  - flammable fluid → explosion proof

  * Dipilih RTD platinum → Rosemount 214C RH
- Suhu furnace preheater (T_3)
  - Untuk memantau suhu furnace karena mempunyai batasan maksimal
  - range : 1200 - 1700 °F → thermocouple tipe K
  - span : 500 °F
  - akurasi : ± 0,01 - 0,2 % span
  - karakteristik linier

  voltage input : 24 VDC
  output : 4 - 20 mA
  proses pembakaran → explosion proof

  * Dipilih thermocouple tipe K → Rosemount 214C TK
- Combustion air flow rate (F_7)
  Untuk memantau combustion air flow rate
  - range : 0 - 60,99 lb/s
  - span : 60,99 lb/s
  - akurasi : ± 1 - 2 % span

  rangeability : 4-8 : 1
  karakteristik linier
  suhu : 270 °F

  Voltage input : 24 VDC
  output : 4 - 20 mA
  non flammable → general purpose

* Dipilih orifice flow meter → Rosemount 3051 CFC

o> Level katalis pada stand pipe (Lsp)

- untuk memantau ketinggian katalis pada stand pipe

- range : 0 - 20 ft	voltage input : 24 VDC
- span : 20 ft	output : 4-20 mA
- akurasi : ±0,01 - 0,2% span	proses pembakaran → explosion proof
- karakteristik linier	

* Dipilih differential pressure level transmitter → Rosemount 3051L

5) P & ID

- Keluaran transmitter berupa sinyal listrik 4-20 mA sehingga sinyal ditransmisikan ke control room dengan sinyal elektrik
- Control valve menggunakan sistem pneumatic, oleh karena itu sinyal elektrik dari control room dikonversi menjadi sinyal pneumatic, baru kemudian digunakan untuk mengontrol control valve. sinyal kontrol pneumatic : 3-15 psi dengan supply 20 psi
- Controller menggunakan DCS

o> Equipment tag number

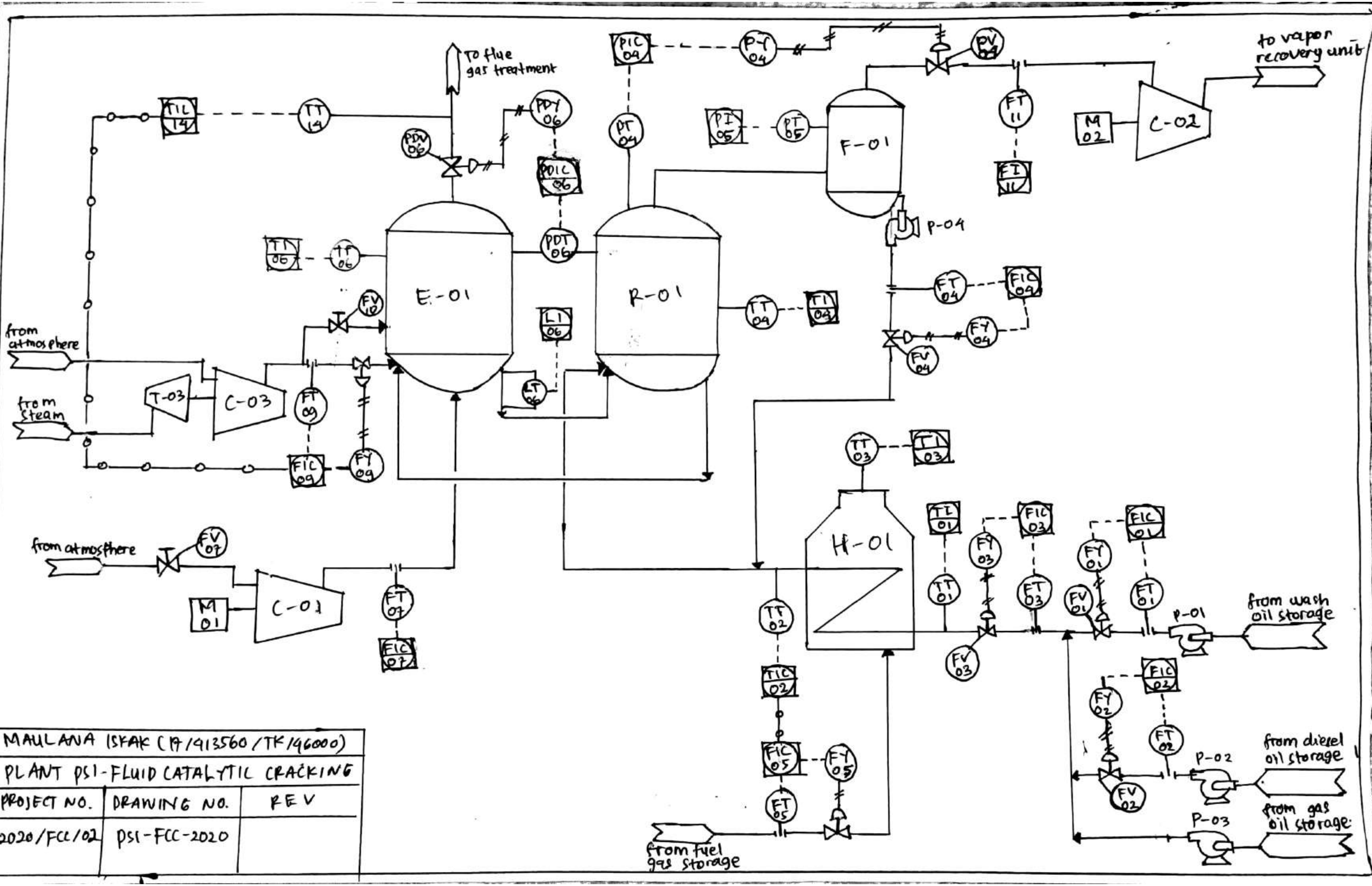
- wash oil pump : P-01	- kompresor wet gas : C-02
- diesel oil pump : P-02	- kompresor combustion air : C-01
- gas oil pump : P-03	- kompresor lift air : C-03
- slurry oil pump : P-04	- Turbin lift air : T-03
- Furnace preheater : H-01	- Motor combustion air : M-01
- Reaktor : R-01	- Motor wet gas : M-02
- Regenerator : E-01	
- Menara fraksinasi : F-01	

o> Instrument tag number

- wash oil flow transmitter : FT-01	- fresh feed temperature transmitter : TT-01
- wash oil flow indicator controller : FIC-01	- fresh feed temperatur indicator : TI-01
- wash oil I/P converter : FY-01	- feed preheat temperatur transmitter : TT-02
- wash oil control valve : FV-01	- feed preheat indicator controller : TIC-02
- diesel oil flow transmitter : FT-02	- furnace temperature transmitter : TT-03
- diesel oil flow indicator controller : FIC-02	- furnace temperature indicator : TI-03
- diesel oil I/P converter : FY-02	- reaktor temperature transmitter : TT-04
- diesel oil control valve : FV-02	- reaktor temperature indicator : TI-04
- fresh feed flow transmitter : FT-03	- reaktor pressure transmitter : PT-04
- fresh feed flow indicator controller : FIC-03	- reaktor pressure indicator contr. : PIC-04
- fresh feed I/P converter : FY-03	- reaktor I/P converter : PY-04
- fresh feed control valve : FV-03	- wet gas control valve : PV-04
- slurry oil flow transmitter : FT-04	- wet gas flow transmitter : FT-11
- slurry oil flow indicator controller : FIC-04	- wet gas flow indicator : FI-11
- slurry oil I/P converter : FY-04	- fractionation pressure transmit : PT-05
- slurry oil control valve : FV-04	- fractionation pressure indicator : PI-05
- fuel gas flow transmitter : FT-05	- reactor-regenerator diff. pressure transmitter : PDT-06
- fuel gas flow indicator controller : FIC-05	
- fuel gas I/P converter : FY-05	- reactor-regenerator diff. pressure indicator controller : PDIC-06
- fuel gas control valve : FV-05	

- reactor-regenerator I/P converter : PDY-06
- flue gas control valve : PDV-06
- regenerator temperature transmitter : TT-06
- regenerator temperature indicator : TI-06
- flue gas temperature transmitter : TT-14
- flue gas temp. indicator controller : TIC-14
- spill air control valve : FV-10

- lift air flow transmitter : FT-09
- lift air flow indicator controller : FIC-09
- lift air I/P converter : FY-09
- lift air control valve : FV-09
- combustion air flow transmitter : FT-07
- combustion air control valve : FV-07
- catalyst level transmitter : LT-06
- catalyst level indicator : LI-06



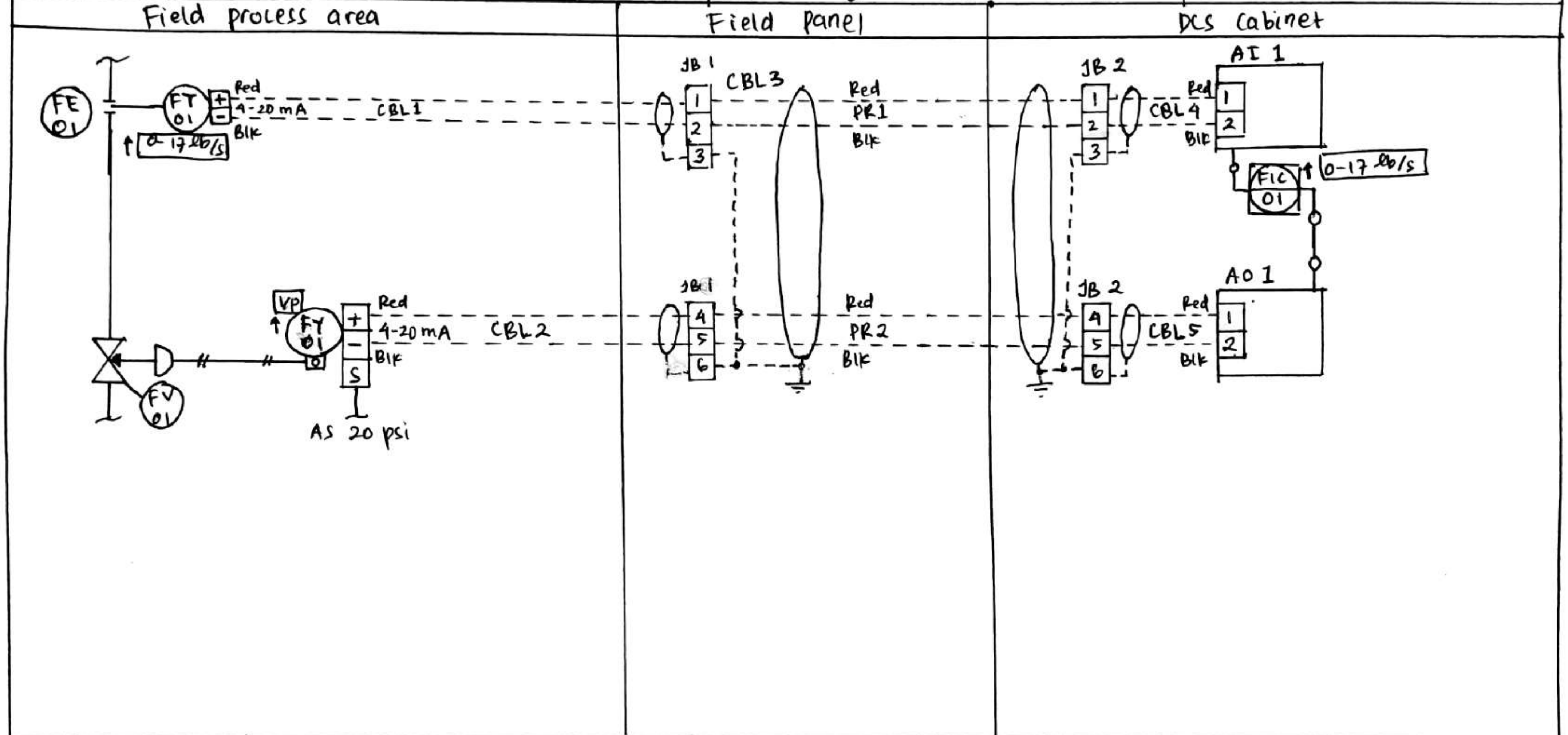
MAULANA ISKAK (19/413560/TK/46000)		
PLANT PSI-FLUID CATALYTIC CRACKING		
PROJECT NO.	DRAWING NO.	REV
2020/FCC/02	PSI-FCC-2020	

6) Analisis Instrument Loop Diagram

- Semua input tegangan pada transmitter di soal no. 2 adalah tegangan DC 24 V → menggunakan 2 wire transmitter. Kemudian controller berupa DCS, menyuplai arus 4-20 mA dan dikoneksikan dengan transmitter melalui 2 wires connection. Selain itu, 2 wire connection dipilih karena kondisi plant FCC yang banyak mengandung flammable fluid, dengan 24V DC input dan 2 wire tidak akan terjadi percikan yang dapat memicu kebakaran / ledakan
- Karena tegangan dan arus sinyal transmitter yang kecil (24 V, 4-20mA) maka rawan terkena noise akibat medan listrik maupun medan magnet yang menyebabkan sinyal transmitter tidak dapat diterima controller (DCS) dengan baik. Oleh karena itu dipakai kabel dengan jenis twisted and shielded cable agar terhindar dari noise. Setiap kabel dari transmitter ke junction box field menggunakan twisted, shielded cable. Sedangkan kabel dari junction box field menuju junction box DCS cabinet adalah semua twisted cable dijadikan satu kemudian diberi satu shield pada kumpulan kabel tersebut (5 twisted cable dengan 1 shield) → 1 junction box : 16 port (6 port ground)
- Konfigurasi pada control valve sama dengan konfigurasi transmitter
- Konfigurasi berlaku untuk semua transmitter dan control valve FCC

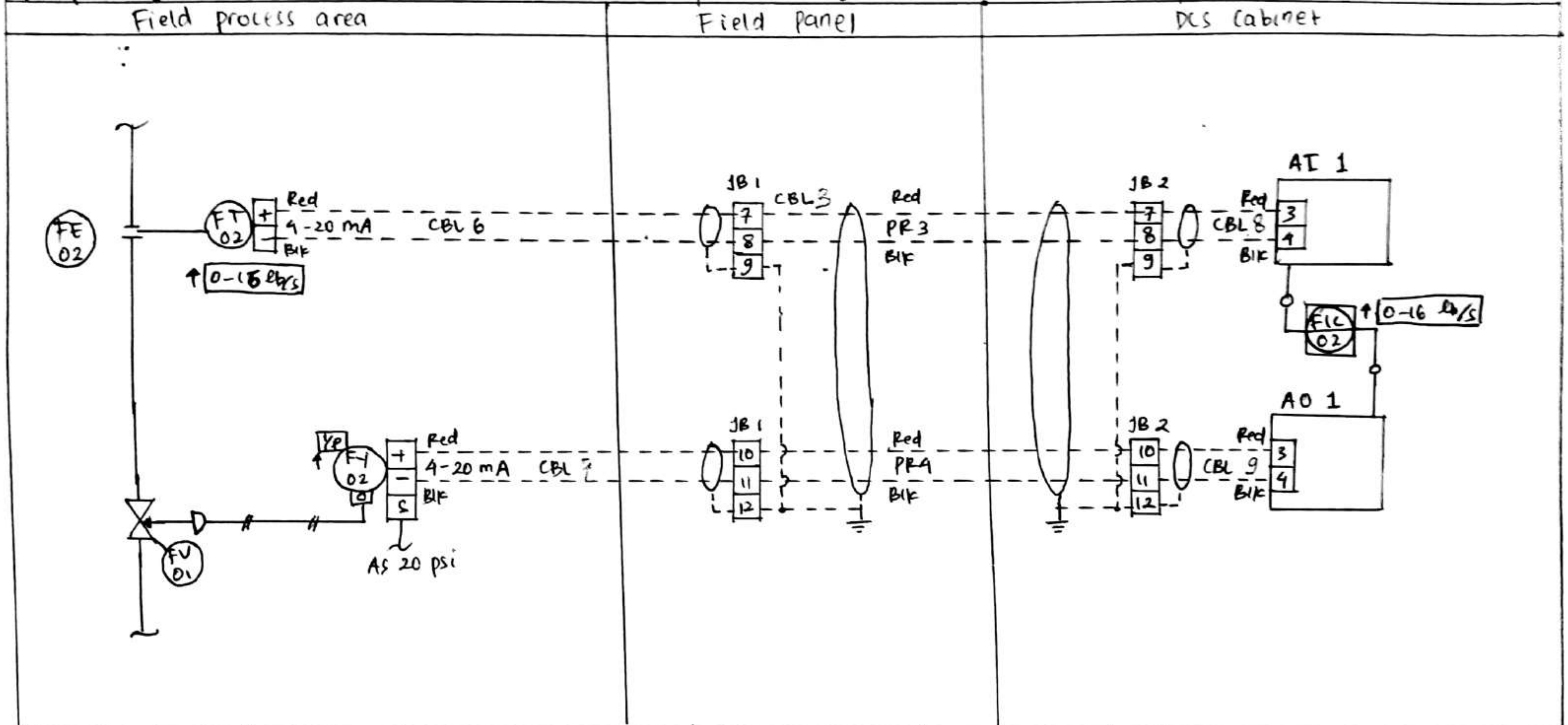
VISION

Loop diagram : wash oil flow rate control	Revised by : Maulana Iskak	Date : October 22, 2020



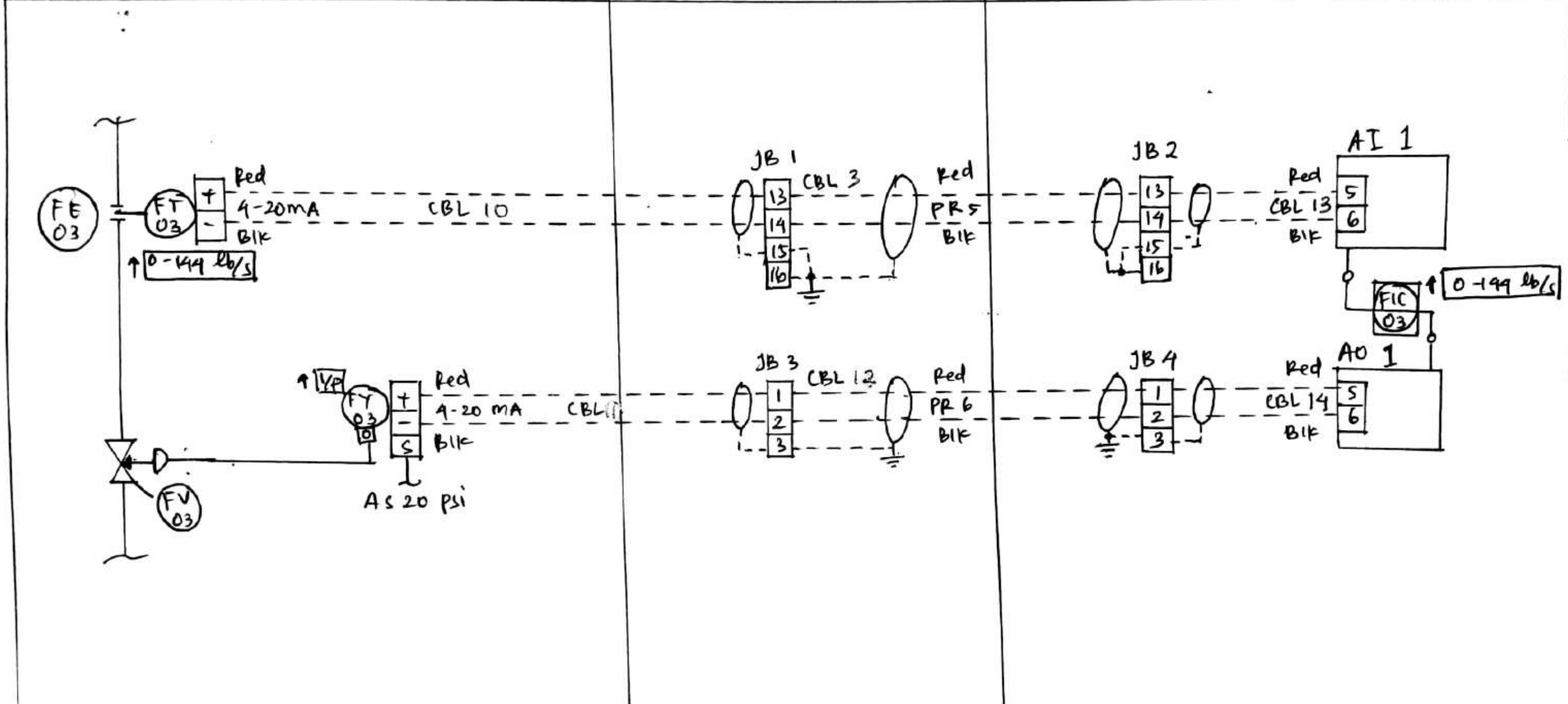
Tag number	Description	Input range	Output range	Notes
FE-01	Orifice plate	0-17 lb/s	0-100 " WC	
FT-01	wash oil flow transmitter	0-100 " WC	4-20 mA	
FIC-01	wash oil flow controller	4-20 mA	4-20 mA	
FY-01	current to pressure converter	4-20 mA	3-15 psi	
FV-01	wash oil flow control valve	3-15 psi	0-100 %	Air to open

Loop diagram : diesel oil flow rate control | Revised by : Maulana Iskak | Date : October 22 2020



Tag number	Description	Input range	Output range	Notes
FE-02	orifice plate	0-17 lb/s	0-100 " WC	
FT-02	diesel oil flow transmitter	0-100 " WC	4-20 mA	
FIC-02	diesel oil flow controller	4-20 mA	4-20 mA	
FY-02	current to pressure converter	4-20 mA	3-15 psi	
FV-02	diesel oil flow control valve	3-15 psi	0-100 %	Air to open

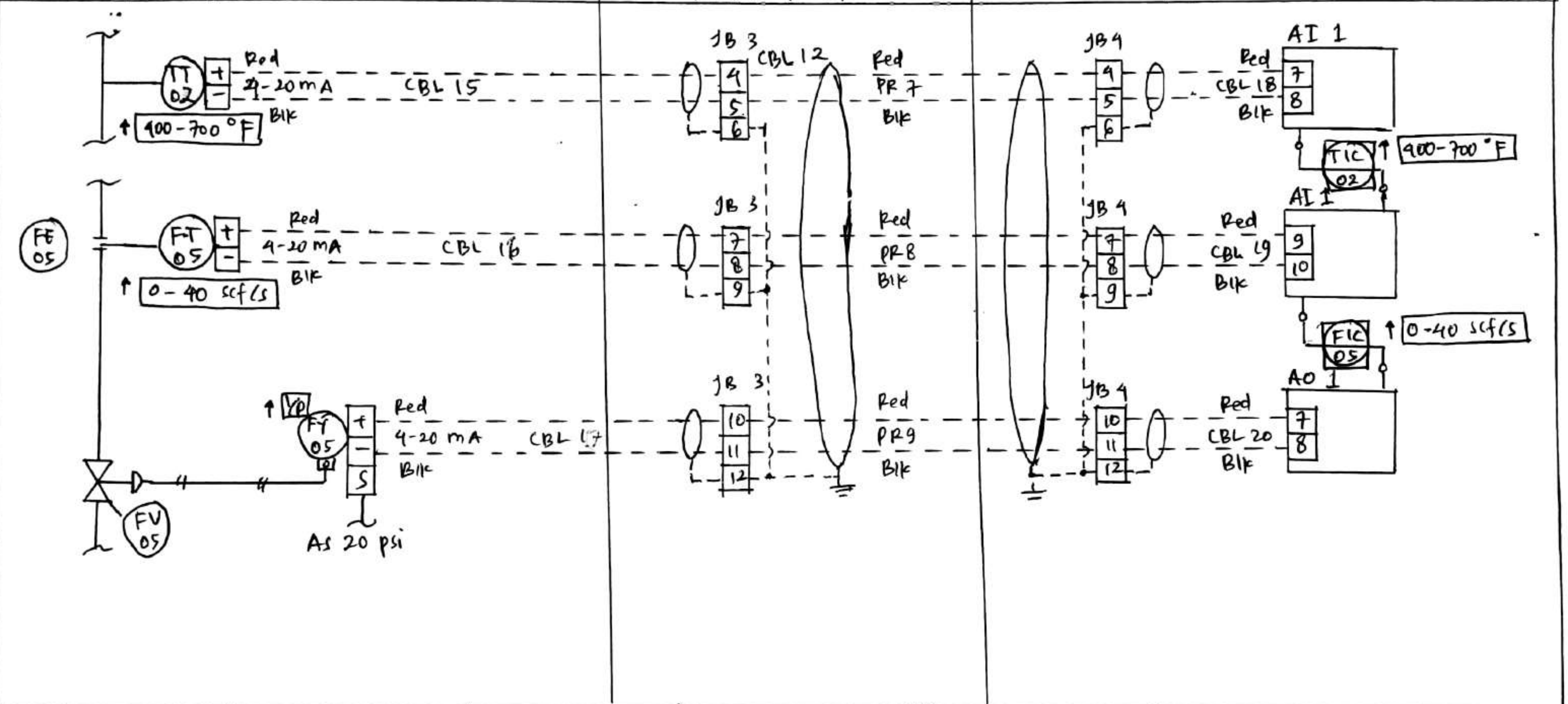
Loop diagram : Gas oil flow rate control	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
FE - 03	orifice plate	0-144 lb/s	0-100" WC	
FT - 03	gas oil flow transmitter	0-100" WC	4-20 mA	
FIC - 03	gas oil flow controller	4-20 mA	4-20 mA	
FY - 03	current to pressure converter	4-20 mA	3-15 psi	
FV - 03	gas oil flow control valve	3-15 psi	0-100 %	Air to open

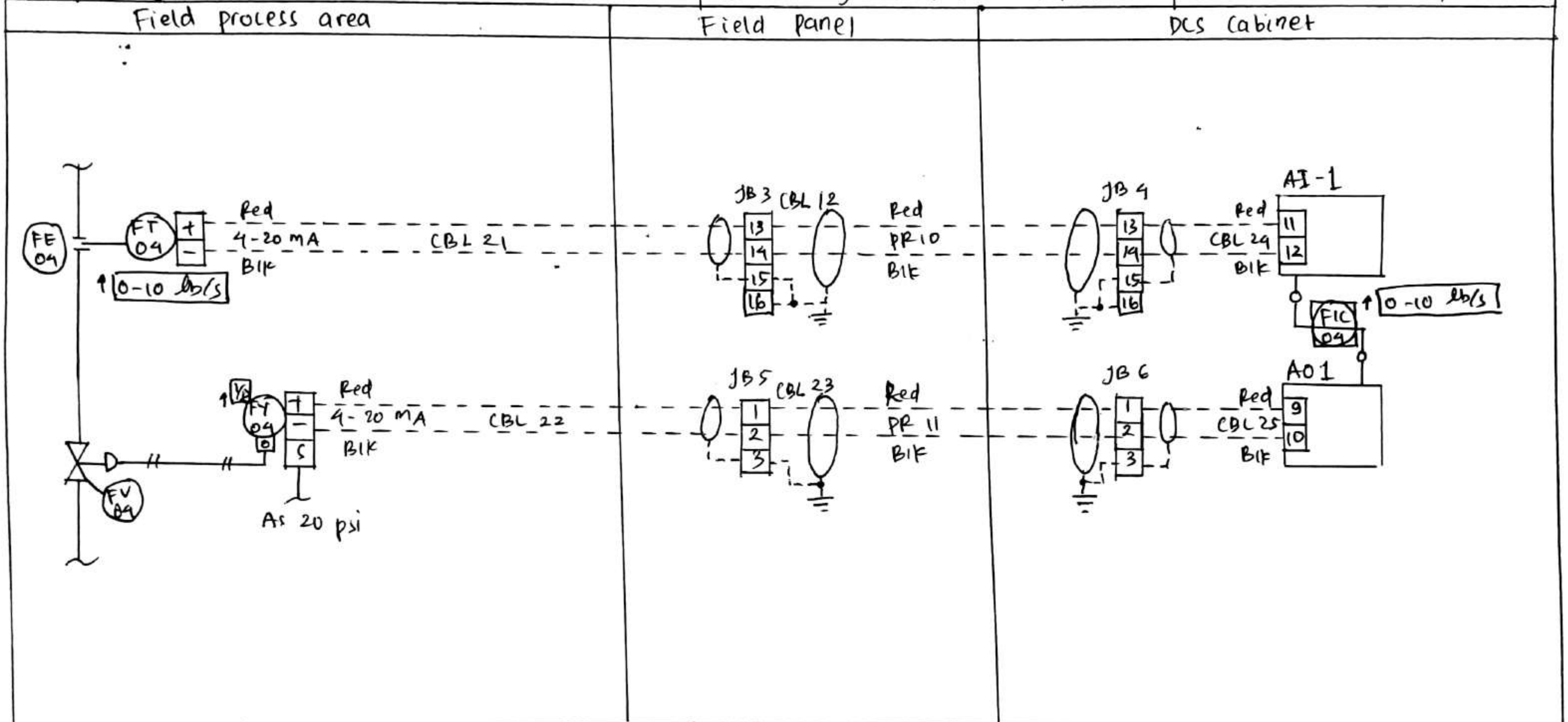
Loop diagram : Fuel gas flow rate control | Revised by : Maulana Iskak | Date : October 22, 2020

Field process area | Field panel | DCS cabinet



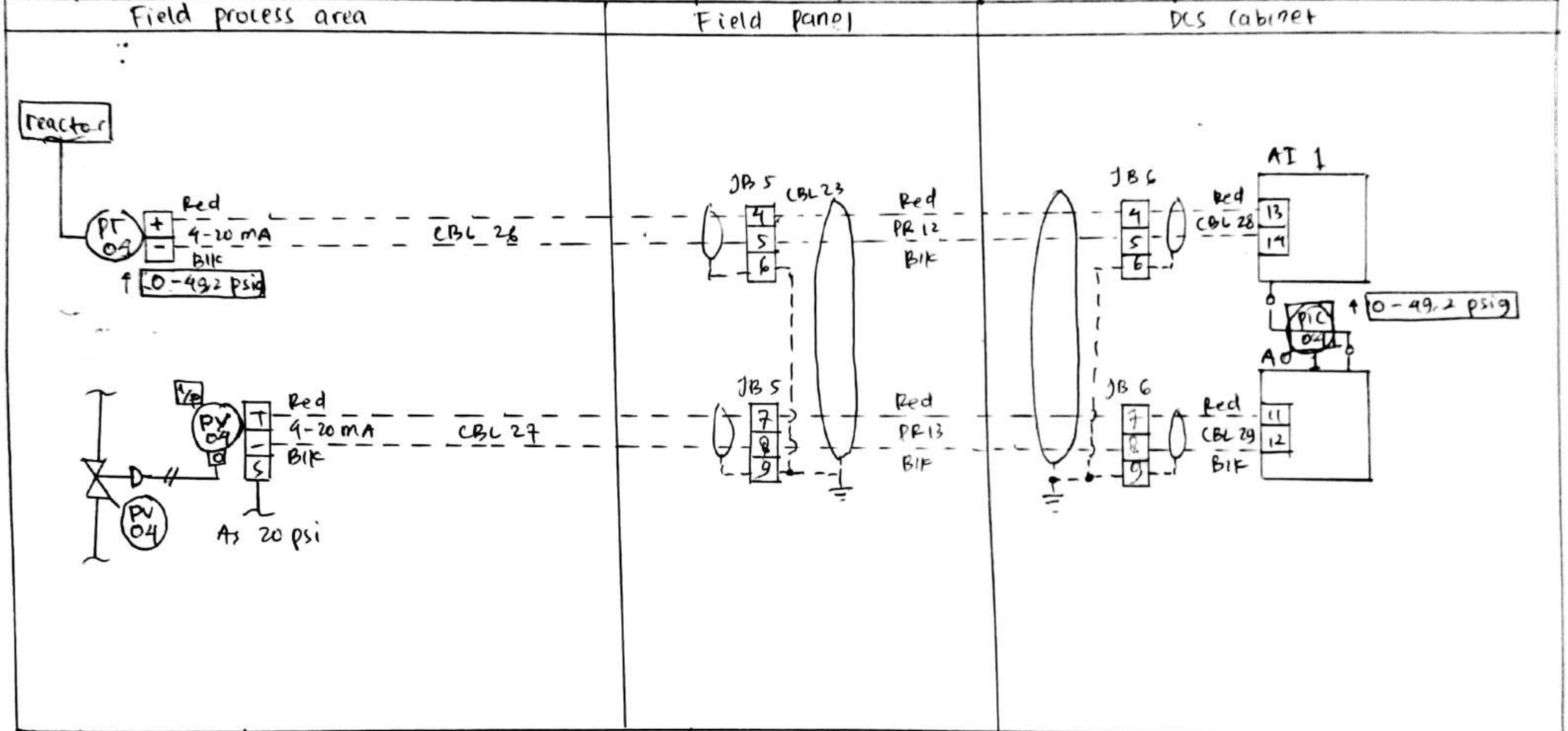
Tag number	Description	Input range	Output range	Notes
TT-02	feed preheater temperature transmitter	400-700 °F	4-20 mA	
TIC-02	feed preheater temperature controller	4-20 mA	4-20 mA	
FE-05	orifice plate	0-40 scf/s	0-100 "WC	
FT-05	fuel gas flow transmitter	0-100 " WC	4-20 mA	
FIC-05	fuel gas flow controller	4-20 mA	4-20 mA	
FY-05	current to pressure converter	4-20 mA	3-15 psi	
FV-05	fuel gas flow control valve	3-15 psi	0-100 %	Air to open

Loop diagram : slurry flow rate control	Revised by : Maulana Iskak	Date : October 22, 2020



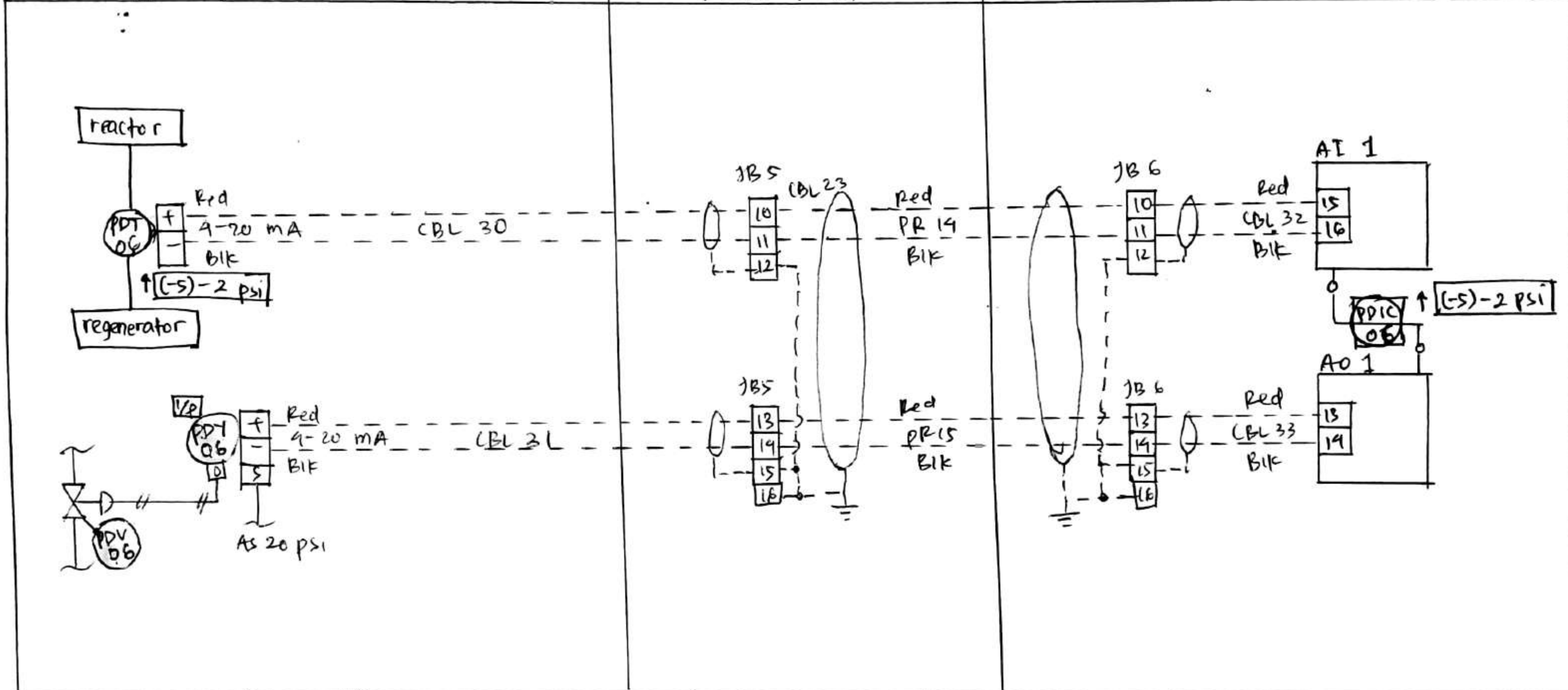
Tag number	Description	Input range	Output range	Notes
FE-04	orifice plate	0-10 lb/s	0-100 WC	
FT-04	slurry oil flow transmitter	0-100" WC	4-20 mA	
FIC-04	slurry oil flow controller	4-20 mA	4-20 mA	
FY-04	current to pressure converter	4-20 mA	3-15 psi	
FV-04	slurry oil flow control valve	3-15 psi	0-100 %	air to open

Loop diagram : reactor pressure control	Revised by : Maulana Iskak	Date : October 22, 2020



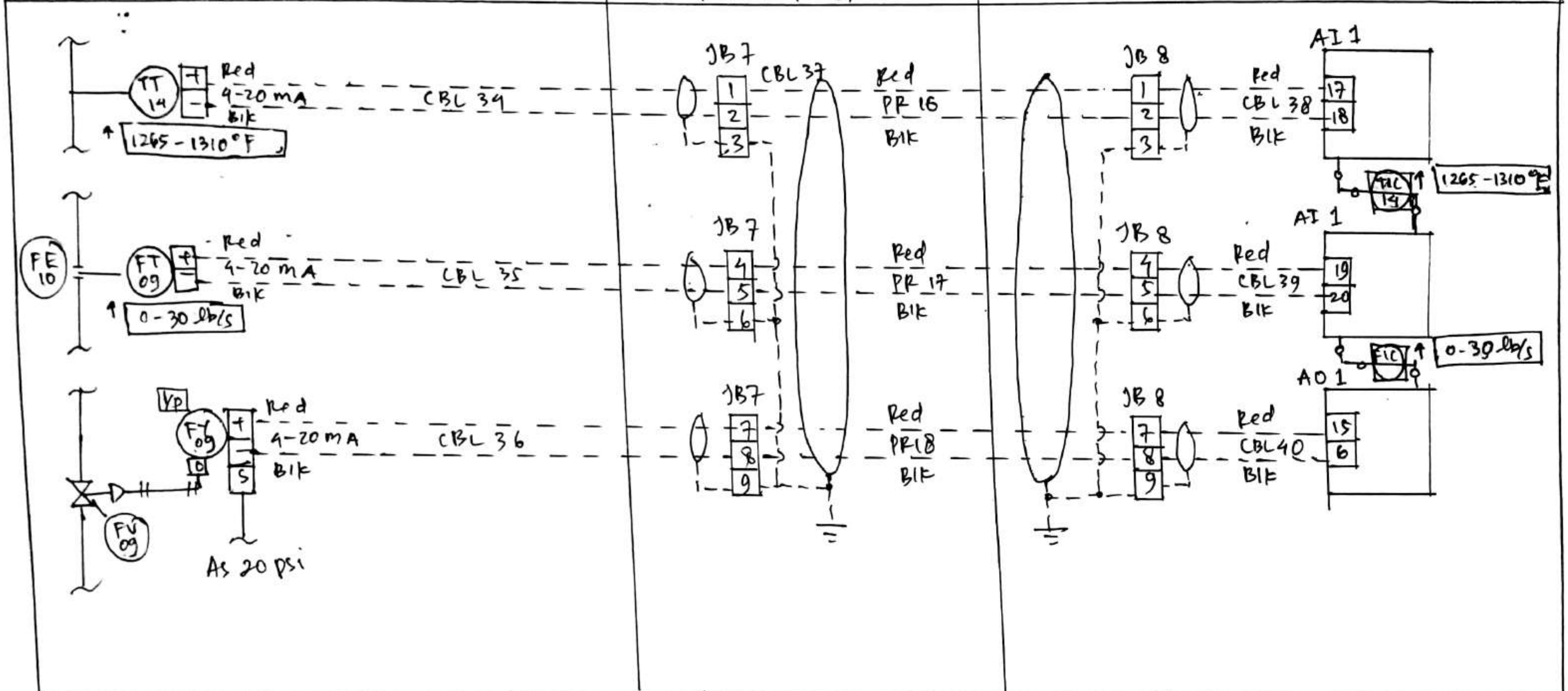
Tag number	Description	Input range	Output range	Notes
PT-04	reactor pressure transmitter	0-49,2 psig	4-20 mA	
PIC-04	reactor pressure controller	4-20 mA	4-20 mA	
PY-04	current to pressure converter	4-20 mA	3-15 psi	
PV-04	wet gas flow control valve	3-15 psi	100-0 %	air to close

Loop diagram : reactor-regenerator diff. press. cont.	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



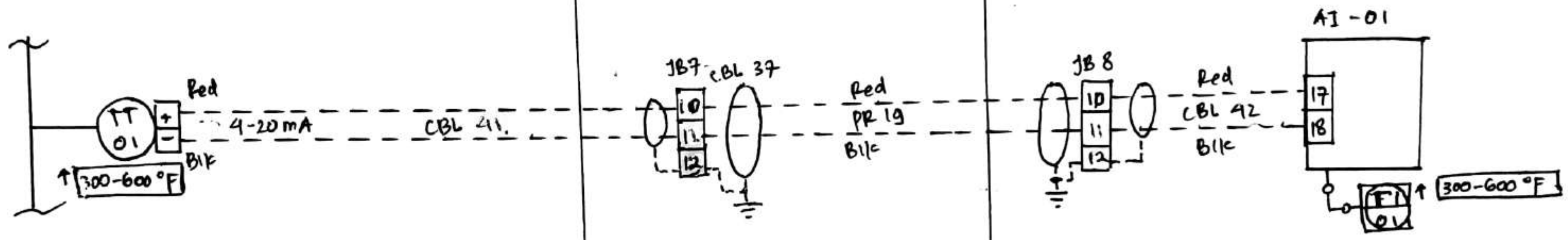
Tag number	Description	Input range	Output range	Notes
PDT-06	reactor-regenerator differential pressure transmitter	(-5) - 2 psi	4-20 mA	
PDIC-06	reactor-regenerator differential pressure controller	4-20 mA	4-20 mA	
PDY - 06	current to pressure converter	4-20 mA	3-15 psi	
PDV- 06	flue gas flow control valve	3-15 psi	100-0 %	air to close

Loop diagram : oxygen concentration control	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
TT - 14	flue gas temperature transmitter	1265 - 1310 °F	4-20 mA	
TIC - 14	flue gas temperature controller	4-20 mA	4-20 mA	
FE - 09	orifice plate	0-30 lb/s	0-100 " WC	
FT - 09	lift air flow transmitter	0 - 100 " WC	4-20 mA	
FIC - 09	lift air flow controller	4 - 20 mA	4- 20 mA	
FY - 09	current to pressure converter	4 - 20 mA	3-15 psi	
FV - 09	lift air flow control valve	3 -15 psi	0 -100 %	air to open

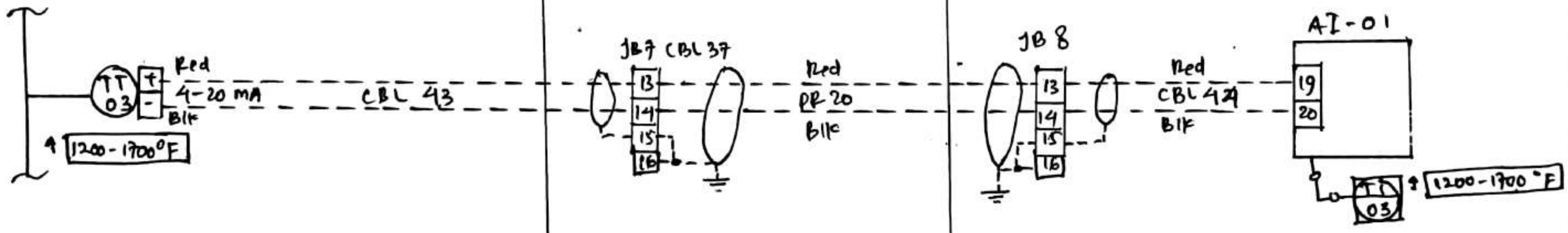
Loop diagram : fresh feed temperature indicator	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
TT-01	fresh feed temperature transmitter	300-600 °F	4-20 mA	
TI-01	fresh feed temperature	4-20 mA	300-600 °F	

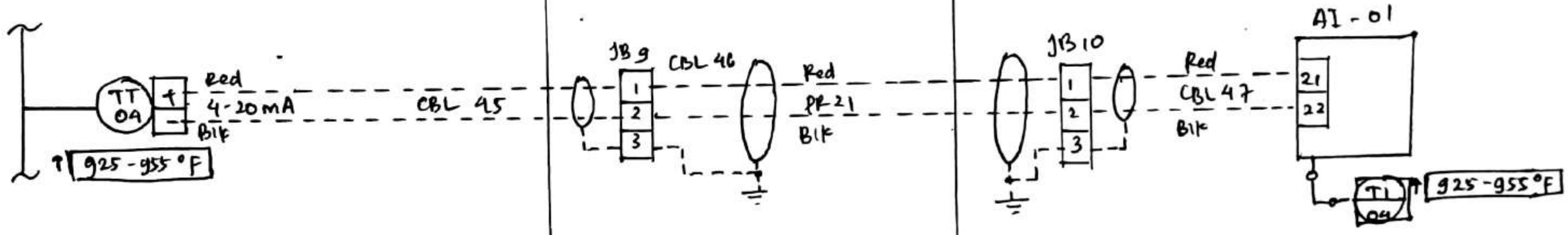
Loop diagram : furnace preheater temperature indicator	Revised by : Maulana Iskak	Date : October 22, 2020

Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
TT-03	furnace temperature transmitter	1200-1700 °F	4-20 mA	
TI-03	furnace temperature indicator	4-20 mA	1200-1700 °F	

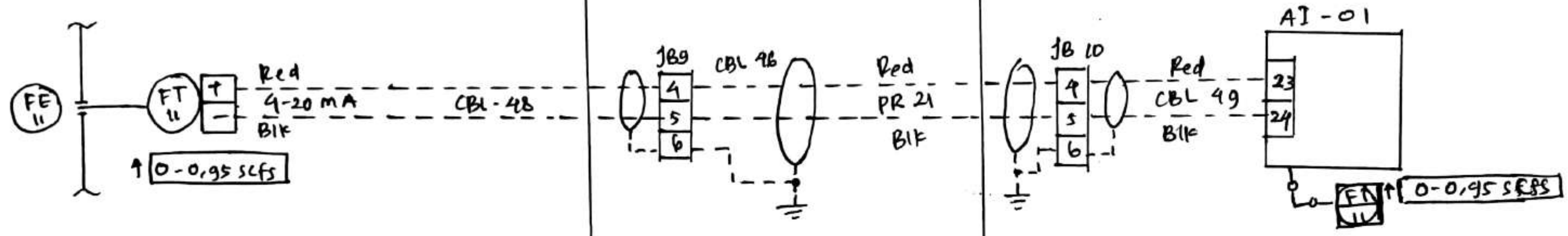
Loop diagram : reactor temperature indicator	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
TT-04	reactor temperature transmitter	925 - 995 °F	4 - 20 mA	
TI-04	reactor temperature indicator	4-20 mA	925 - 995 °F	

Loop diagram : wet gas flow indicator	Revised by : Maulana Iskak	Date : October 22, 2020

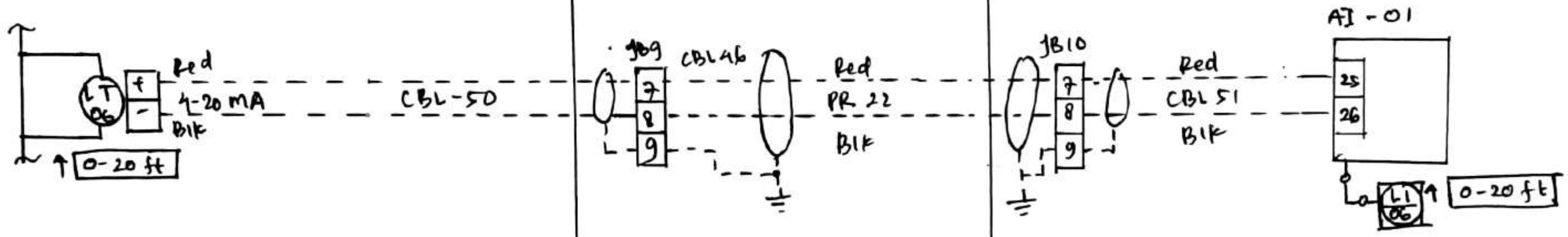
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
FE-11	orifice plate	0-0,95 scfs	0-100" Wc	
FT-11	wet gas flow transmitter	0-100" W	4-20 mA	
FI-11	wet gas flow indicator	4-20 mA	0-0,95 scfs	

Loop diagram : catalyst level indicator	Revised by : Maulana Iskak	Date : October 22, 2020

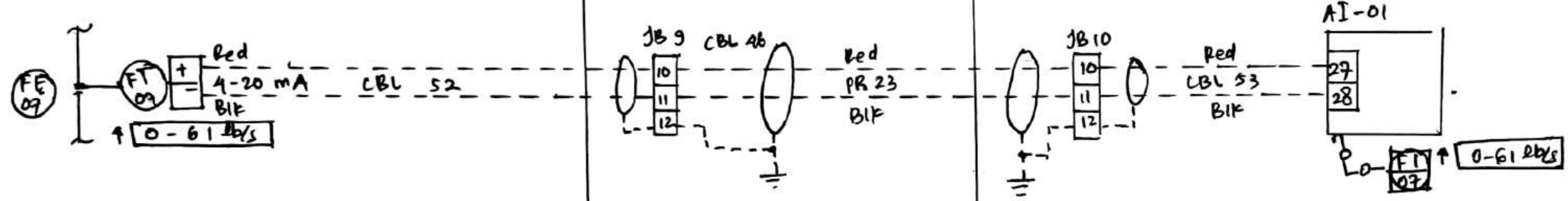
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
LT-06	catalyst level transmitter	0-20 ft	4-20 mA	
LI-06	catalyst level indicator	4-20 mA	0-20 ft	

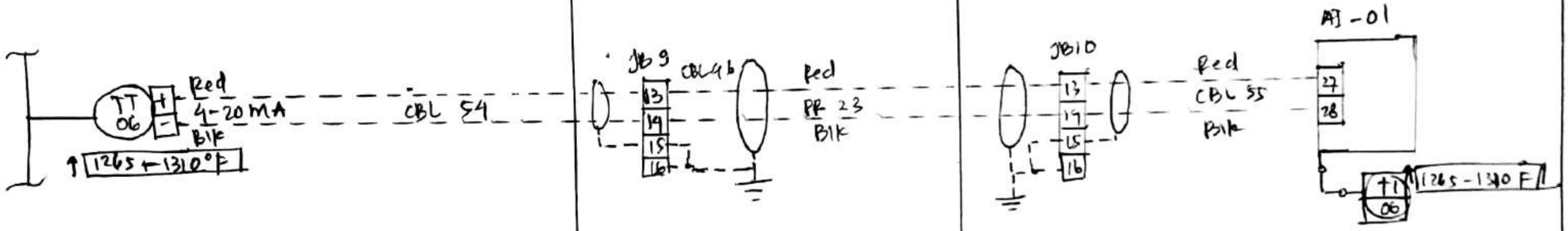
Loop diagram : combustion air flow indicator	Revised by : Maulana Iskak	Date : October 22, 2020

Field process area	Field panel	DCS cabinet



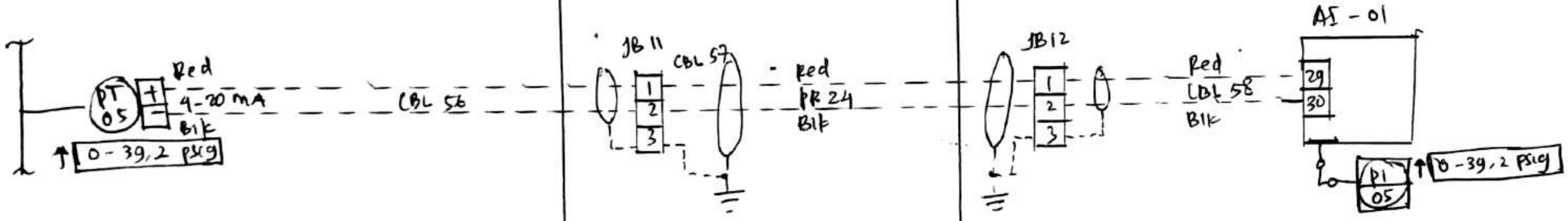
Tag number	Description	Input range	Output range	Notes
FE-07	orifice plate	0-61 lb/s	0-100" WC	
FT-07	combustion air flow transmitter	0-100" WC	4-20 mA	
FI-07	combustion air flow indicator	4-20 mA	0-61 lb/s	

Loop diagram : regenerator temperature indicator	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS cabinet



Tag number	Description	Input range	Output range	Notes
TT-06	regenerator temperature transmitter	1265 - 1310 °F	4 - 20 mA	
TI -06	regenerator temperature indicator	4 - 20 mA	1265 - 1310 °F	

Loop diagram : fractionation pressure indicator	Revised by : Maulana Iskak	Date : October 22, 2020
Field process area	Field panel	DCS Cabinet



Tag number	Description	Input range	Output range	Notes
PT-05	fractionation pressure transmitter	0-39,2 psig	4-20 mA	
PI-05	fractionation pressure indicator	4-20 mA	0-39,2 psig	

NAMA **: MAULAN ISKAK**

NIM **: 17/413560/TK/46000**

TUGAS MINGGU KE-6

Bagian B

1. Apakah yang dimaksud dengan sistem linier?

   Jawab :

   Sistem linear adalah sistem yang memiliki sifat penting dari superposisi. Misalkan $y_1(t)$ adalah respons dari sistem kontinu dengan input $x_1(t)$, dan $y_2(t)$ adalah respons untuk input $x_2(t)$ maka sistem bersifat linier jika [1]:

   a. Respons dari $x_1(t) + x_2(t)\ adalah\ y_1(t) + y_2(t)$ atau :

      $\boldsymbol{f\{x_1(t) + x_2(t)\} = f(x_1(t)\} + f\{x_2(t)\} \rightarrow aditif}$

   b. Respons dari $ax_1(t)\ adalah\ ay_1(t)$ atau :

      $\boldsymbol{f\{ax(t)\} = af\{x(t)\} \rightarrow homogen}$

2. Apakah yang dimaksud dengan sistem invarian-waktu?

   Jawab :

   Sistem invarian-waktu adalah sistem yang perilaku dan karakteristiknya akan tetap sama dari waktu ke waktu. Sistem bersifat invarian-waktu jika pergeseran waktu dalam sinyal input menghasilkan pergeseran waktu yang identik dalam sinyal output. Misalkan y(t) adalah output saat x(t) maka sistem bersifat invarian-waktu jika [1]:

   $$\boldsymbol{f\{x(t - t_0)\} = y(t - t_0)}$$

3. Apakah semua sistem yang bersifat linier pasti juga bersifat invarian-waktu?

   Jawab :

   Tidak. Terdapat pula sistem linier yang varian-waktu. Misalnya adalah sebagai berikut

   $y(t) = 3t\, x(t) \rightarrow f\{x(t)\} = 3t\, x(t)$

   $f\{x_1(t)\} = 3t\, x_1(t)$

   $f\{x_2(t)\} = 3t\, x_2(t)$

   $f\{x_1(t) + x_2(t)\} = 3t\, \{x_1(t) + x_2(t)\}$

   $= 3t\, x_1(t) + 3t\, x_2(t)$

   $\boldsymbol{f\{x_1(t) + x_2(t)\} = f\{(x_1(t)\} + f\{x_2(t)\} \rightarrow sistem\ aditif}$

$$f\{ax(t)\} = 3t\,\{ax(t)\}$$
$$= a\,3t\,x(t)$$
$$\boldsymbol{f\{ax(t)\} = af\{x(t)\} \rightarrow sistem\ homogen}$$
$$\boldsymbol{sistem\ aditif\ dan\ homogen \rightarrow sistem\ linier}$$

$$y(t) = 3t\,x(t)$$
$$y(t - t_0) = 3(t - t_0)\,x(t - t_0)$$
$$f\{x(t - t_0)\} = 3t\,x(t - t_0)$$
$$\boldsymbol{f\{x(t - t_0)\} \neq y(t - t_0) \rightarrow sistem\ varian - waktu}$$
$$\boldsymbol{sistem\ tersebut\ linier\ varian - waktu}$$

4. Apakah semua sistem yang bersifat invarian-waktu pasti juga bersifat linier?

   Jawab :

   Tidak. Terdapat pula sistem non-linier yang invarian-waktu misalnya sebagai berikut :

   $y(t) = 3x(t) + 2 \rightarrow f\{x(t)\} = 3x(t) + 2$

$$f\{x_1(t)\} = 3x_1(t) + 2$$
$$f\{x_2(t)\} = 3x_2(t) + 2$$
$$f\{x_1(t) + x_2(t)\} = 3\{x_1(t) + x_2(t)\} + 2$$
$$= 3x_1(t) + 3x_1(t) + 2$$
$$\boldsymbol{f\{x_1(t) + x_2(t)\} \neq f\{(x_1(t)\} + f\{x_2(t)\} \rightarrow sistem\ non - aditif}$$

$$f\{ax(t)\} = 3\{ax(t)\} + 2$$
$$= a3x(t) + 2$$
$$\boldsymbol{f\{ax(t)\} \neq af\{x(t)\} \rightarrow sistem\ non - homogen}$$
$$\boldsymbol{sistem\ non - aditif\ dan\ non - homogen \rightarrow sistem\ non - linier}$$

$$y(t) = 3x(t) + 2$$
$$y(t - t_0) = 3x(t - t_0) + 2$$
$$f\{x(t - t_0)\} = 3x(t - t_0) + 2$$
$$\boldsymbol{f\{x(t - t_0)\} = y(t - t_0) \rightarrow sistem\ invarian - waktu}$$
$$\boldsymbol{sistem\ tersebut\ non - linier\ \ invarian - waktu}$$

5. Apakah sistem instrumentasi yang anda rancang saat UTS adalah sebuah sistem linier?

Jawab :

Ya. Pada sistem kontrol yang saya rancang, output akan mempunyai nilai yang proporsional dengan input bergantung pada karakteristik sistem tersebut. Terdapat dua jenis sistem kontrol yaitu *feedback control system* dan *cascade control system*. Dalam perancangan *feedback control system* ataupun *cascade control system*, akan didapatkan fungsi transfer yang merupakan factor pengali dari input pada domain frekuensi. Selain itu, pada sistem instrumentasi yang saya rancang, komponen-komponen penyusun sistem kontrol instrumentasi seperti *transmitter, control valve*, dan *current to pressure converter* haruslah bersifat linier invarian-waktu agar sistem kontrol dapat stabil dan dapat mengendalikan proses dengan baik. Dikarenakan komponen penyusun sistem kontrol linier invarian-waktu, maka sistem kontrol (*feedback/cascade control system*) juga bersifat linier invarian-waktu.

a. Misalkan *control valve, transmitter,* dan proses diasumsikan sebagai sistem orde satu:

$$\frac{Y(s)}{Y_d(s)} = \frac{K}{\tau s + 1}$$

Jika dilakukan inverse Laplace :

$$\frac{y(t)}{y_d(t)} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}$$

$$y(t) = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_d(t) \rightarrow f\{\, y_d(t)\} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_d(t)$$

$$f\{\, y_{d1}(t)\} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_{d1}(t)$$

$$f\{\, y_{d2}(t)\} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_{d2}(t)$$

$$f\{\, y_{d1}(t) +\ y_{d2}(t)\} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\{\, y_{d1}(t) +\ y_{d2}(t)\}$$

$$= \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_{d1}(t) + \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_{d2}(t)$$

$$\boldsymbol{f\{\, y_{d1}(t) +\ y_{d2}(t)\} = f\{\, y_{d1}(t)\} + f\{\, y_{d2}(t)\} \rightarrow sistem\ aditif}$$

$$f\{a\, y_d(t)\} = \frac{K}{\tau}\ e^{-\frac{t}{\tau}}\,\{a y_d(t)\}$$

$$= a\frac{K}{\tau}\ e^{-\frac{t}{\tau}}\, y_d(t)$$

$$\boldsymbol{f\{a\, y_d(t)\} = af\{\, y_d(t)\} \rightarrow sistem\ homogen}$$

***control valve* dan *transmitter* merupakan sistem linier**

b. Misalkan diasumsikan *controller* yang digunakan adalah *controller* PI didapatkan :

$$\boldsymbol{y(t) = Kp\, e(t) + Ki \int_0^t e(t)\, dt \rightarrow f\{e(t)\} = Kp\, e(t) + Ki \int_0^t e(t)\, dt}$$

$$f\{e_1(t)\} = Kp\, e_1(t) + Ki \int_0^t e_1(t)\, dt$$

$$f\{e_2(t)\} = Kp\, e_2(t) + Ki \int_0^t e_2(t)\, dt$$

$$f\{e_1(t) + e_2(t)\} = Kp\, \{e_1(t) + e_2(t)\} + Ki \int_0^t \{e_1(t) + e_2(t)\} dt$$

$$= \; Kp\, e_1(t) + Ki \int_0^t e_1(t)\, dt + \; Kp\, e_2(t) + Ki \int_0^t e_2(t)\, dt$$

$$\boldsymbol{f\{e_1(t) + e_2(t)\} = \; f\{e_1(t)\} + f\{e_2(t)\} \rightarrow sistem\ aditif}$$

$$f\{ae(t)\} = Kp\, \{ae(t\}) + Ki \int_0^t \{ae(t)\} dt$$

$$= \; aKp\, e(t) + aKi \int_0^t e(t)\, dt$$

$$= a\left(Kp\, e(t) + Ki \int_0^t e(t)\, dt\right)$$

$$\boldsymbol{f\{ae(t)\} = af\{e(t)\} \rightarrow sistem\ homogen}$$

***Controller* PI merupakan sistem linier**

Karena *control valve, transmitter,* proses dan *controller* PI merupakan sistem yang linier, maka keseluruhan *feedback/cascade control system* juga merupakan sistem linier

6. Apakah sistem instrumentasi yang anda rancang saat UTS adalah sebuah sistem invarian-waktu?

Jawab :

Ya. Pada sistem kontrol yang saya rancang, output akan mempunyai nilai yang sama meskipun input mengalami pergeseran waktu (*delay*). Kemudian, sama seperti jawaban no. 5, pada sistem instrumentasi yang telah saya rancang, terdapat dua jenis sistem kontrol yaitu *feedback control system* dan *cascade control system*. Kedua sistem kontrol tersebut merupakan sistem linier invarian-waktu.

Dari jawaban no. 5 didapatkan :

a. *Control valve, transmitter,* proses diasumsikan sebagai sistem orde satu, sehingga :

$$\frac{y(t)}{y_d(t)} = \frac{K}{\tau}\, e^{-\frac{t}{\tau}}$$

$$y(t) = \frac{K}{\tau}\, e^{-\frac{t}{\tau}}\, y_d(t) \rightarrow f\{\, y_d(t)\} = \frac{K}{\tau}\, e^{-\frac{t}{\tau}}\, y_d(t)$$

$$y(t - t_0) = \frac{K}{\tau}\, e^{-\frac{t-t_0}{\tau}}\, y_d(t - t_0)$$

$$f\{\, y_d(t - t_0)\} = \frac{K}{\tau}\, e^{-\frac{t-t_0}{\tau}}\, y_d(t - t_0)$$

$$\boldsymbol{f\{\, y_d(t - t_0)\} = y(t - t_0) \rightarrow sistem\ invarian - waktu}$$

***Control valve, transmitter,* dan proses merupakan sistem yang invariant waktu**

b. Untuk *controller* PI :

$$y(t) = Kp\, e(t) + Ki \int_0^t e(t)\, dt \rightarrow f\{e(t)\} = Kp\, e(t) + Ki \int_0^t e(t)\, dt$$

$$y(t - t_0) = Kp\, e(t - t_0) + Ki \int_0^t e(t - t_0)\, dt$$

$$f\{e(t - t_0)\} = Kp\, e(t - t_0) + Ki \int_0^t e(t - t_0)\, dt$$

$$\boldsymbol{f\{e(t - t_0)\} = y(t - t_0) \rightarrow sistem\ invarian - waktu}$$

***Controller* PI merupakan sistem yang invarian-waktu**

Karena *control valve, transmitter,* proses dan *controller* PI merupakan sistem yang invarian-waktu, maka keseluruhan *feedback/cascade control system* juga merupakan sistem yang invarian-waktu.

Berdasarkan jawaban no 5 dan 6 dapat diketahui bahwa sistem instrumentasi yang didesain merupakan sistem linier invarian-waktu

TUGAS MINGGU KE-7

1. Uraikan dengan jelas dan sistematis bagaimana sebuah sistem dapat dikarakterisasi di domain waktu.

   Jawab :

   Langkah-langkah untuk karakterisasi sistem di domain waktu adalah sebagai berikut [1]:

   a. Mengidentifikasi input dan output dari sistem

   b. Membuat pemodelan matematis dari sistem

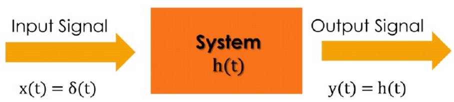
   c. Karakteristik dari sistem pada domain waktu dapat diketahui dengan melihat respons impuls dari sistem tersebut. Teori konvolusi menyatakan bahwa karakteristik dari sistem LTI ditentukan oleh respon impuls dari sistem tersebut sesuai dengan persamaan berikut :

$$y(t) = x(t) * h(t)$$

$$jika\ x(t) = \delta(t), maka:$$

$$y(t) = \delta(t) * h(t)$$

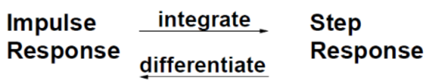
$$y(t) = h(t) \rightarrow h(t)\ merupakan\ karakteristik\ dari\ sistem$$



Gambar 1 karakteristik sistem

   d. Untuk analisis secara praktikal, karakteristik sistem, h(t), dapat diketahui dengan cara memberikan input berupa sinyal impuls pada sistem. Keluaran dari sistem tersebut merupakan karakteristik dari sistem.

   e. Untuk analisis secara matematis, memberikan input berupa sinyal impuls pada domain waktu cukup sulit dilakukan. Oleh karena itu dilakukan analisis dengan input berupa sinyal step. Setelah didapatkan solusi dari analisis respon step maka solusi tersebut kemudian diturunkan untuk mendapatkan karakteristik sistem (respon impuls)



Gambar 2 hubungan respon impuls dan respon step

   f. Pada analisis matematis, ketika menggunakan teori konvolusi terkadang perhitungan cukup kompleks untuk dilakukan. Oleh karena itu dapat dilakukan transformasi

Laplace ataupun Fourier pada persamaan matematis sistem sehingga akan didapatkan fungsi transfer sistem, H(s), yang merupakan hasil pembagian output dengan input pada domain frekuensi. Setelah itu dilakukan operasi inverse Laplace/Fourier pada fungsi transfer untuk mendapatkan fungsi karakteristik sistem pada domain frekuensi, h(t).

g. Setelah didapatkan fungsi karakteristik sistem h(t), fungsi tersebut digambarkan pada suatu grafik h(t) vs t. Dengan grafik tersebut, karakteristik sistem pada domain waktu dapat dinyatakan oleh variabel *time constant* untuk sistem orde satu, dan dapat dinyatakan dalam variabel *settling time, rise time, delay time, overshoot* dan *steady state error* untuk sistem orde dua.

h. Untuk sistem yang terdiri dari beberapa subsistem, hal yang pertama dilakukan adalah mencari fungsi karakteristik, h(t), dari masing-masing subsistem. Setelah itu fungsi h(t) dari masing-masing subsistem digabungkan dengan metode konvolusi. Akan tetapi karena biasanya perhitungan menggunakan metode konvolusi cukup kompleks, dapat dilakukan Langkah seperti poin (f).

i. Untuk sistem pengendali (*control system*), sinyal input biasanya berupa sinyal step. Oleh karena itu untuk mendapatkan karakteristik sistem perlu dilakukan operasi konvolusi antara sinyal input dengan fungsi karakteristik sistem. Setelah itu dapat dicari variabel-variabel yang menyatakan karakteristik sistem pada domain waktu.

2. Lakukan karakterisasi di domain waktu terhadap sistem instrumentasi yang telah anda rancang saat Ujian Tengah Semester. Nyatakan hasil karakterisasi dari sistem instrumentasi anda dalam bentuk persamaan diferensial.

Jawab :

Pada saat Ujian Tengah Semester, saya belum menentukan tujuan perancangan untuk sistem kontrol pada domain waktu. Untuk itu, saya membuatnya pada tugas ini. Berikut ini adalah tujuan perancangan dalam domain waktu :

Tabel 1 Tujuan Perancangan

No.	Variabel	Nilai
1.	*Rise time* (T_r)	$T_r \leq 15\ s$
2.	*Settling time* (T_s)	$T_s \leq 20s$
3.	*Overshoot* ($\%OS$)	$\%OS \leq 20\ \%$
4.	*Steady state error* (e_{ss})	$0 \leq e_{ss} \leq 0{,}05$

a. *Current to pressure converter* dimodelkan hanya sebagai konstanta (*gain*) dalam domain frekuensi

$H_{IP}(s) = K_{IP},$

jika dilakukan inverse transformasi laplace :

$h_{IP}(t) = K_{IP}\delta(t)$

b. Transmitter dan control valve dimodelkan sebagai sistem orde 1

$$\tau\frac{dy(t)}{dt} + y(t) = Kx(t)$$

jika dilakukan transformasi laplace :

$$\tau sY(s) + Y(s) = KX(s)$$

$$(\tau s + 1)Y(s) = KX(s)$$

$$H(s) = \frac{Y(s)}{X(s)} = \frac{K}{\tau s + 1}$$

jika dilakukan inverse transformasi laplace :

$$\boldsymbol{h(t) = \frac{K}{\tau}e^{-\frac{t}{\tau}}}$$

sehingga :

$$transmitter: \boldsymbol{H_m(s) = \frac{K_m}{\tau_m s + 1} \rightarrow h_m(t) = \frac{K_m}{\tau_m}e^{-\frac{t}{\tau_m}}}$$

$$control\ valve: \boldsymbol{H_v(s) = \frac{K_v}{\tau_v s + 1} \rightarrow h_v(t) = \frac{K_v}{\tau_v}e^{-\frac{t}{\tau_v}}}$$

c. *Controller* menggunakan PID *controller*

$$y_c(t) = K_p\, e(t) + K_i\int_0^t e(t)\, dt + K_d\frac{de(t)}{dt}$$

jika dilakukan transformasi laplace :

$$Y_c(s) = K_p\, E(s) + \frac{K_i}{s}\, E(s) + K_d sE(s)$$

$$Y_c(s) = \left(K_p + \frac{K_i}{s} + K_d s\right)E(s)$$

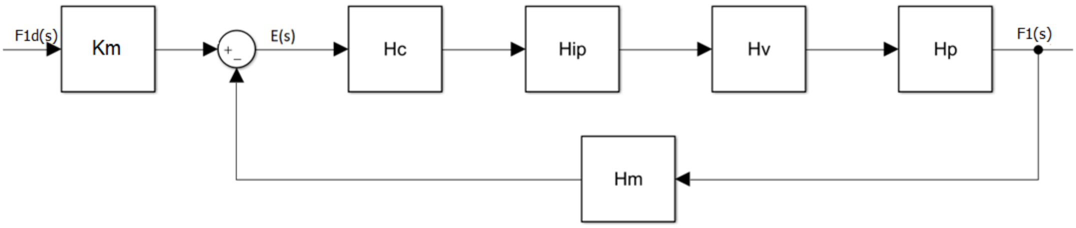
$$\boldsymbol{H_c(s) = \frac{Y_c(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d\, s}$$

jika dilakukan inverse transformasi laplace :

$$h_c(t) = K_p\delta(t) + K_i + K_d\frac{d\delta(t)}{dt}$$

Close loop control system **:**

a. Sistem kontrol w*ash oil flow*



Gambar 3 sistem kontrol wash oil flow

- Proses

  Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses :

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *wash oil flow* (0-17 lb/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(17-0)\ lb/s} = 0{,}94\ {}^{mA}/_{lb/s}$$

$$time\ constant\ (\tau) = 0{,}055\ s$$

  Sehingga :

$$\boldsymbol{H_m(s) = \frac{0,94}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 17,09\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *wash oil flow* (0-17 lb/s)

$$Span\ (K_m) = \frac{(17-0)\ lb/s}{(15-3)psi} = 1{,}45\ {}^{lb/s}/_{psi}$$

  $time\ constant\ (\tau) = 1\ s$

  Sehingga :

$$\boldsymbol{H_v(s) = \frac{1,45}{s+1}}$$

$$\boldsymbol{h_v(t) = 1,45\ e^{-t}}$$

- *Controller*

$$\widetilde{H} = H_{IP}H_vH_PH_m$$

$$= 0{,}75\ x\ \frac{1{,}45}{s+1} x\ 1x\frac{0{,}94}{0{,}055s+1}$$

$$= \frac{1{,}02}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s$$

  $dengan\ metode\ IMC\ Tuning\ relation$ [2]:

$$\tau_c = \frac{\tau_{dom}}{3} = 0{,}333\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{0{,}333\ x\ 1{,}02} = 2{,}944$$

$$\tau_i = \tau_{dom} = 1\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{2{,}944}{1} = 2{,}944$$

$$K_d = 0$$

  $sehingga$:

$$\boldsymbol{H_c(s) = 2,944\ + \frac{2,944}{s}}$$

$$\boldsymbol{h_c(t) = 2,944\delta(t) + 2,944}$$

- Fungsi karakteristik sistem :

$$f_w(t) = e(t) * h_c(t) * h_{IP}(t) * h_v(t) * h_P(t)$$

$$e(t) = f_{wd(t)} - f_w(t) * h_m(t)$$

  persamaan di atas cukup sulit untuk diselesaikan. Oleh karena itu, untuk mencari fungsi karakteristik sistem $h(t)$ terlebih dahulu dicari fungsi transfer sistem pada domain frekuensi, $H(s)$, dan kemudian dilakukan operasi inverse transformasi laplace sehingga didapatkan, $h(t)$.

$$F_1(s) = E(s)H_c(s)H_{IP}(s)H_v(s)H_P(s)$$

$$E(s) = K_m F_{1d}(s) - F_1(s)H_m(s)$$

$$substitusi\ E(s)\ ke\ F_w(s):$$

$$F_1(s) = H_c(s)H_{IP}(s)H_v(s)H_P(s)K_m F_{1d}(s) - H_c(s)H_{IP}(s)H_v(s)H_P(s)F_1(s)H_m(s)$$

$$F_1(s)\left(1 + H_c(s)H_{IP}(s)H_v(s)H_P(s)H_m(s)\right) = H_c(s)H_{IP}(s)H_v(s)H_P(s)K_m F_{1d}(s)$$

$$\frac{F_{1d}(s)}{F_1(s)} = H(s) = \frac{K_m H_c(s)H_{IP}(s)H_v(s)H_P(s)}{1 + H_c(s)H_{IP}(s)H_v(s)H_P(s)H_m(s)}$$

didapatkan :

$$H(s) = \frac{0{,}1655\ s^2 + 3.175\ s + 3{,}01}{0{,}055\ s^3 + 1{,}055\ s^2 + 4{,}01\ s + 3{,}01}$$

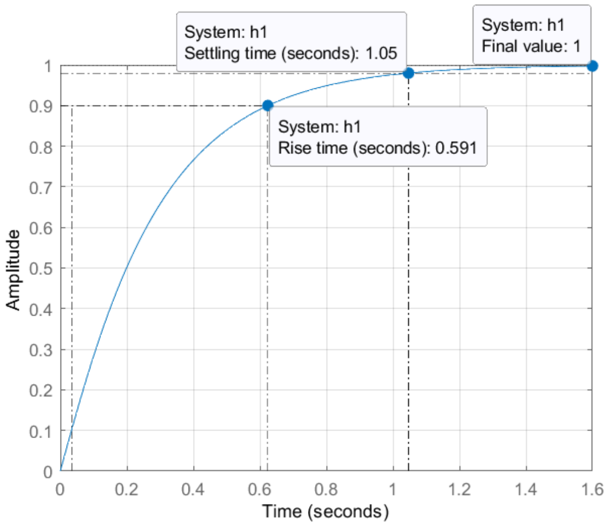
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{1{,}0838}{s + 14{,}3755} + \frac{4{,}0934}{s + 3{,}8064} + \frac{5{,}6788x10^{-16}}{s + 1}$$

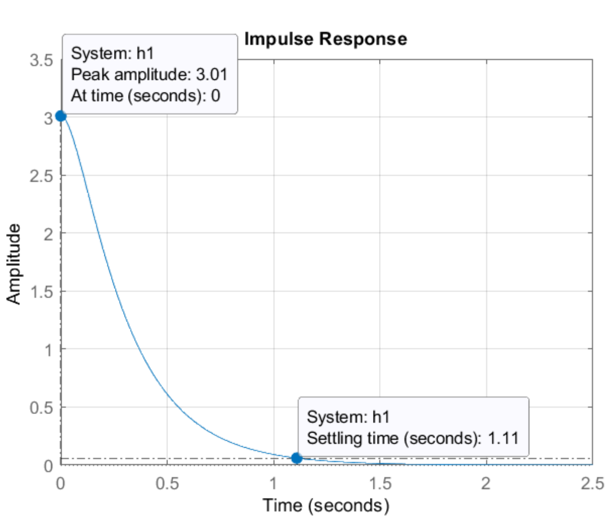
inverse transformasi laplace :

$$\boldsymbol{h(t) = -1{,}0838\ e^{-14{,}3755t} + 4{,}0934\ e^{-3{,}8064t} + 5{,}6788x10^{-16}e^{-t}}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 5 respon step sistem wash oil



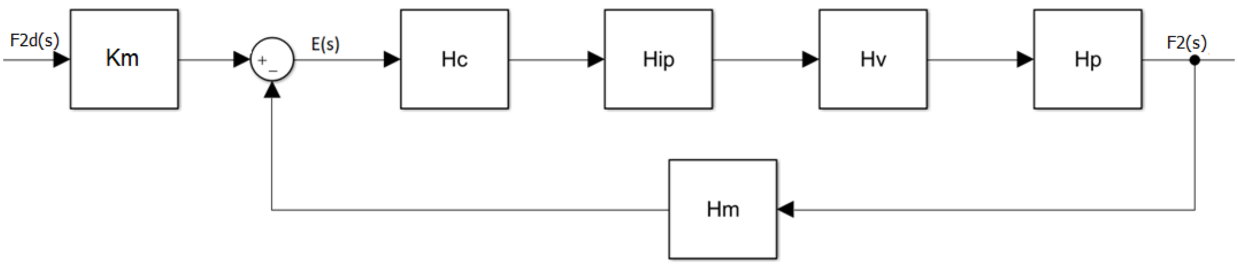
Gambar 4 respon impuls sistem wash oil

Berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,591 s
2.	*Settling time* (T_s)	1,05 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah memenuhi tujuan perancangan.

b. *Sistem kontrol diesel oil flow*



Gambar 6 sistem kontrol diesel oil flow

- Proses

  Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *diesel oil flow* (0-16 lb/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(16-0)\ lb/s} = 1\ {}^{mA}/_{lb/s}$$

$$time\ constant\ (\tau) = 0{,}055\ s$$

Sehingga :

$$\boldsymbol{H_m(s) = \frac{1}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 18,18\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,77}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *diesel oil flow* (0-17 lb/s)

$$Span\ (K_m) = \frac{(16-0)\ lbs}{(15-3)psi} = 1{,}3\ {}^{lb/s}/_{psi}$$

  $time\ constant\ (\tau) = 1\ s$

  Sehingga :

$$\boldsymbol{H_v(s) = \frac{1,3}{s+1}}$$

$$\boldsymbol{h_v(t) = 1,3\ e^{-t}}$$

- *Controller*

$$\widetilde{H} = H_{IP}H_vH_PH_m$$

$$= 0{,}75\ x\ \frac{1{,}3}{s+1}\ x\ 1x\frac{1}{0{,}055s+1}$$

$$= \frac{0{,}975}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s$$

$dengan\ metode\ IMC\ Tuning\ relation$:

$$\tau_c = \frac{\tau_{dom}}{3} = 0{,}333\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{0{,}333\ x\ 0{,}975} = 3{,}08$$

$$\tau_i = \tau_{dom} = 1\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{3{,}08}{1} = 3{,}08$$

$$K_d = 0$$

$sehingga$:

$$\boldsymbol{H_c(s) = 3,08 + \frac{3,08}{s}}$$

$$\boldsymbol{h_c(t) = 3,08\delta(t) + 3,08}$$

- Fungsi transfer sistem :

$$H(s) = \frac{K_m H_c(s) H_{IP}(s) H_v(s) H_P(s)}{1 + H_c(s) H_{IP}(s) H_v(s) H_P(s) H_m(s)}$$

$$\boldsymbol{H(s) = \frac{0,1652\ s^2 + 3,168\ s + 3,003}{0,055\ (s)^3 + 1,055\ (s)^2 + 4,003\ s + 3,003}}$$

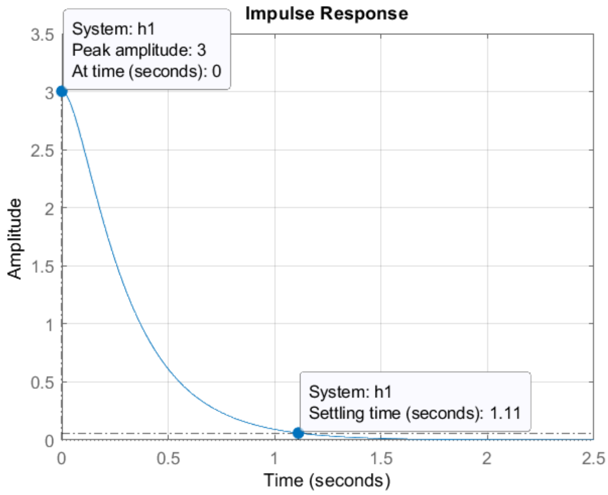
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{1,076}{s + 14,387} + \frac{4,079}{s + 3,7952} - \frac{5,697x10^{-16}}{s + 1}$$

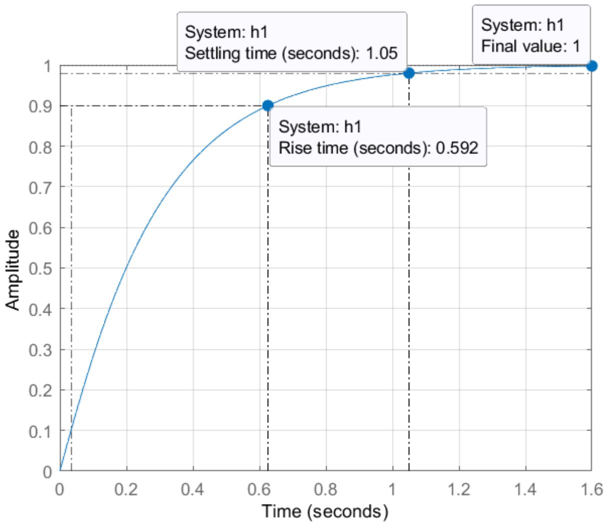
Inverse transformasi laplace :

$$\boldsymbol{h(t) = -1,076\ e^{-14,387t} + 4,079\ e^{-3,7952t} - 5,697x10^{-16} e^{-t}}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 8 respon impulse sistem diesel oil



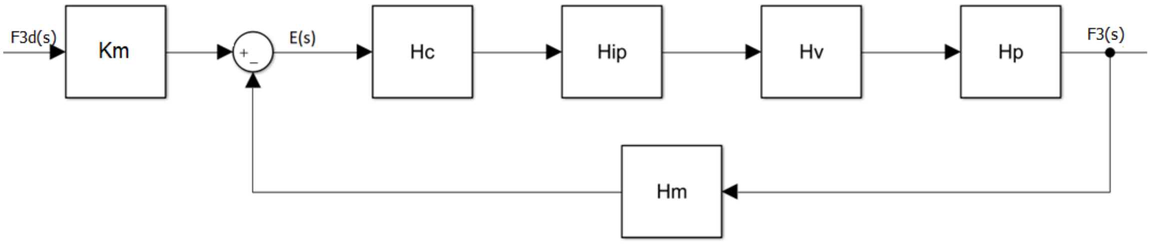
Gambar 7 respon step sistem diesel oil

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,592 s
2.	*Settling time* (T_s)	1,05 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah memenuhi tujuan perancangan

c. Sistem kontrol *total fresh feed flow*



Gambar 9 sistem kontrol total fresh feed flow

- Proses

  Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses :

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *total fresh feed flow* (0-144 lb/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(144-0)\ lbs} = 0{,}1\ {}^{mA}/_{lb/s}$$

  $time\ constant\ (\tau) = 0{,}055\ s$

  Sehingga :

$$\boldsymbol{H_m(s) = \frac{0,1}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 1,818\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *total fresh feed flow* (0-144 lb/s)

$$Span\ (K_m) = \frac{(144-0)\ lb/s}{(15-3)psi} = 12\ {}^{lb/s}/_{psi}$$

$time\ constant\ (\tau) = 1\ s$

Sehingga :

$$\boldsymbol{H_v(s) = \frac{12}{s+1}}$$

$$\boldsymbol{h_v(t) = 12\ e^{-t}}$$

- *Controller*

$$\widetilde{H} = H_{IP}H_vH_PH_m$$

$$= 0{,}75\ x\ \frac{12}{s+1} x\ 1x \frac{0{,}1}{0{,}055s+1}$$

$$= \frac{0{,}9}{(2s+1)(0{,}1s+1)} \rightarrow \tau_{dom} = 1\ s$$

$dengan\ metode\ IMC\ Tuning\ relation:$

$$\tau_c = \frac{\tau_{dom}}{3} = 0{,}333\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{0{,}333\ x\ 0{,}9} = 3{,}337$$

$$\tau_i = \tau_{dom} = 1\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{3{,}337}{1} = 3{,}337$$

$$K_d = 0$$

$sehingga:$

$$\boldsymbol{H_c(s) = 3{,}337 + \frac{3{,}337}{s}}$$

$$\boldsymbol{h_c(t) = 3{,}337\delta(t) + 3{,}337}$$

- Fungsi transfer sistem :

$$\boldsymbol{H(s) = \frac{0{,}1652\ s^2 + 3{,}168\ s + 3{,}003}{0{,}055\ s^3 + 1{,}055\ s^2 + 4{,}003\ s + 3{,}003}}$$

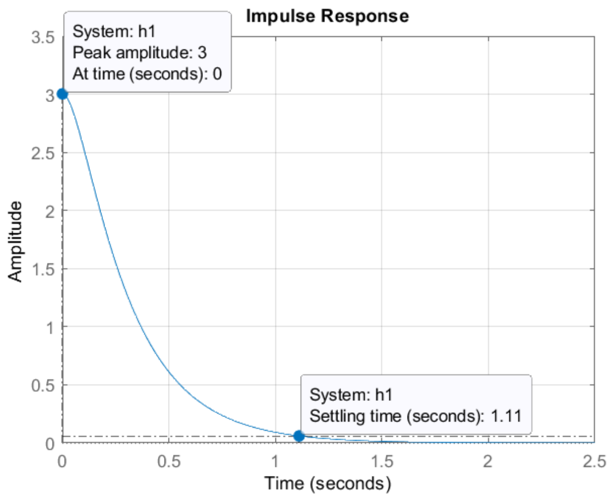
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{1{,}076}{s+14{,}387} + \frac{4{,}079}{s+3{,}7952} - \frac{5{,}697x10^{-16}}{s+1}$$

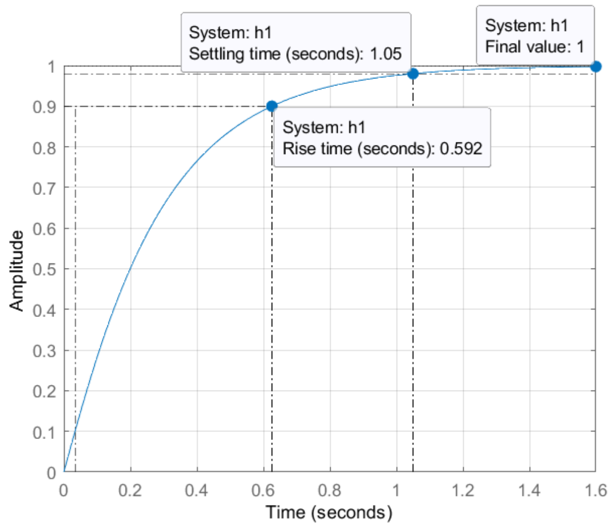
Inverse Transformasi laplace :

$$\boldsymbol{h(t) = -1{,}076\ e^{-14{,}387t} + 4{,}079\ e^{-3{,}7952t} - 5{,}697x10^{-16}e^{-t}}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 10 respon impuls sistem fresh feed



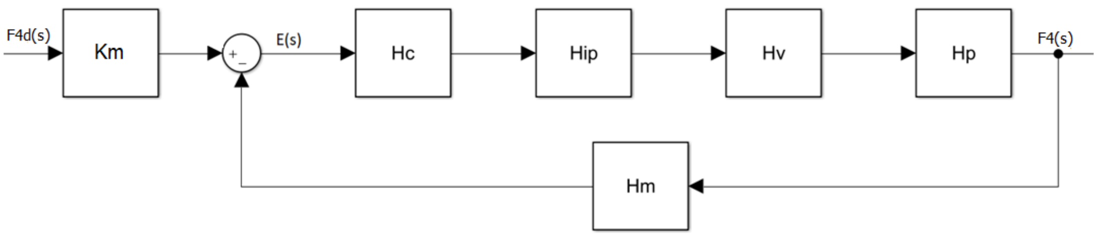
Gambar 11 respon step sistem fresh feed

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,592 s
2.	*Settling time* (T_s)	1,05 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah memenuhi tujuan perancangan

d. Sistem kontrol *slurry oil flow*



Gambar 12 sistem kontrol slurry oil flow

- Proses

Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses :

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *slurry oil flow* (0-10 lb/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(10-0)\ lbs} = 1{,}6\ {}^{mA}/_{lb/s}$$

  $time\ constant\ (\tau) = 0{,}055\ s$

  Sehingga :

$$\boldsymbol{H_m(s) = \frac{1,6}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 29,09\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *slurry oil flow* (0-10 lbs)

$$Span\ (K_m) = \frac{(10-0)\ lb/s}{(15-3)psi} = 0{,}83\ {}^{lb/s}/_{psi}$$

  $time\ constant\ (\tau) = 1\ s$

  Sehingga :

$$\boldsymbol{H_v(s) = \frac{0,83}{s+1}}$$

$$\boldsymbol{h_v(t) = 0,83\ e^{-t}}$$

- *Controller*

$$\begin{aligned}\widetilde{H} &= H_{IP}H_vH_PH_m \\ &= 0{,}75\ x\ \frac{0{,}83}{s+1} x\ 1x \frac{1{,}6}{0{,}055s+1} \\ &= \frac{0{,}996}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s\end{aligned}$$

  $dengan\ metode\ IMC\ Tuning\ relation:$

$$\tau_c = \frac{\tau_{dom}}{3} = 0{,}333\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{0{,}333\ x\ 0{,}996} = 3{,}015$$

$$\tau_i = \tau_{dom} = 2\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{3{,}015}{1} = 3{,}015$$

$$K_d = 0$$

$$sehingga :$$

$$\boldsymbol{H_c(s) = 3,015 + \frac{3,015}{s}}$$

$$\boldsymbol{h_c(t) = 3,015\delta(t) + 3,015}$$

- Fungsi karakteristik sistem :

$$\boldsymbol{H(s) = \frac{0,1652\ s^2 + 3,168\ s + 3,003}{0,055\ s^3 + 1,055\ s^2 + 4,003\ s + 3,003}}$$

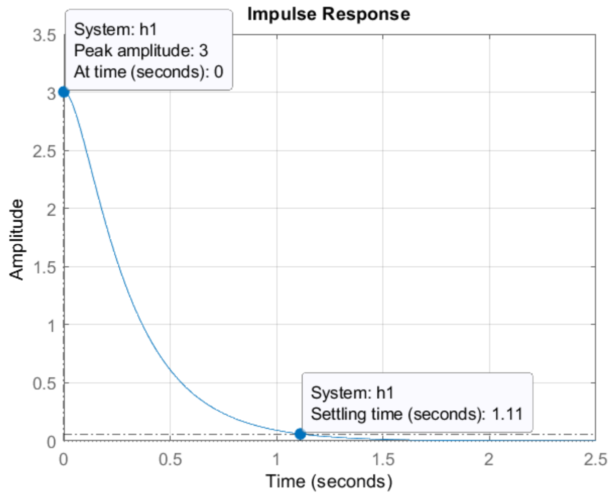
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{1{,}076}{s + 14{,}387} + \frac{4{,}079}{s + 3{,}7952} - \frac{5{,}697x10^{-16}}{s + 1}$$

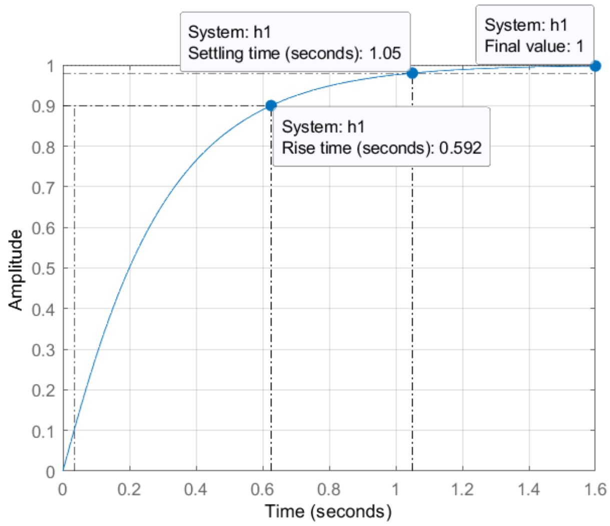
Inverse Transformasi laplace

$$\boldsymbol{h(t) = -1,076\ e^{-14,387t} + 4,079\ e^{-3,7952t} - 5,697x10^{-16}e^{-t}}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 14 respon impuls sistem slurry oil



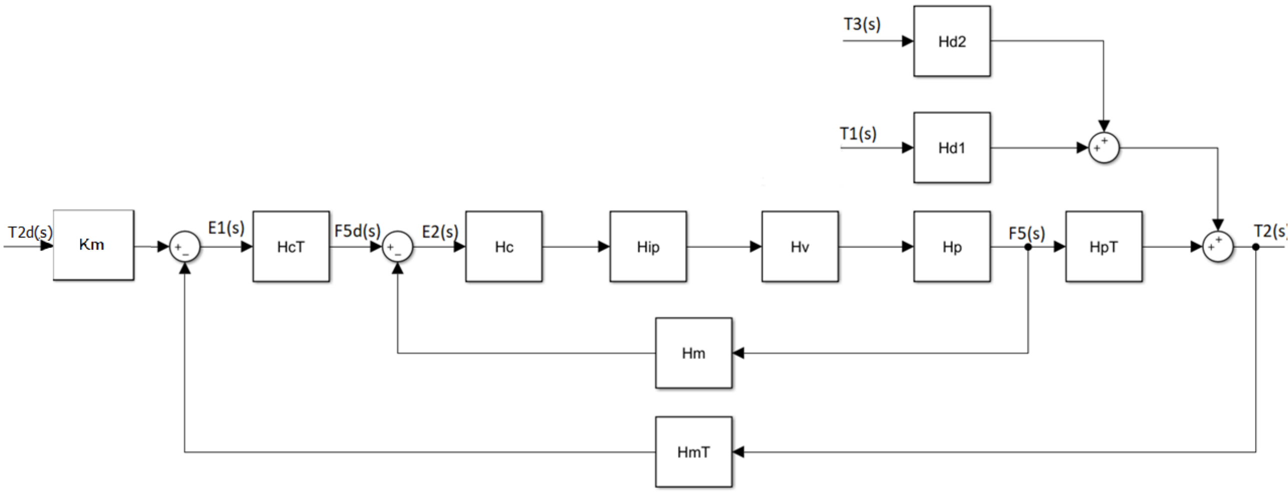
Gambar 13 respon step sistem slurry oil

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,592 s
2.	*Settling time* (T_s)	1,05 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah memenuhi tujuan perancangan

e. Temperatur *feed* keluar *preheater*



Gambar 15 sistem kontrol temperatur feed keluar preheater

a. Slave

- Proses

  Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses :

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *fuel gas flow* (0-40 scf/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(40-0)\ scf/s} = 0{,}4\ {}^{mA}/_{scf/s}$$

  $time\ constant\ (\tau) = 0{,}055\ s$

  Sehingga :

$$\boldsymbol{H_m(s) = \frac{0,4}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 7,27\, e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *fuel gas flow* (0-40 scf/s)

$$Span\ (K_m) = \frac{(40-0)\ scf/s}{(15-3)psi} = 3{,}3\ {}^{scf/s}/_{psi}$$

  $time\ constant\ (\tau) = 1\ s$

  Sehingga :

$$\boldsymbol{H_v(s) = \frac{3,3}{s+1}}$$

$$\boldsymbol{h_v(t) = 3,3\, e^{-t}}$$

- *Controller*

$$\widetilde{H} = H_{IP}H_vH_PH_m$$

$$= 0{,}75\ x\ \frac{3{,}3}{s+1} x\ 1x \frac{0{,}4}{0{,}055s+1}$$

$$= \frac{0{,}99}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s$$

  Pada bagian slave biasanya digunakan kontrol proporsional :

$$K_c = \frac{1}{K} = \frac{1}{0{,}99} = 1{,}01$$

$$\boldsymbol{H_c(s) = 1,01}$$

$$\boldsymbol{h_c(t) = 1,01\delta(t)}$$

- Fungsi transfer sistem :

$$\boldsymbol{H(s)} = \frac{\mathbf{0,1375s + 2,5}}{\mathbf{0,055\, s^2 + 1,055\, s + 2}}$$

$$= \frac{1{,}25(0{,}055s + 1)}{(0{,}469s + 1)(0{,}059s + 1)}$$

jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{7{,}597x10^{-16}}{s + 1} - \frac{2{,}689}{s + 14{,}387} + \frac{10{,}196}{s + 3{,}795}$$

inverse transformasi laplace :

$$\boldsymbol{h(t) = -7,597x10^{-16}e^{-t} - 2,689e^{-14,387t} + 10,196e^{-3,795t}}$$

b. Master

- Proses

$$\tau_{fo}\frac{dT_2(t)}{dt} + T_2(t) = T_1(s) + \frac{\Delta H_{fu}}{F_{3,ss}}F_5(t) - \frac{\tau_{fb}}{F_{3,ss}}\frac{dT_3(t)}{dt} - \frac{a_1 F_{5,ss}}{F_{3,ss}}T_3(t) - \frac{a_2}{F_{3,ss}}$$

$$60\frac{dT_2(t)}{dt} + T_2(t) = T_1(s) + 6{,}94F_5(t) - 1{,}38\,\frac{dT_3(t)}{dt} - 0{,}042T_3(t) - 1{,}429$$

$Transformasi\ Laplace$:

$$60sT_2(s) + T_2(s) = T_1(s) + 6{,}94F_5(s) - 1{,}38s\,T_3(s) - 0{,}042T_3(s)$$

$$(60s + 1)T_2(s) = T_1(s) + 6{,}94F_5(s) - (1{,}38s + 0{,}042)T_3(s)$$

$Disturbance, T_1(s) = T_3(s) = 0$:

$$(60s + 1)T_2(s) = 6{,}94F_5(s)$$

$$\boldsymbol{H_{PT}(s) = \frac{T_2(s)}{F_5(s)} = \frac{6,94}{(60s + 1)}}$$

$$\boldsymbol{h_{PT}(t) = 0,116\, e^{-\frac{t}{60}}}$$

- *Temperature transmitter*

Input : Temperatur *feed* keluar *preheater* ((-58) - 842) °F

Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20 - 4)mA}{(842 - (-58))\ lbs} = 0{,}017\ {}^{mA}/_{°F}$$

$time\ constant\ (\tau) = 13.36\ s$

Sehingga :

$$\boldsymbol{H_{mT}(s) = \frac{0,017}{13.36s + 1}}$$

$$\boldsymbol{h_m(t) = 0,0013\, e^{-0,0749t}}$$

- *Controller*

$$\widetilde{H_T} = HH_{PT}H_{mT}$$

$$= \frac{1{,}25(0{,}055s+1)}{\mathbf{(0,469s+1)(0,059s+1)}} \text{x} \frac{\mathbf{6,94}}{\mathbf{(60s+1)}} \text{x} \frac{\mathbf{0,017}}{\mathbf{13.36s+1}}$$

$$= \frac{0{,}147(0{,}055s+1)}{(0{,}469\boldsymbol{s}+\mathbf{1})(\mathbf{0,059s+1})(\mathbf{60s+1})(\mathbf{13.36s+1})}$$

pendekatan :

$$\frac{0{,}147(\boldsymbol{s}+\mathbf{0,055})}{\mathbf{(13.36s+1)(60s+1)}} \rightarrow \tau_{dom} = 60\ s$$

$dengan\ metode\ IMC\ Tuning\ relation$:

$$\tau_c = \frac{\tau_{dom}}{3} = 20\ s$$

$$K_pK = \frac{\tau_1+\tau_2-\tau_3}{\tau_c} \rightarrow K_p = \frac{60+13{,}36-0{,}055}{20\ x\ 0{,}147} = 24{,}934$$

$$\tau_i = \tau_1+\tau_2-\tau_3 = 73{,}305\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{24{,}934}{73{,}305} = 0{,}34$$

$$\tau_d = \frac{\tau_1\tau_2-(\tau_1+\tau_2-\tau_3)\tau_3}{\tau_1+\tau_2-\tau_3} = 10{,}879 \rightarrow K_d = K_p\tau_d = 271{,}256$$

$sehingga$:

$$\boldsymbol{H_{cT}(s)} = \mathbf{24,934} + \frac{\mathbf{0,34}}{\boldsymbol{s}} + \mathbf{271,256}\ \boldsymbol{s}$$

$$\boldsymbol{h_c(t)} = \mathbf{24,934}\boldsymbol{\delta(t)} + \mathbf{0,34} + \mathbf{271,256}\ \boldsymbol{\delta'(t)}$$

- Fungsi transfer sistem :

$$\boldsymbol{H_T(s)} = \frac{58{,}78s^4+1079s^3+178{,}8s^2+8{,}699s+0{,}1}{44{,}09\ s^{\mathbf{6}} + 893{,}8\ s^5 + 2530s^4+1833s^3+230{,}2s^2+9{,}359s+0{,}1}$$

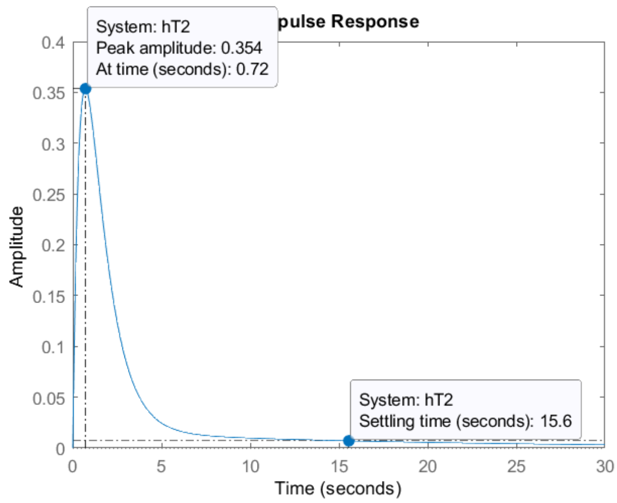
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = \frac{0{,}006}{s+17{,}049} - \frac{1{,}118}{s+2{,}175} + \frac{1{,}095}{s+0{,}904} - \frac{0{,}0001}{s+0{,}074} + \frac{0{,}017}{s+0{,}055} + \frac{7{,}51x10^{-6}}{s+0{,}017}$$

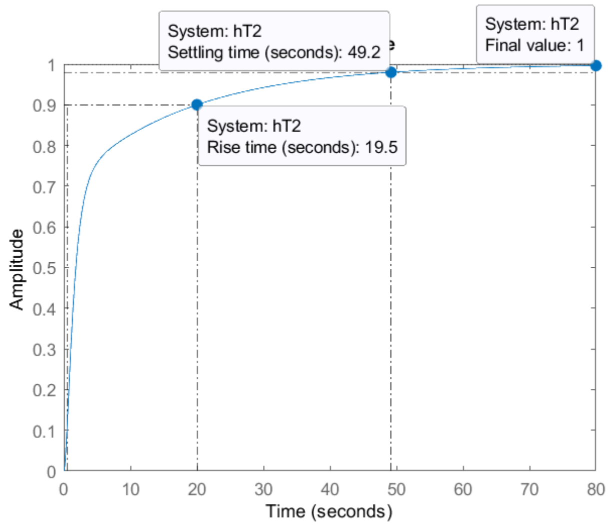
inverse transformasi laplace :

$$h(t) = 0{,}006e^{-17{,}049t} - 1{,}118e^{-2{,}175t} + 1{,}095e^{-0{,}904t} - 0{,}0001e^{0{,}074t} + 0{,}017e^{-0{,}055t} + 7{,}5x10^{-6}e^{-0{,}017t}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 17 Respon impuls sistem kontrol suhu keluar preheater



Gambar 16 Respon step sistem kontrol suhu keluar preheater

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	19,5 s
2.	*Settling time* (T_s)	49,2 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Terlihat bahwa *rise time* dan *settling time* tidak memenuhi tuntutan perancangan, sehingga dilakukan penyesuaian pada PID pada master. Dengan trial & error didapatkan Kp = 30,934, Ki = 0,34 dan Kd = 271,256. Sehingga fungsi transfer menjadi :

$$\boldsymbol{H_T(s)} = \frac{58{,}78s^4 + 1080s^3 + 202{,}5s^2 + 10{,}47s + 0{,}1}{44{,}09\, s^6 + 893{,}8\, s^5 + 2530s^4 + 1833s^3 + 230{,}2s^2 + 11{,}3s + 0{,}1}$$

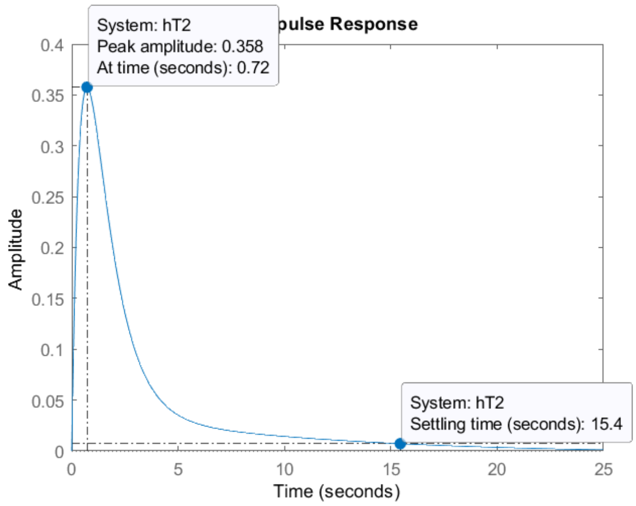
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = \frac{0{,}006}{s + 17{,}049} - \frac{1{,}107}{s + 2{,}174} + \frac{1{,}063}{s + 0{,}907} + \frac{0{,}019 + 0{,}011i}{s + 0{,}066 - 0{,}039i} + \frac{0{,}019 - 0{,}011i}{s + 0{,}066 + 0{,}039i} + \frac{0{,}0008}{s + 0{,}012}$$

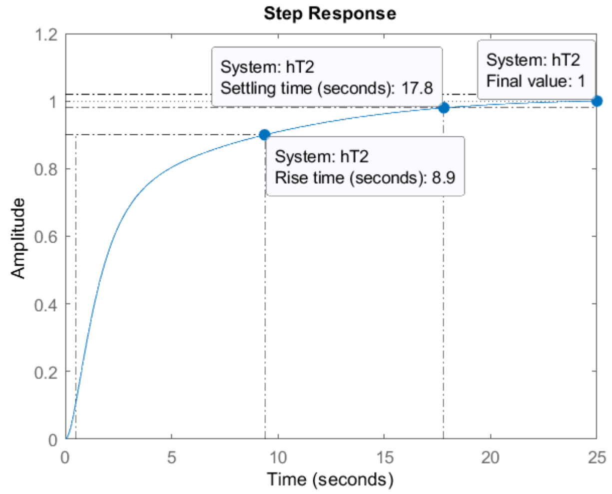
inverse transformasi laplace :

$$h(t) = 0{,}006e^{-17{,}049t} - 1{,}107e^{-2{,}174t} + 1{,}063e^{-0{,}907t} - (0{,}019 + 0{,}011i)e^{-(0{,}066-0{,}039i)t} + (0{,}019 - 0{,}011i)e^{-(0{,}066+0{,}039i)t} + 0{,}0008e^{-0{,}012t}$$

respon untuk masukan berupa impuls dan step 1 satuan



Gambar 19 Respon impuls sistem kontrol suhu keluar preheater



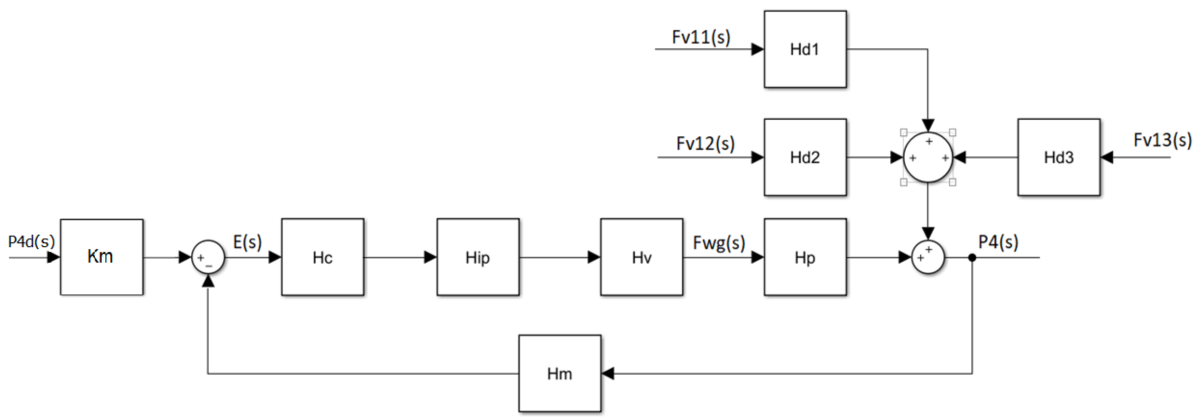
Gambar 18 Respon step sistem kontrol suhu keluar preheater

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan

No.	Parameter	Nilai
1.	*Rise time* (T_r)	8,9 s
2.	*Settling time* (T_s)	17,8 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah sesuai dengan tujuan perancangan

f. Sistem kontrol Tekanan reaktor



Gambar 20 Sistem kontrol tekanan reaktor

- Proses

  Output : Tekanan reaktor

  Input : *wet gas flow*

  Persamaan tekanan reaktor :

  $P_4(t) = P_5(t) + \Delta P_{frac}$

  $\frac{dP_4(t)}{dt} = \frac{dP_5(t)}{dt}$

  $\frac{dP_4(t)}{dt} = 0.833\left(F_{wg}(t) - F_{v11}(t) - F_{v12}(t) + F_{v13}(t)\right)$

  Transformasi laplace :

  $$sP_4(s) = 0.833\left(F_{wg}(s) - F_{v11}(s) - F_{v12}(s) + F_{v13}(s)\right)$$

  diasumsikan kompresor jauh dari titik surge dan *control system* berfungsi sempurna→tidak aliran ada flare sehingga :

  $$F_{v11}(s) = F_{v12}(s) = F_{v13}(s) = 0$$

  $$s\,P_4(s) = 0.833F_{wg}(s)$$

  $$\boldsymbol{H_P(s) = \frac{P_4(s)}{F_{wg}(s)} = \frac{0,833}{s}}$$

- *Pressure transmitter*

  Input : tekanan reaktor {0-150 psia}

  Output : *current* (4-20 mA)

  $$Span\ (K_m) = \frac{(20-4)mA}{(150-0))\ psia} = 0{,}106\ {}^{mA}/_{psia}$$

  $time\ constant\ (\tau) = 0{,}085\ s$

  Sehingga :

  $$\boldsymbol{H_m(s) = \frac{0,106}{0,085s+1}}$$

  $$\boldsymbol{h_m(t) = 1,247\ e^{-11,765t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

  $$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

  $$\boldsymbol{H_{IP}(s) = 0,75}$$

  $$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *wet gas flow* (0-1,375 mol/s)

$$Span\ (K_m) = \frac{(1{,}375 - 0)\ mol/s}{(15-3)psia} = 0{,}115\ {}^{mol/s}/_{psia}$$

  $time\ constant\ (\tau) = 1\ s$

  Sehingga :

$$\boldsymbol{H_v(s) = \frac{0,115}{s+1}}$$

$$\boldsymbol{h_v(t) = 0,115\ e^{-t}}$$

- *Controller*

  Pada fungsi transfer proses terdapat pole pada titik (0,0) sehingga saya menduga sistem ini hanya membutuhkan *controller* proporsional.

$$\widetilde{H} = H_{IP}H_vH_PH_m$$

$$= 0{,}75x\,\boldsymbol{\frac{0,115}{s+1}\,x\,\frac{0,833}{s}\,x\,\frac{0,106}{0,085s+1}}$$

$$= \frac{0{,}008}{s(s+1)(0{,}085s+1)}$$

  dengan

$$K_p = \frac{1}{K} = \frac{1}{0{,}008} = 125$$

$$\boldsymbol{H_c(s) = 125 \rightarrow h_c(t) = 125\delta(t)}$$

- Fungsi transfer sistem :

$$\boldsymbol{H(s) = \frac{0,081s + 0,952}{0,085s^3 + 1,085s^2 + s + 0,952}}$$

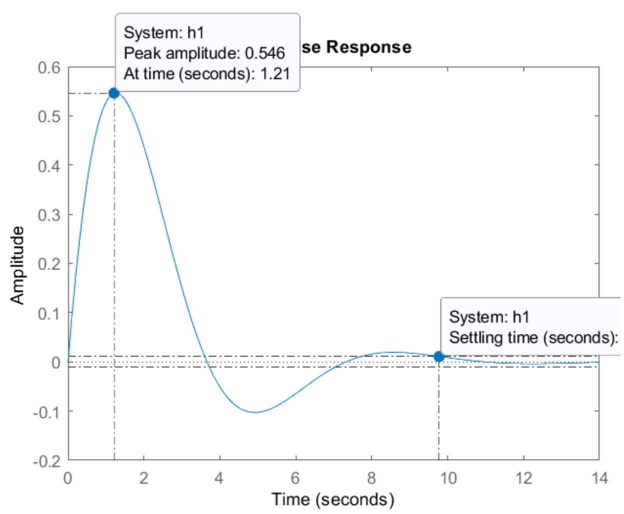
  jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{-0{,}0006}{s+11{,}852} + \frac{0{,}0003 - 0{,}55i}{s + 0{,}457 - 0{,}858i} + \frac{0{,}0003 + 0{,}55i}{s + 0{,}457 + 0{,}858i}$$

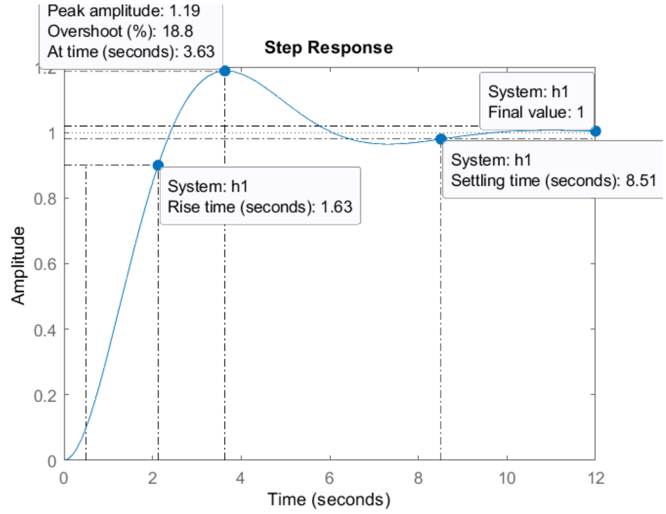
  inverse transformasi laplace :

$$\boldsymbol{h(t) = -0,0006\ e^{-11,852t} + (0,0003 - 0,55i)e^{-(0,457-0,858i)t} + (0,0003 + 0,55i)\ e^{-(0,457+0,858i)t}}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 22 respon impuls sistek kontrol tekana reaktor



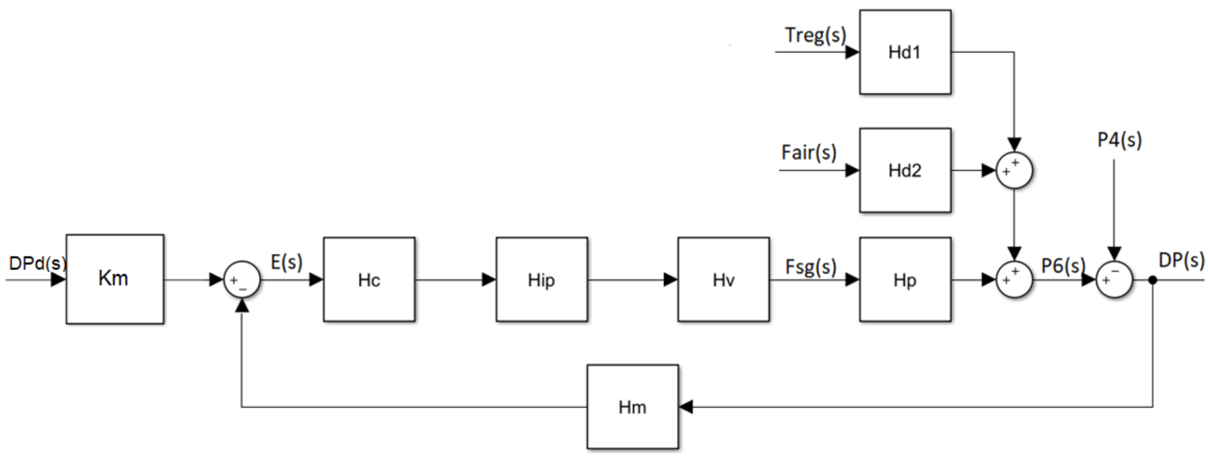
Gambar 21 respon step sistem kontrol tekanan reaktor

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	1,63 s
2.	*Settling time* (T_s)	8,51 s
3.	*Overshoot* ($\%OS$)	18,8 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah sesuai tujuan perancangan

g. Beda tekanan reaktor-regenerator



Gambar 23 Sistem kontrol beda tekanan reaktor-regenerator

- Proses

  Output : beda tekanan reaktor-regenerator

  Input : *flue gas flow*

  Persamaan beda tekanan reaktor-regenerator :

  $\Delta P_{RR}(t) = P_6(t) - P_4(t)$

  $$\frac{dP_6(t)}{dt} = \frac{1}{V_{reg,g}}\left\{R\left(n(t)\frac{dT_{reg}(t)}{dt} + \left(T_{reg} + 459.6\right)\right)\frac{dn(t)}{dt}\right\}$$

  $$sP_6(s) = \frac{1}{V_{reg,g}}\left\{R\left(N(s)\, s\, T_{reg}(s) + \left(T_{reg}(s) + 459.6\right)\right) s\, N(s)\right\}$$

  $untuk\ disturbance, T_{reg}(s) = 0, maka:$

  $$sP_6(s) = \frac{459.6\ R\ s}{V_{reg,g}} N(s)$$

  $$= \frac{459.6\ R\ s}{V_{reg,g}}\left(F_{air}(s) - F_{sg}(s)\right)$$

  $untuk\ disturbance, F_{air}(s) = 0, maka:$

  $$s\ P_6(s) = -\frac{459.6\ R\ s}{V_{reg,g}} F_{sg}(s)$$

  $$\boldsymbol{H_{P_6}(s) = \frac{P_6(s)}{F_{sg}(s)} = -\frac{459.6\ R}{V_{reg,g}} = -0,214}$$

  $sehingga:$

  $$\boldsymbol{\Delta P_{RR}(s) = P_6(s) - P_4(s)}$$

  $$\boldsymbol{= -0,214\ F_{sg}(s) - \frac{0,833}{s} F_{wg}(s)}$$

  $P_4 : disturbance, karena\ MV\ (flue\ gas) berhubungan\ langsung\ dengan\ P_6$

  $maka:$

  $$\boldsymbol{H_{\Delta P_{RR6}}(s) = \frac{\Delta P_{RR}(s)}{F_{sg}(s)} = -0,214 \rightarrow h_{\Delta P_{RR6}}(t) = -0,214\delta(t)}$$

  $$\boldsymbol{H_{\Delta P_{RR4}}(s) = \frac{\Delta P_{RR}(s)}{F_{wg}(s)} = -\frac{0,833}{s} \rightarrow h_{\Delta P_{RR4}}(t) = -0,833}$$

- *Differential pressure transmitter*

  Input : tekanan reaktor {0-30 psia}

  Output : *current* (4-20 mA)

  $$Span\ (K_m) = \frac{(20-4)mA}{(30-0)\ psia} = 0,53\ {}^{mA}/_{lbs}$$

  $time\ constant\ (\tau) = 0,055\ s$

Sehingga :

$$\boldsymbol{H_m(s) = \frac{0,53}{0,055s + 1}}$$

$$\boldsymbol{h_m(t) = 9,63\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75}$$

$$\boldsymbol{h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *flue gas flow* (0-4,262 mol/s)

$$Span\ (K_m) = \frac{(4{,}262-0)\ mol/s}{(15-3)psi} = 0{,}355\ {}^{mol/s}/_{psi}$$

$time\ constant\ (\tau) = 1\ s$

Sehingga :

$$\boldsymbol{H_v(s) = \frac{0,355}{s+1}}$$

$$\boldsymbol{h_v(t) = 0,355e^{-t}}$$

- *Controller*

$$H = H_{IP}H_vH_{\Delta P_{RR6}}H_m$$

$$= 0{,}75\ x\ \frac{0{,}355}{s+1} x\ (-0{,}214) x \frac{0{,}53}{0{,}055s+1}$$

$$= \frac{-0{,}03}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s$$

$dengan\ metode\ IMC\ Tuning\ relation:$

$$\tau_c = \frac{\tau_{dom}}{3} = 0{,}333\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{0{,}333\ x\ (-0{,}03)} = -100{,}1$$

$$\tau_i = \tau_{dom} = 1\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{-100{,}1}{1} = -100{,}1$$

$K_d = 0, \qquad sehingga:$

$$H_c(s) = -100,1 - \frac{100,1}{s}$$

$$h_c(t) = -100,1\delta(t) - 100,1$$

- Fungsi transfer sistem :

$$H(s) = \frac{0,1663\ s^2 + 3,189\ s + 3,023}{0,055\ s^3 + 1,055\ s^2 + 4,023\ s + 3,023}$$

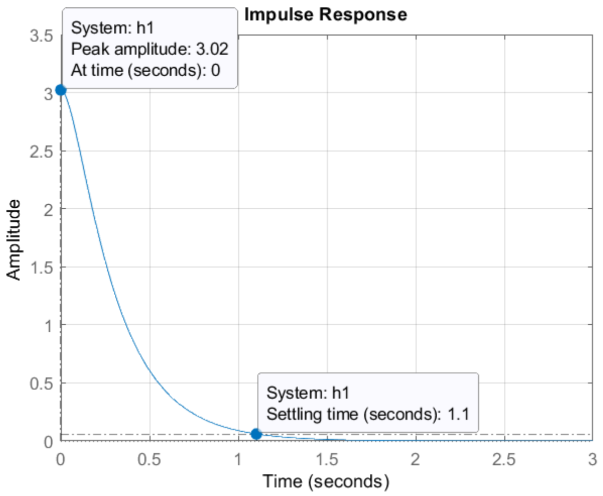
jika H(s) diuraikan dengan *partial fraction* :

$$H(s) = -\frac{1,1}{s + 14,353} + \frac{4,123}{s + 3,829} + \frac{1,88x10^{-16}}{s + 1}$$

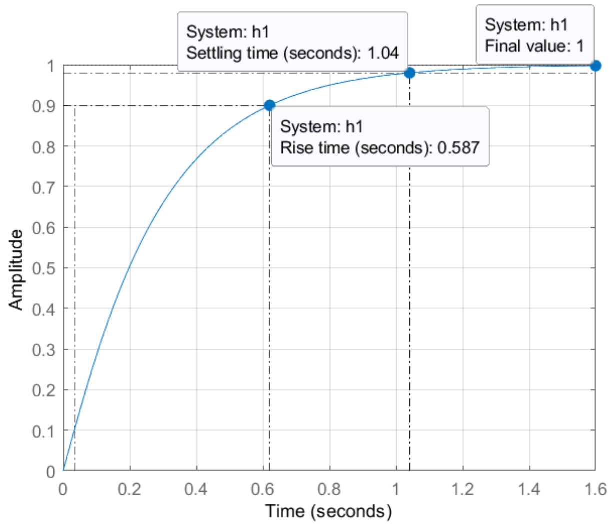
inverse transformasi laplace :

$$h(t) = -1,1\ e^{-14,353t} + 4,123\ e^{-3,829t} + 1,88x10^{-16}e^{-t}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 25 respon impuls



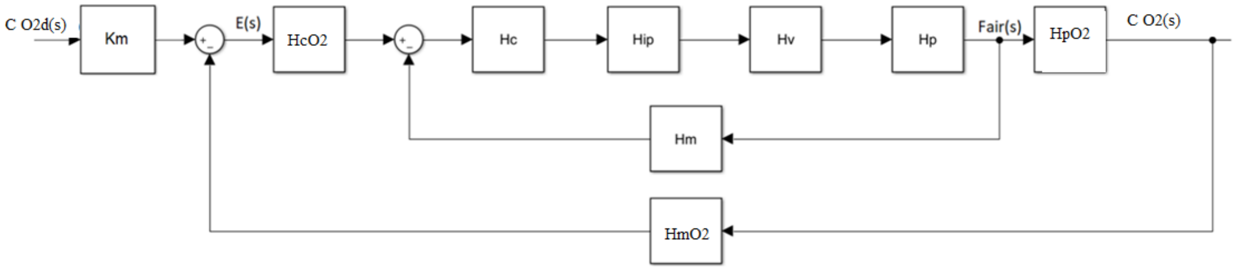
Gambar 24 respon step

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,587 s
2.	*Settling time* (T_s)	1,04 s
3.	*Overshoot* $(\%OS)$	0 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah sesuai tujuan perancangan

h. Sistem kontrol konsentrasi oksigen pada *flue gas*



Gambar 26 diagram blok sistem

Pada perancangan, saya menggunakan sensor suhu untuk mengukur konsentrasi oksigen pada *flue gas*. Akan tetapi, saya kesulitan untuk mendapatkan antara suhu *flue gas* dengan *lift air.* Oleh karena itu saya menggantinya dengan *oxygen analyzer*.

a. *Slave*

- Proses

  Karena transmitter langsung mengukur *flow* pada pipa, maka fungsi transfer dari proses :

$$\boldsymbol{H_P(s) = 1}$$

$$\boldsymbol{h_P(t) = \delta(t)}$$

- *Flow transmitter*

  Input : *lift air flow* (0-83,164 lb/s)

  Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(83{,}164-0)\ lbs} = 0{,}192\ {}^{mA}/_{lb/s}$$

  $time\ constant\ (\tau) = 0{,}055\ s$

  Sehingga :

$$\boldsymbol{H_m(s) = \frac{0,192}{0,055s+1}}$$

$$\boldsymbol{h_m(t) = 3,49\ e^{-18,18t}}$$

- I/P *converter*

  Input : *current* (4-20 mA)

  Output : *pressure* (3-15 psi)

$$K_{IP} = \frac{(15-3)psi}{(20-4)mA} = 0{,}75\ {}^{psi}/_{mA}$$

$$\boldsymbol{H_{IP}(s) = 0,75 \rightarrow h_{IP}(t) = 0,75\delta(t)}$$

- *Control valve*

  Input : *pressure* (3-15 psi)

  Output : *lift air flow* (0-83,164 lb/s)

$$Span\ (K_m) = \frac{(83{,}164 - 0)\ lb/s}{(15-3)psi} = 6{,}93\ {}^{lb/s}/_{psi}$$

$time\ constant\ (\tau) = 1\ s$

Sehingga :

$$\boldsymbol{H_v(s) = \frac{6,93}{s+1}}$$

$$\boldsymbol{h_v(t) = 6,93\ e^{-t}}$$

- *Controller*

$$\begin{aligned} H &= H_{IP}H_vH_PH_m \\ &= 0{,}75\ x\ \frac{6{,}93}{s+1} x\ 1x \frac{0{,}192}{0{,}055s+1} \\ &= \frac{0{,}998}{(s+1)(0{,}055s+1)} \rightarrow \tau_{dom} = 1\ s \end{aligned}$$

Pada bagian slave biasanya digunakan kontrol proporsional :

$$\boldsymbol{K_c = \frac{1}{K} = \frac{1}{0,998} = 1,002}$$

- Fungsi transfer sistem :

$$\begin{aligned} \boldsymbol{H(s)} &= \boldsymbol{\frac{0,286s + 5,208}{0,055s^2 + 1,055\ s + 2}} \\ &= \frac{5{,}2(s+18{,}21)}{(s+2{,}133)(s+17{,}049)} \\ &= \frac{2{,}754(0{,}055s+1)}{(s+0{,}467)(s+0{,}059)} \end{aligned}$$

b. Master

- Proses

  Konsentrasi oksigen pada *flue gas :*

$$C_{O_2,sg}(t) = \frac{100F_{air}(t)\ \chi_{O_2}}{F_{sg,ss}} \rightarrow asumsi\ \chi_{O_2} = 21\%\ dan\ F_{sg,ss} = 2{,}6 \frac{mol}{s}$$

$$C_{O_2,sg}(t) = 8{,}077\ F_{air}(t)$$

Transformasi laplace :

$$C_{O_2,sg}(s) = 8{,}077\ F_{air}(s)$$

$$\boldsymbol{H_{PO_2}(s) = 8,077}$$

$$\boldsymbol{h_{PO_2}(t) = 8,077\delta(t)}$$

- *Transmitter*

Input : *oxygen concentration* (0-100 %)

Output : *current* (4-20 mA)

$$Span\ (K_m) = \frac{(20-4)mA}{(100-0)\ \%} = 0,16\ {}^{mA}/_{\%}$$

$time\ constant\ (\tau) = 3\ s$

Sehingga :

$$\boldsymbol{H_{mO_2}(s) = \frac{0,16}{3s+1}}$$

$$\boldsymbol{h_{mO_2}(t) = 0,053\ e^{-3t}}$$

- *Controller*

$$\widetilde{H_T} = HH_{PO_2}H_{mO_2}$$

$$= \frac{2,754(0,055s+1)}{(0,467s+1)(0,059s+1)} \text{x} 8,077x\frac{0,16}{3s+1}$$

$$= \frac{3,559(0,055s+1)}{(0,467s+1)(0,059s+1)(3s+1)}$$

pendekatan :

$$\frac{3,559}{(3s+1)} \rightarrow\ \tau_{dom} = 3\ s$$

$$dengan\ metode\ IMC\ Tuning\ relation:$$

$$\tau_c = \frac{\tau_{dom}}{3} = 1\ s$$

$$K_pK = \frac{\tau_{dom}}{\tau_c} \rightarrow K_p = \frac{1}{1\ x\ 3,559} = 0,281$$

$$\tau_i = \tau_{dom} = 2\ s \rightarrow K_i = \frac{K_p}{\tau_i} = \frac{0,281}{3} = 0,094$$

$$K_d = 0$$

$$sehingga:$$

$$\boldsymbol{H_{cO_2}(s) = 0,281\ + \frac{0,094}{s}}$$

$$\boldsymbol{h_c(t) = 0,281\delta(t) + 0,094}$$

Fungsi transfer sistem :

$$\boldsymbol{H_{O_2}(s) = \frac{0,312s^3 + 5,882s^2 + 3,824s + 0,634}{0,165s^4 + 3,22s^3 + 7,159s^2 + 3,926s + 0,633}}$$

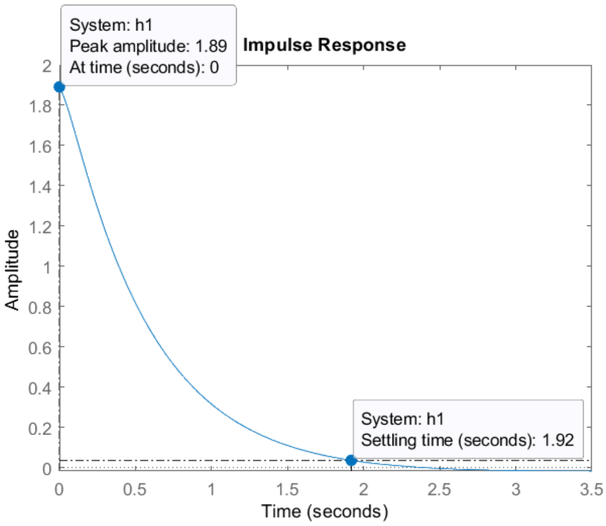
jika H(s) diuraikan dengan *partial fraction* :

$$H_{O_2}(s) = -\frac{0,14}{s + 17,052} + \frac{2,105}{s + 1,745} - \frac{0,091}{s + 0,337} + \frac{0,017}{s + 0,349}$$

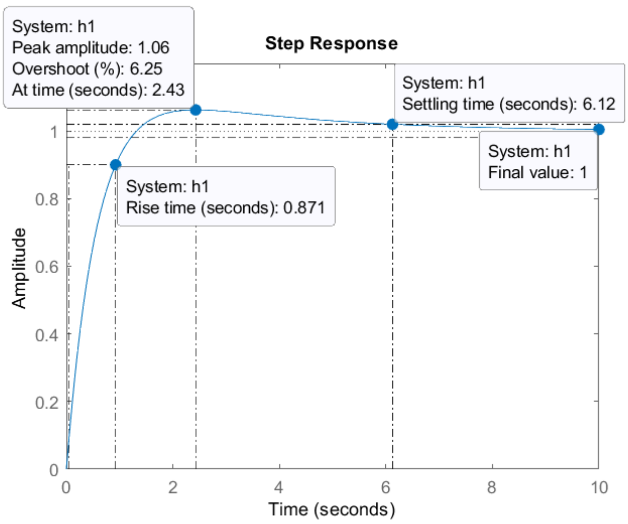
inverse transformasi laplace :

$$h(t) = -0,14e^{-17,052t} + 2,105e^{-1,745t} - 0,091e^{-0,337} + 0,017e^{-0,349}$$

respon untuk masukan berupa impuls dan step 1 satuan :



Gambar 28 respon impuls



Gambar 27 respon step

berdasarkan hasil dari respon sistem dengan input step 1 satuan didapatkan :

No.	Parameter	Nilai
1.	*Rise time* (T_r)	0,871 s
2.	*Settling time* (T_s)	6,12 s
3.	*Overshoot* $(\%OS)$	6,25 %
4.	*Steady state error* (e_{ss})	0

Parameter respon waktu sudah sesuai tujuan perancangan

TUGAS MINGGU KE-8

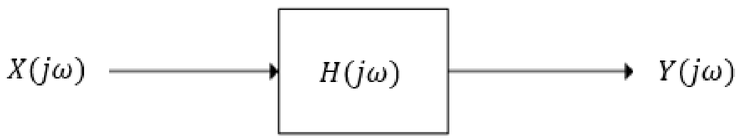
1. Uraikan dengan jelas dan sistematis bagaimana sebuah sistem dapat dikarakterisasi di domain waktu.

   Jawab :

   Langkah-langkah untuk karakterisasi sistem di domain waktu adalah sebagai berikut [1],[2]:

   a. Mengidentifikasi input dan output dari sistem
   b. Membuat pemodelan matematis dari sistem
   c. Karakteristik dari sistem pada domain frekuensi dapat diketahui dengan melihat fungsi transfer dari sistem tersebut. Fungsi transfer merupakan pembagian antara output dengan input pada domain frekuensi.

$$Y(j\omega) = H(j\omega)X(j\omega)$$
$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$



   d. Untuk analisis secara praktikal, fungsi transfer, $H(j\omega)$, didapatkan dengan cara memberikan input berupa sinyal *white noise* pada sistem. Keluaran dari sistem tersebut merupakan karakteristik dari sistem.
   e. Untuk analisis secara matematis, karakteristik sistem dalam domain frekuensi dapat diketahui dengan cara mencari persamaan fungsi transfer *open loop*, $H_{OL}(j\omega)$, kemudian melakukan *plotting gain* dan fase fungsi transfer terhadap frekuensi atau yang biasa disebut diagram bode. Sistem *close loop* akan stabil jika mempunyai nilai *gain open loop* kurang dari 0 dB dan fase *open loop* lebih dari -180^o. Nilai tersebut biasanya dinyatakan dalam *gain margin* (GM) dan *phase margin* (PM), dimana :

$$GM = 0 - G \rightarrow GM > 0 : stabil$$
$$PM = \phi - (-180°) \rightarrow PM > 0 : stabil$$

   semakin besar nilai *gain margin* dan *phase margin,* maka sistem semakin stabil.

2. Lakukan karakterisasi di domain waktu terhadap sistem instrumentasi yang telah anda rancang saat Ujian Tengah Semester. Nyatakan hasil karakterisasi dari sistem instrumentasi anda dalam bentuk persamaan diferensial.

   Jawab :

   Pada saat Ujian Tengah Semester, saya belum menentukan tujuan perancangan untuk sistem kontrol pada domain frekuensi. Untuk itu, saya membuatnya pada tugas ini. Berikut ini adalah tujuan perancangan dalam domain frekuensi

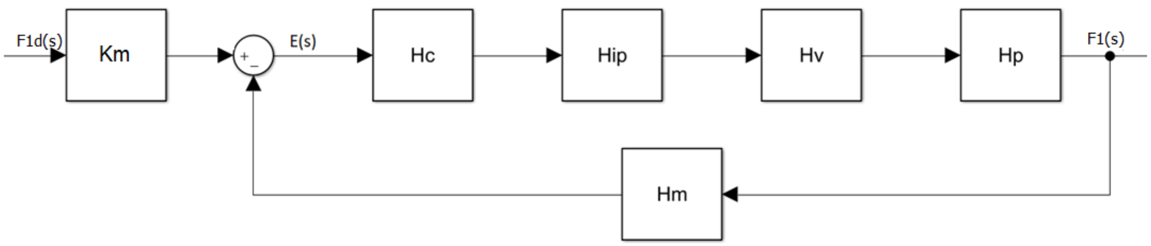
Tabel 2 Tujuan Perancangan

No.	Variabel	Nilai
1.	*Gain margin* (GM)	$G_M > 20\ dB$
2.	*Phase margin* (PM)	$, P_M > 40°$

Fungsi transfer untuk seluruh sistem telah didapatkan di tugas 7. Untuk mendapatkan karakteristik respon frekuensi maka domain s (Laplace) pada fungsi transfer pada tugas 7 disubstitusi menjadi domain $j\omega$ (Fourier).

Fungsi transfer untuk seluruh sistem telah didapatkan di tugas 7. Untuk mendapatkan karakteristik respon frekuensi maka domain s (Laplace) pada fungsi transfer pada tugas 7 disubstitusi menjadi domain $j\omega$ (Fourier).

a. *Wash oil flow control*



Gambar 29 diagram blok

$$H_P(j\omega) = 1$$

$$H_m(j\omega) = \frac{0{,}94}{0{,}055j\omega + 1}$$

$$H_{IP}(j\omega) = 0{,}75$$

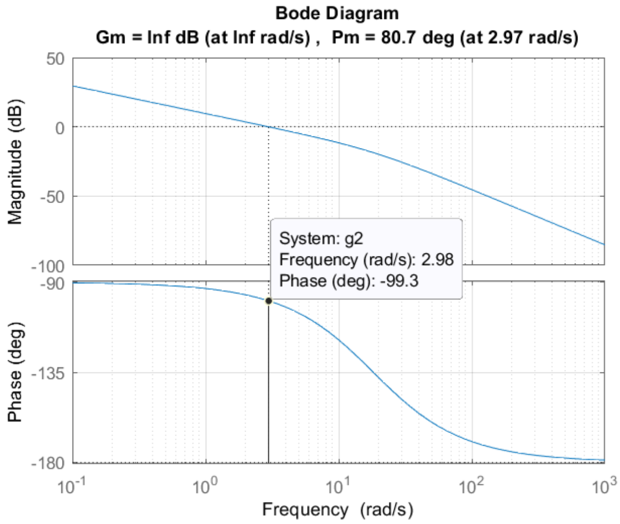
$$H_v(j\omega) = \frac{1{,}45}{j\omega + 1}$$

$$H_c(j\omega) = 2{,}944 + \frac{2{,}944}{j\omega}$$

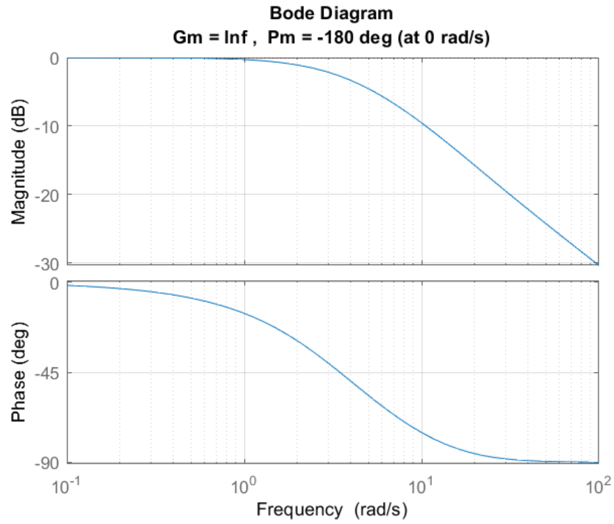
$$H_{OL}(j\omega) = H_c(j\omega)H_{IP}(j\omega)H_v(j\omega)H_P(j\omega)H_m(j\omega)$$

$$= \frac{3{,}01j\omega + 3{,}01}{0{,}055(j\omega)^3 + 1{,}055(j\omega)^2 + j\omega}$$

$$\boldsymbol{H(j\omega) = \frac{0,1655\,(j\omega)^2 + 3.175\,j\omega + 3,01}{0,055\,(j\omega)^3 + 1,055\,(j\omega)^2 + 4,01\,j\omega + 3,01}}$$



Gambar 31 bode plot open loop



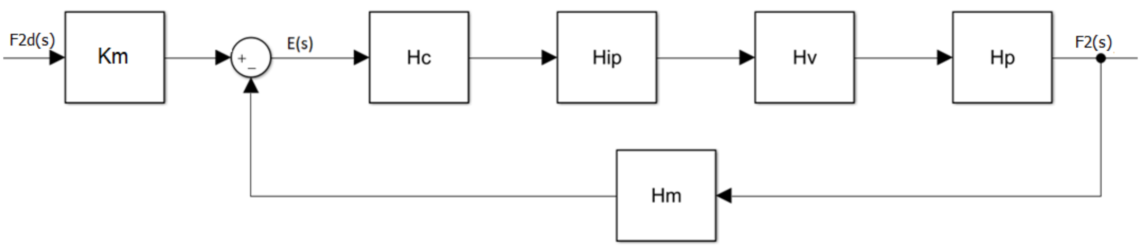
Gambar 30 bode plot close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	80,7º

Sistem kontrol *wash oil flow* sudah stabil.

b. ***Diesel oil flow control***



Gambar 32 diagram blok

$$H_P(j\omega) = 1$$

$$H_m(j\omega) = \frac{1}{0{,}055j\omega + 1}$$

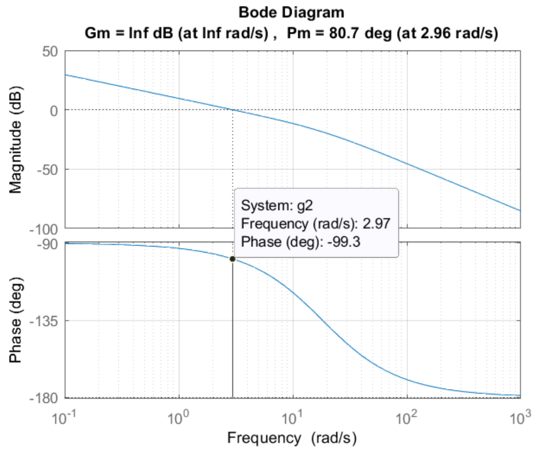
$$H_{IP}(j\omega) = 0{,}75$$

$$H_v(j\omega) = \frac{1{,}3}{j\omega + 1}$$

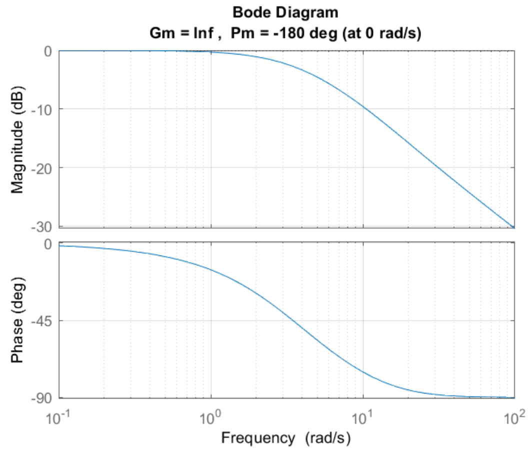
$$H_c(j\omega) = 3{,}08 + \frac{3{,}08}{j\omega}$$

$$H_{OL}(j\omega) = \frac{3{,}003j\omega + 3{,}003}{0{,}055(j\omega)^3 + 1{,}055(j\omega)^2 + j\omega}$$

$$\boldsymbol{H(j\omega) = \frac{0,1652\ (j\omega)^2 + 3,168\ j\omega + 3,003}{0,055\ (j\omega)^3 + 1,055\ (j\omega)^2 + 4,003\ j\omega + 3,003}}$$



Gambar 34 bode plot open loop



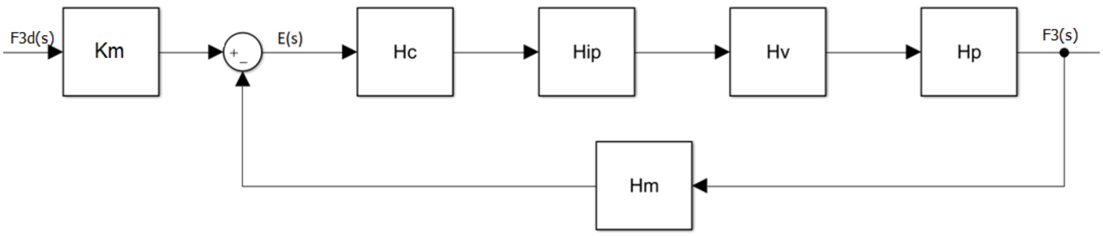
Gambar 33 bode plot close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	80,7º

Sistem kontrol *diesel oil flow* sudah stabil.

c. *Total fresh feed flow control*



Gambar 35 diagram blok

$$H_P(j\omega) = 1$$

$$H_m(j\omega) = \frac{0{,}1}{0{,}055j\omega + 1}$$

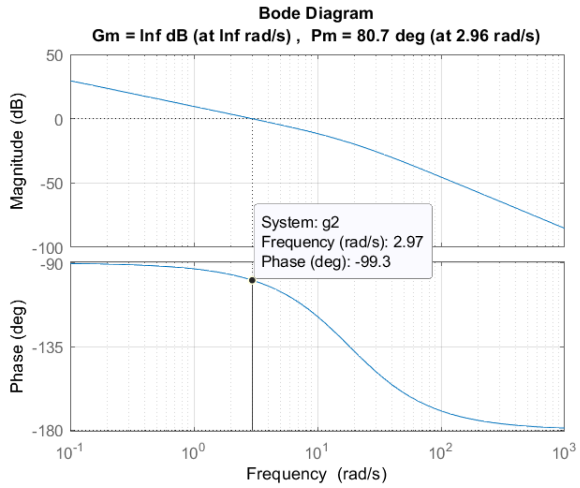
$$H_{IP}(j\omega) = 0{,}75$$

$$H_v(j\omega) = \frac{12}{j\omega + 1}$$

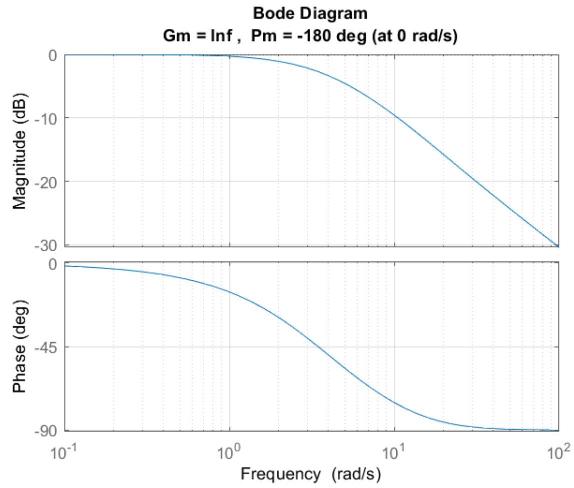
$$H_c(j\omega) = 3{,}337 + \frac{3{,}337}{j\omega}$$

$$H_{OL}(j\omega) = \frac{3,003j\omega + 3,003}{0,055(j\omega)^3 + 1,055(j\omega)^2 + j\omega}$$

$$H(j\omega) = \frac{0,1652\,(j\omega)^2 + 3,168\,j\omega + 3,003}{0,055\,(j\omega)^3 + 1,055\,(j\omega)^2 + 4,003\,j\omega + 3,003}$$



Gambar 37 bode plot open loop



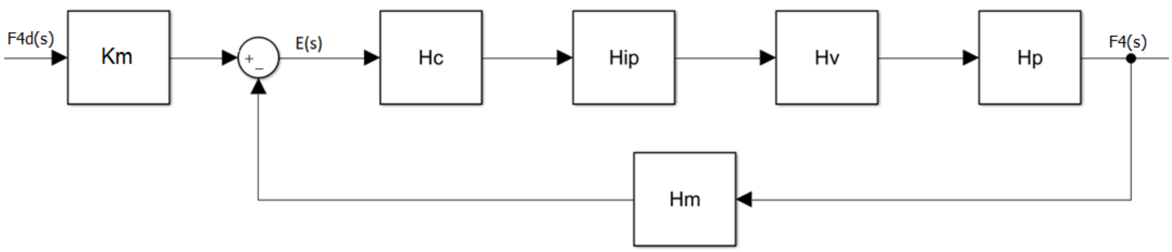
Gambar 36 bode plot close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	80,7°

Sistem kontrol *total fresh feed flow* sudah stabil.

d. *Slurry oil control*



$$H_P(j\omega) = 1$$

$$H_v(j\omega) = \frac{0,83}{j\omega + 1}$$

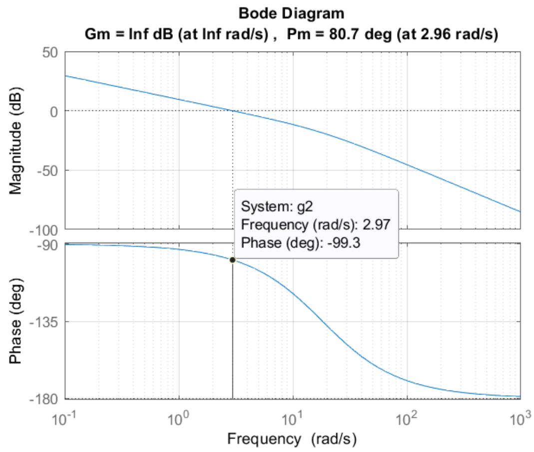
$$H_m(j\omega) = \frac{1,6}{0,055j\omega + 1}$$

$$H_c(j\omega) = 3,015 + \frac{3,015}{j\omega}$$

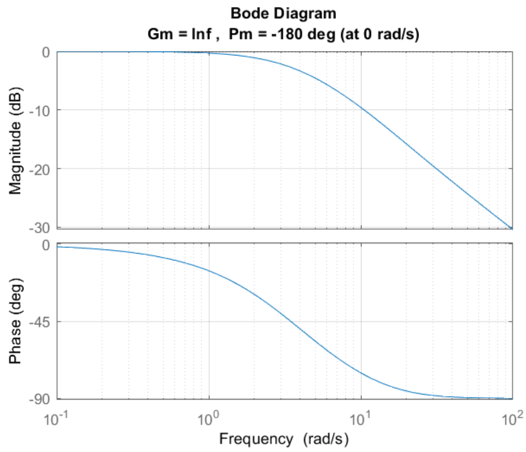
$$H_{IP}(j\omega) = 0,75$$

$$H_{OL}(j\omega) = \frac{3,003j\omega + 3,003}{0,055(j\omega)^3 + 1,055(j\omega)^2 + j\omega}$$

$$H(j\omega) = \frac{0,1652\,(j\omega)^2 + 3,168\,j\omega + 3,003}{0,055\,(j\omega)^3 + 1,055\,(j\omega)^2 + 4,003\,j\omega + 3,003}$$



Gambar 39 bode plot open loop



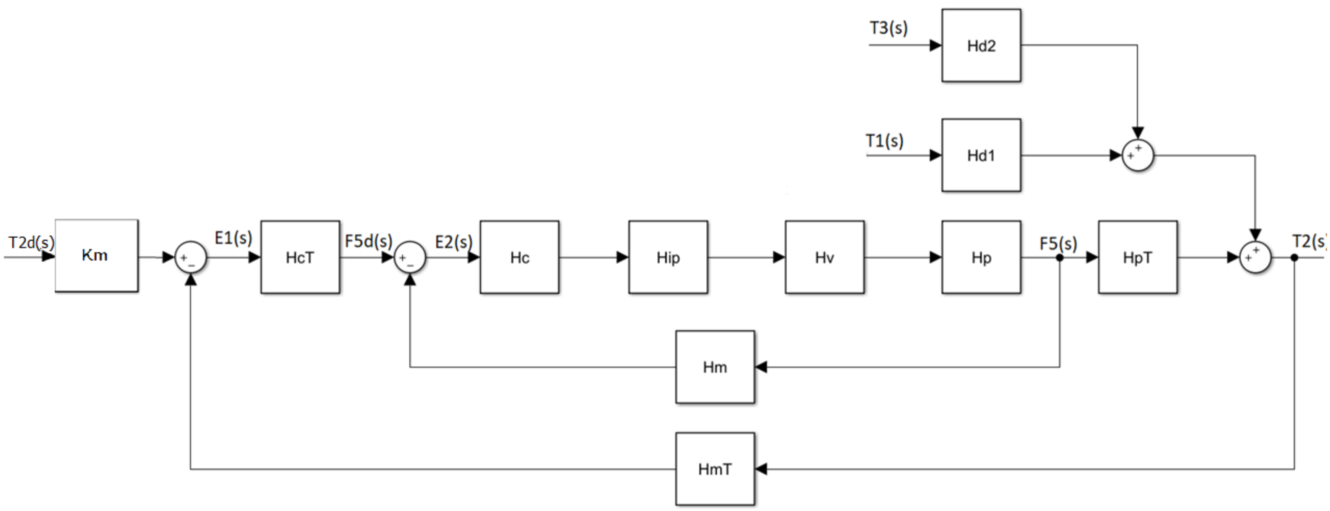
Gambar 38 bode plot close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	80,7º

Sistem kontrol *slurry oil flow* sudah stabil.

e. Temperatur *feed* keluar *preheater control*



Gambar 40 diagram blok

c. Slave

$$H_P(j\omega) = 1$$

$$H_m(j\omega) = \frac{0{,}4}{0{,}055j\omega + 1}$$

$$H_{IP}(j\omega) = 0{,}75$$

$$H_v(j\omega) = \frac{3{,}3}{j\omega + 1}$$

$$H_c(j\omega) = 1{,}01$$

$$\boldsymbol{H(j\omega)} = \frac{\mathbf{0,1375}\boldsymbol{j\omega} + \mathbf{2,5}}{\mathbf{0,055}\ (\boldsymbol{j\omega})^2 + \mathbf{1,055}\ \boldsymbol{j\omega} + \mathbf{2}}$$

$$= \frac{1{,}25(0{,}055s + 1)}{(0{,}469s + 1)(0{,}059s + 1)}$$

d. Master

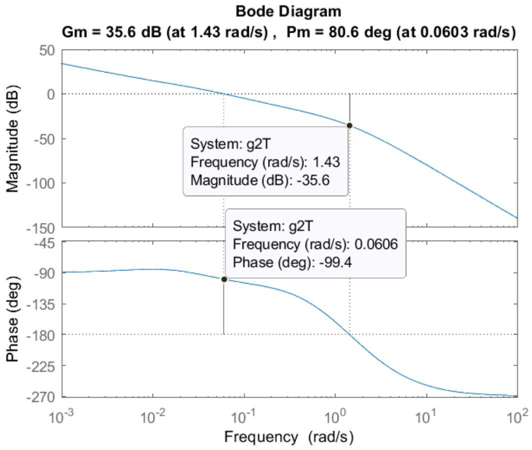
$$H_{PT}(j\omega) = \frac{6{,}94}{(60j\omega + 1)}$$

$$H_{mT}(j\omega) = \frac{0{,}017}{13.36j\omega + 1}$$

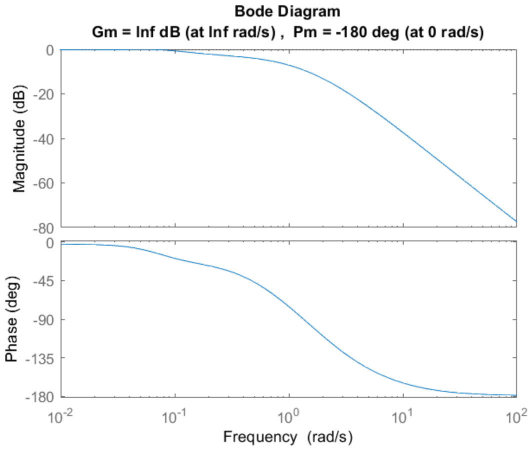
$$H_{cT}(j\omega) = 24{,}934 + \frac{0{,}34}{j\omega} + 271{,}256\, j\omega$$

$$\boldsymbol{H_{OLT}(j\omega)} = \frac{\mathbf{4,4}\ (\boldsymbol{j\omega})^3 + \mathbf{80,5}\ (\boldsymbol{j\omega})^2 + \mathbf{9,129}\ \boldsymbol{j\omega} + \mathbf{0.1}}{\mathbf{44,09}\ (\boldsymbol{j\omega})^6 + \mathbf{893,8}\ (\boldsymbol{j\omega})^5 + \mathbf{2530}\ (\boldsymbol{j\omega})^4 + \mathbf{1828}\ (\boldsymbol{j\omega})^3 + \mathbf{149,8}\ (\boldsymbol{j\omega})^2 + \mathbf{2}\boldsymbol{j\omega}}$$

$$\boldsymbol{H_T(j\omega)} = \frac{\mathbf{58,78}(\boldsymbol{j\omega})^4 + \mathbf{1079}(\boldsymbol{j\omega})^3 + \mathbf{178,8}(\boldsymbol{j\omega})^2 + \mathbf{8,699}\boldsymbol{j\omega} + \mathbf{0,1}}{\mathbf{44,09}\ (\boldsymbol{j\omega})^6 + \mathbf{893,8}\ (\boldsymbol{j\omega})^5 + \mathbf{2530}(\boldsymbol{j\omega})^4 + \mathbf{1833}(\boldsymbol{j\omega})^3 + \mathbf{230,2}(\boldsymbol{j\omega})^2 + \mathbf{9,359}\boldsymbol{j\omega} + \mathbf{0,1}}$$



Gambar 42 bode plot open loop



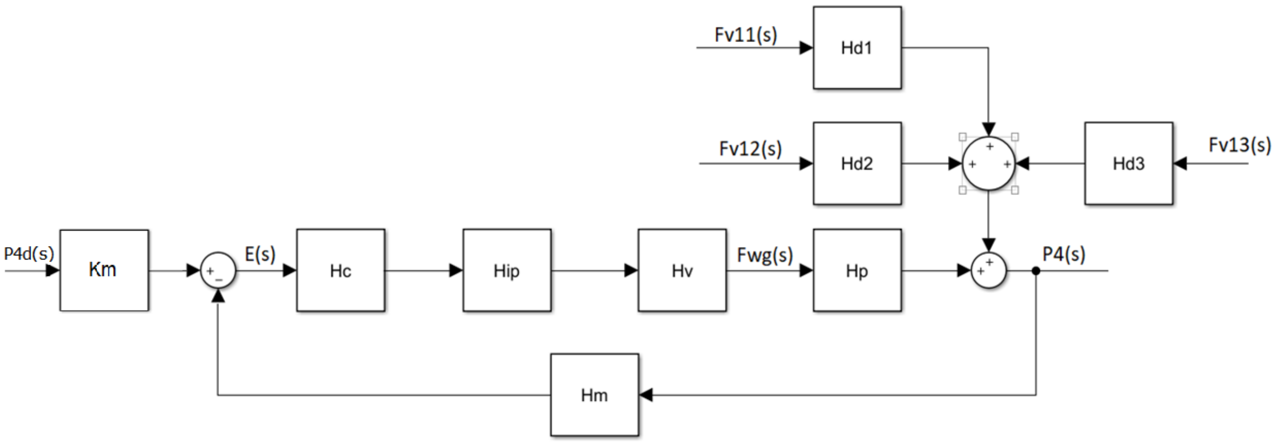
Gambar 41 bode close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	35,6 dB
2.	*Phase Margin* (PM)	80,6º

Sistem kontrol *slurry oil flow* sudah stabil.

f. Tekanan reaktor



Gambar 45 diagram blok

$$H_P(j\omega) = \frac{0,833}{j\omega}$$

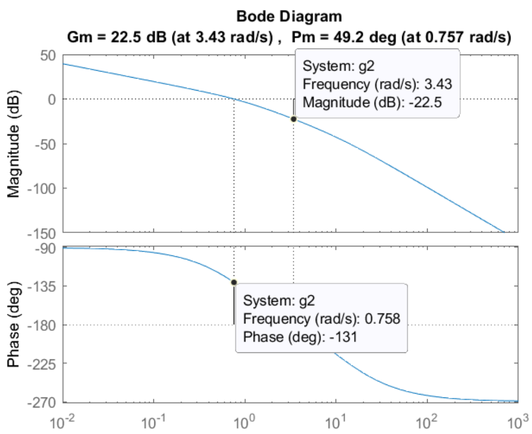
$$H_m(j\omega) = \frac{0,106}{0,085j\omega + 1}$$

$$H_{IP}(j\omega) = 0,75$$

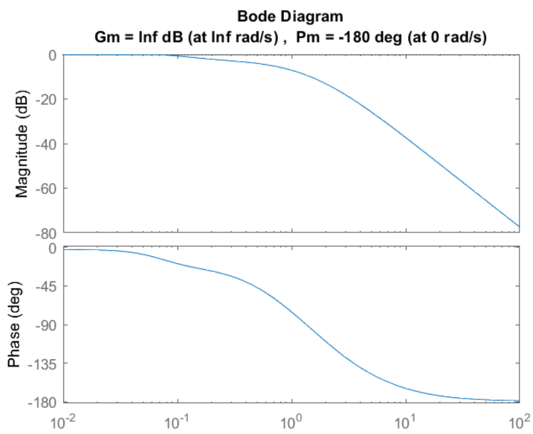
$$H_v(j\omega) = \frac{0,115}{j\omega + 1}$$

$$H_c(j\omega) = 125$$

$$H_{OL}(j\omega) = \frac{0,952}{0,085(j\omega)^3 + 1,085(j\omega)^3 + j\omega}$$



Gambar 44 bode plot open loop



Gambar 43 bode plot close loop

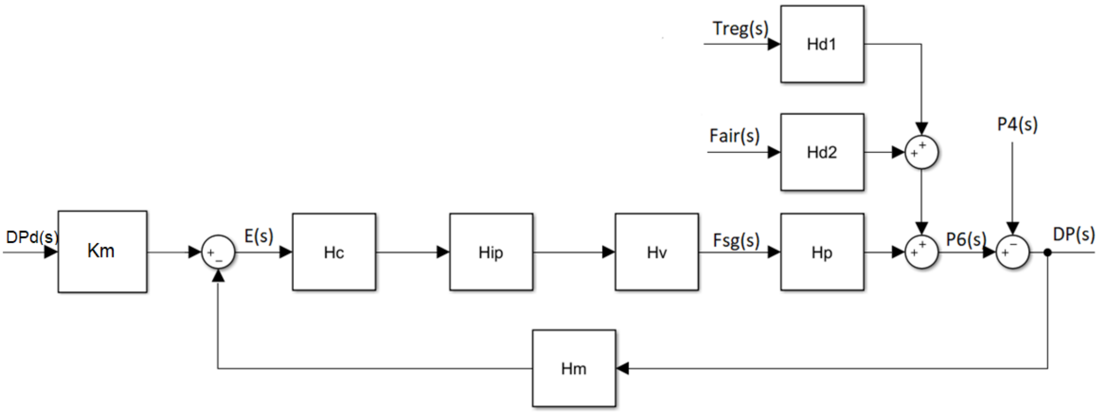
$$\boldsymbol{H(j\omega)} = \frac{0{,}081j\omega + 0{,}952}{0{,}085(j\omega)^3 + 1{,}085(j\omega)^2 + s + 0{,}952j\omega}$$

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	22,5 dB
2.	*Phase Margin* (PM)	49,2º

Sistem kontrol tekanan reaktor sudah stabil

g. Beda tekanan reaktor-regenerator



Gambar 48 diagram blok

$$H_{\Delta P_{RR6}}(j\omega) = -0{,}214$$

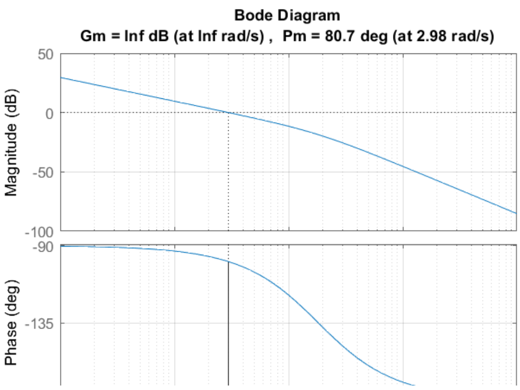
$$H_m(j\omega) = \frac{0{,}53}{0{,}055j\omega + 1}$$

$$H_{IP}(j\omega) = 0{,}75$$

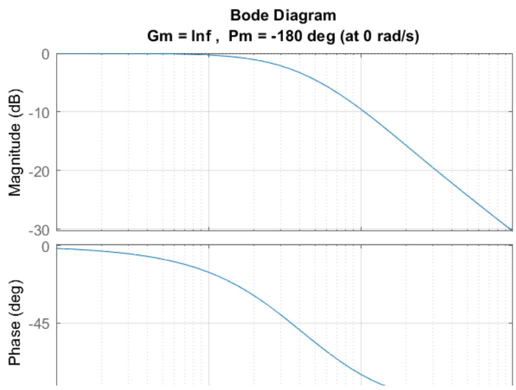
$$H_v(j\omega) = \frac{0{,}355}{j\omega + 1}$$

$$H_c(j\omega) = -100{,}1 - \frac{100{,}1}{j\omega}$$

$$H_{OL}(j\omega) = \frac{3{,}023j\omega + 3{,}023}{0{,}055(j\omega)^3 + 1{,}055(j\omega)^2 + j\omega}$$



Gambar 46 bode plot open loop



Gambar 47 bode plot close loop

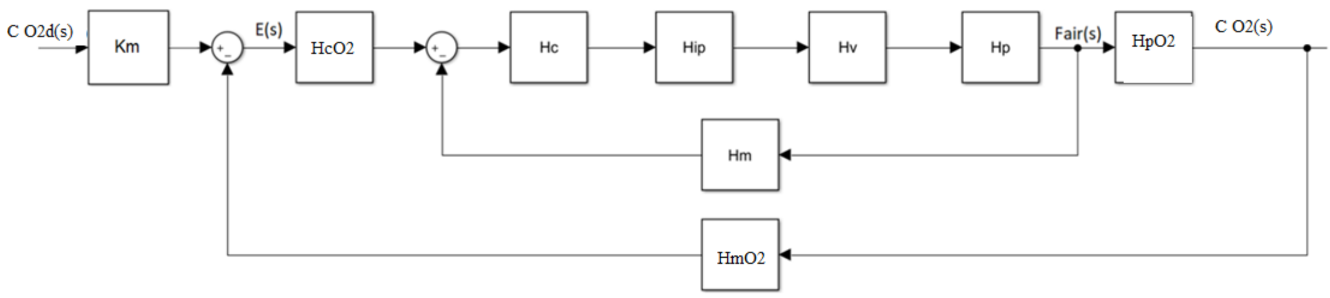
$$\boldsymbol{H(j\omega) = \frac{0,1663\,(j\omega)^2 + 3,189\,j\omega + 3,023}{0,055\,(j\omega)^3 + 1,055\,(j\omega)^2 + 4,023\,j\omega + 3,023}}$$

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	80,7º

Sistem kontrol beda tekanan reaktor-regenerator sudah stabil

h. Sistem kontrol konsentrasi oksigen pada *flue gas*



Gambar 49 diagram blok

- Slave

$$H_P(j\omega) = 1$$

$$H_m(j\omega) = \frac{0{,}192}{0{,}055j\omega + 1}$$

$$H_{IP}(j\omega) = 0{,}75$$

$$H_v(j\omega) = \frac{6{,}93}{j\omega + 1}$$

$$H_c(j\omega) = 1{,}002$$

$$\boldsymbol{H(j\omega) = \frac{0,286j\omega + 5,208}{0,055(j\omega)^2 + 1,055\,j\omega + 2}}$$

- Master

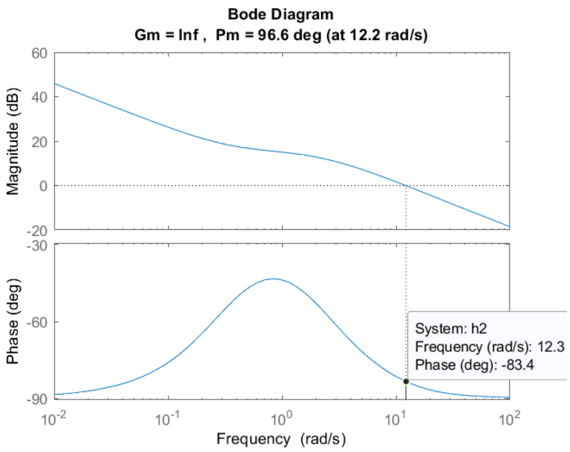
$$H_{PO_2}(j\omega) = 8{,}077$$

$$H_{mO_2}(j\omega) = \frac{0{,}16}{3j\omega + 1}$$

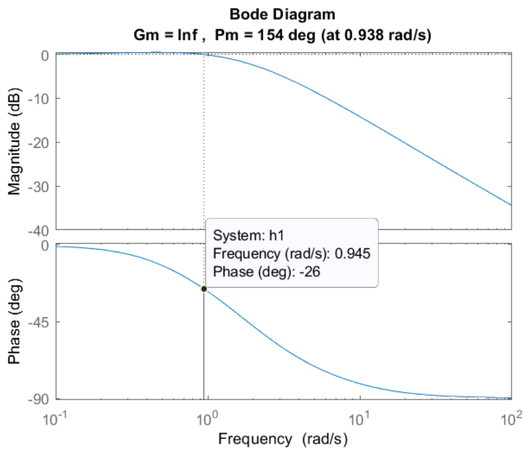
$$H_{cO_2}(j\omega) = 0{,}281 + \frac{0{,}094}{s}$$

$$\boldsymbol{H_{OL,O_2}(j\omega) = \frac{0,65(j\omega)^2 + 12,04j\omega + 3,954}{0,055(j\omega)^3 + 1,055(j\omega)^2 + 2j\omega}}$$

$$\boldsymbol{H_{O_2}(j\omega) = \frac{0,312(j\omega)^3 + 5,882(j\omega)^2 + 3,824j\omega + 0,634}{0,165(j\omega)^4 + 3,22(j\omega)^3 + 7,159(j\omega)^2 + 3,926j\omega + 0,633}}$$



Gambar 50 bode plot open loop



Gambar 51 bode plot close loop

berdasarkan hasil dari bode plot open loop didapatkan :

No.	Parameter	Nilai
1.	*Gain margin* (GM)	Tak hingga
2.	*Phase Margin* (PM)	96,6º

Sistem kontrol *konsentrasi oksigen pada* flue gas sudah stabil.

TUGAS MINGGU KE-9

1. Uraikan dengan jelas dan sistematis bagaimana proses perancangan di domain waktu dan proses perancangan di domain frekuensi akan selalu saling mempengaruhi.

   Jawab :

   Pada domain frekuensi, terdapat syarat yang harus dipenuhi berupa *gain margin* dan *phase margin open loop* agar sistem stabil. Apabila syarat tersebut tidak dipenuhi, maka sistem menjadi tidak stabil. Pada domain waktu, karakteristik sistem berupa *settling time, overshoot, rise time,* dan *steady state error* harus dipenuhi sesuai tujuan perancangan. Ketidakstabilan pada domain waktu ditunjukkan dengan respon sistem yang berosilasi dan tidak pernah mencapai kondisi *steady state* [3].

   Suatu sistem dapat stabil dan memenuhi karakteristik yang diinginkan pada domain waktu dan tidak berosilasi akan tetapi sistem tersebut tidak memenuhi kriteria kestabilan di domain frekuensi. Hal tersebut akan membuat sistem rawan menuju ketidakstabilan ketika input mempunyai frekuensi tertentu. Oleh karena itu dalam perancangan diharuskan meninjau domain waktu dan frekuensi secara bersamaan agar didapatkan karakteristik dan kestabilan yang sesuai pada kedua domain tersebut sehingga tujuan perancangan terpenuhi .

2. Pastikan bahwa sistem instrumentasi yang anda rancang memiliki karakter yang memenuhi semua tujuan perancangan yang perlu dipenuhi, baik di domain waktu maupun di domain frekuensi. Jika tujuan perancangan tersebut belum terpenuhi, maka silakan melakukan rekayasa terhadap sistem instrumentasi yang telah anda rancang sebelumnya, sedemikian sehingga tujuan perancangan tersebut dapat terpenuhi.

   Jawab :

   Tujuan perancangan :

$$\boldsymbol{domain\ waktu:\ T_r \leq 15\ s, T_s \leq 20\ s, \%OS \leq 20\%, e_{ss} = 0 - 0,05}$$

$$\boldsymbol{domain\ frekuensi: close\ loop\ stabil \rightarrow G_M > 20\ dB, P_M > 40^\circ}$$

__Tabel 2 Karakteristik sistem__

Sistem kontrol	$T_r(s)$	$T_s(s)$	$\%OS$	e_{ss}	$G_M(dB)$	$P_M(°)$
Wash oil flow	0,591	1,05	0	0	∞	80,7
Diesel oil flow	0,592	1,05	0	0	∞	80,7
Total fresh feed flow	0,592	1,05	0	0	∞	80,7
Slurry oil flow	0,592	1,05	0	0	∞	80,7
Preheater outlet feed temperature	8,9	17,8	0	0	35,6	80,6
Reactor pressure	1,63	8,51	18,8	0	22,5	49,2
Reactor-regenerator pressure diff.	0,587	1,04	0	0	∞	80,7
Flue gas oxygen concentration	0,871	6,12	6,25	0	∞	96,6

3. Setelah anda selesai dengan langkah kedua di atas, silakan nyatakan karakter sistem instrumentasi anda, baik di domain waktu maupun di domain frekuensi. Tunjukkan bahwa karakter-karakter tersebut telah memenuhi semua tujuan perancangan yang perlu dipenuhi.
   Jawab :

   Seperti yang ditunjukkan Tabel 2, semua sistem kontrol sudah memenuhi semua tujuan perancangan baik pada domain waktu maupun pada domain frekuensi, yaitu $T_r \leq 15\ s,\ T_s \leq 20\ s,\ \%OS \leq 20\%, e_{ss} = 0 - 0{,}05,\ G_M > 20\ dB, P_M > 40°$

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[3] N. S. Nise, *Control Systems Engineering, 7th Edition*, 7th edition. Wiley, 2014.